Dark-portal interactions of high-energy astrophysical neutrinos

Working notes

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Contents

1 Summary **3**

2 The model and its implementation **3**
- 2.1 Where the physics is defined . . . 3
- 2.2 From the equations to the code . . . 5
- 2.3 Potential textures . . . 5
- 2.4 Physical inputs . . . 5

3 The transport equation in full **6**
- 3.1 What is followed . . . 6
- 3.2 The cosmological phase . . . 7
- 3.3 The Galactic emission . . . 9
- 3.4 From the Sun to the detector . . . 9
- 3.5 The other transport: dp_si . . . 10
- 3.6 How the code solves it . . . 10

4 Numerical method **11**
- 4.1 Resolution and defaults . . . 11
- 4.2 Thermal absorption rates . . . 11
- 4.3 Cosmological mixing . . . 12
- 4.4 Coherent propagation . . . 12
- 4.5 Earth steps . . . 12
- 4.6 Parallelism and cost . . . 13

5 Validation **13**
- 5.1 The four benchmark points . . . 13
- 5.2 Record of validated changes . . . 14

6 Statistical analysis: current state **14**
- 6.1 Likelihood and parameters . . . 14
- 6.2 Scans and limits . . . 14
- 6.3 Thin excluded bands with fitted source compositions (2026-10-21) . . . 17
- 6.4 Where the fitted-composition regions end: missing tau neutrinos (2026-10-21) . . . 18

7 Secret neutrino interactions **19**
- 7.1 Model and cross sections . . . 19
- 7.2 Dirac neutrinos and a vector mediator . . . 20
- 7.3 Transport . . . 22
- 7.4 Validation . . . 23

7.5 The flavor-triangle planes 24
7.6 Four models 26
7.7 Secret interactions below 1 keV (2026-10-25) 28

8 Arrival directions and Galactic sources 37

9 Track rates with IceTracks-DR2 43

10 Starting events with HESE 44

11 Matter charges and textures: the scope of the dark-portal paper 45

12 The physics audit of 2026-10-14 47

13 Planned work 50
13.1 Event rates at IceCube 50
13.2 Systematic uncertainties 50
13.3 Statistical soundness 51
13.4 Computation 52
13.5 Relic motion in the collision transport (2026-10-18) 52
13.6 Dirac relics in motion: their helicity (2026-10-21) 53
13.7 Soft scatterings on the relics (2026-10-22) 54
13.8 Relic mass changes, neutrino decay, and where the models hold (2026-10-24) . . . 56
13.9 Processes below 1 keV: the catalogue (2026-10-24) 64

14 Open items 65

A Corrections 66
A.1 The relic-neutrino temperature (2026-10-12) 66
A.2 The cosmological and Galactic neutron densities (2026-10-12) 67
A.3 Neutron fractions along Earth chords (2026-10-12) 67
A.4 Point sources and cold disc gas (2026-10-12) 68
A.5 The length of the Earth chords (2026-10-12) 68
A.6 The Earth's own potential (2026-10-12) 69
A.7 The zenith average of the Earth chords (2026-10-12) 69
A.8 The cosmological cascade (2026-10-12) 71
A.9 The normalization of the resonant absorption (2026-10-12) 71
A.10 The unitarity of the spatial propagator (2026-10-12) 72
A.11 Both secondaries of every absorption (2026-10-13) 72
A.12 The transit to the Earth (2026-10-13) 73
A.13 The sources' redshift weighting (2026-10-13) 74
A.14 The Galactic gas at $N_{\mathrm{side}} = 64$ (2026-10-14) 76
A.15 The mixing-parameter priors (found 2026-10-12) 76
A.16 The thermal kernels' grid (2026-10-21) 76
A.17 Two rounding failures below 1 keV (2026-10-24) 77
A.18 Earlier corrections 77

1 Summary

What the code computes, and where it stands. We compute the diffuse flux of high-energy astrophysical neutrinos at IceCube in the dark-portal model of the theory notes (`notes/main.pdf`, "Notes on the theoretical framework for generalized new neutrino interactions"). A dark vector boson Z_D couples to active neutrinos through their mixing with a sterile state and to matter through Z–Z_D mass mixing. Two effects follow: neutrinos scatter resonantly on the relic (cosmic) neutrino background, which absorbs and regenerates them, and neutrons source a long-range coherent potential that changes their flavor in flight. The solver integrates both across cosmological distances, the Milky Way and the Earth; it is validated at four benchmark parameter points and runs in 0.71 s at its production defaults on 8 cores (1.04 s on one). The statistical analysis fits the predicted flavor composition to a projected 2050 measurement; it carries no explicit systematic uncertainties yet. The next stage, agreed on 2026-10-12, adds IceCube event rates, systematic uncertainties and coverage tests (Sec. 13). On 2026-10-12 we corrected the relic-neutrino temperature, every neutron number that sources the potential and the length of the Earth chords, and we now compute the Earth's own potential as the full Yukawa integral over its neutrons (Appendix A). The same day we removed the chord-sampling floor of the Earth transit, which had left $\sim 10^{-2}$ of noise in the predicted flavor fractions (Appendix A.7). Late that day we found three errors in the solver itself, present since its first version, and corrected them: the cosmological cascade created neutrinos without bound, the resonant absorption did not depend on the size of the couplings, and the propagator of the Milky Way and Earth transits was not unitary (Appendices A.8–A.10); on 2026-10-13 we made every absorption feed both of its secondaries, a neutrino and an antineutrino, where only one had been kept (Appendix A.11). Also on 2026-10-13 we redesigned the transit from the cosmological arrival to the Earth (Appendix A.12): adiabatic through the Galaxy, at its Yukawa-smoothed density, and coherent through the Sun's potential and the Earth's exterior, with eigenvectors that stay accurate at potentials 10^{40} times the vacuum term; the cosmological potential no longer stops growing below 3×10^{-13} eV. Every result of the statistical analysis, and every effect size quoted for the earlier corrections, was obtained before these fixes. Also on 2026-10-13 the project widened to secret neutrino interactions in general (Sec. 7): a scalar mediator with the full cross sections of arXiv:2107.13568, validated against that paper and run through the flavor-triangle analysis.

Table 1 lists the state of each component. Every number in these notes comes from a run of the code, identified by the commit that contains it or by the test that checks it; anything not yet traced is marked [**VERIFY**]. Corrections to the code and to earlier statements are kept in Appendix A, because they are the reason the remaining numbers can be trusted. The agreed order of work is also kept, in shorter form, in `docs/dev/WORK_PLAN.md`.

2 The model and its implementation

2.1 Where the physics is defined

The theory notes define the framework; we cite their equations as "Eq. (Tn)". The neutrino ensemble is a 3×3 density matrix $\rho(E_0, z)$, where E_0 is the energy observed at Earth and z the redshift. It obeys the quantum kinetic equation (T3). The Hamiltonian (T4) drives its coherent part; the collision operator (T6) is its incoherent part. The coherent part contains the standard matter potential (T5) and the long-range dark-portal potential (T9) for neutrinos, (T10) for antineutrinos. The incoherent part contains the attenuation rate (T7) and the regeneration kernel (T8), evaluated with a Breit–Wigner resonance of width (T11) and coupling (T12). The

Table 1: **State of each component, as of 2026-10-12.** "Validated" means checked at the four benchmark points of Sec. 5 with the acceptance rules stated there.

Component	State	Section
Flux solver	Validated; cascade, absorption normalization and propagator corrected 2026-10-12; 0.71 s on 8 cores	4, 5
Thermal absorption rates	Semi-analytic kernel; relic temperature corrected 2026-10-12	4.2, A.1
Earth transit	Phase-exact stepping with a step-count certificate; 25 steps; the Earth's potential as its Yukawa integral	4.5, A.6
Flavor-ratio analysis	Results predate the solver corrections and are void; being rebuilt	6
IceCube event rates	Planned; data sets and folding method chosen	13.1
Systematics and coverage	Planned; list and procedure agreed	13.2, 13.3
Neutron densities	All corrected 2026-10-12	2.4, A
Earth-chord sampling	Floor removed by the phase-aware zenith average (2026-10-12)	4.5, A.7
Transit to the Earth	Redesigned 2026-10-13: adiabatic Galaxy, coherent Sun and Earth exterior; checked against brute force	A.12
Secret interactions	Scalar mediator, Majorana neutrinos, full cross sections (2026-10-13); validated; flavor-triangle planes. Dirac and vector next	7

sources follow a power law with a free flavor composition (T18), distributed in redshift as the star-formation rate (T19, T20). The thermal (Debye) mass that screens the mediator is (T23).

Parameters.—The dark sector is set by the mediator mass m_{Z_D}, the dark gauge coupling g_D, the Z–Z_D mass mixing ϵ', and the active–sterile mixing magnitudes $u_\alpha \equiv |U_{\alpha 4}|$ ($\alpha = e, \mu, \tau$). Resonant absorption on a relic neutrino of mass m_i peaks near $E_{\rm res} = m_{Z_D}^2/(2m_i)$, so a mediator of mass $\sim$ 1–10 MeV places the absorption features inside the IceCube energy range. Table 2 lists every parameter with its default value and configuration key. The defaults live in one place, `src/dp_config.py` (`DEFAULT_CONFIG`); the analysis configurations carry their own copies of the physical constants.

Table 2: **Parameters and their defaults.** Keys refer to the solver's configuration sections; the analysis configurations use the same names under `physics_constants`. Masses and energies in eV unless stated.

Symbol	Meaning	Default	Key
m_{Z_D}	Dark mediator mass	2×10^6	`dark_portal.m_Zd_eV`
g_D	Dark gauge coupling	10^{-4}	`dark_portal.g_D`
ϵ'	Z–Z_D mass mixing	0.1	`dark_portal.epsilon_prime`
u_e, u_μ, u_τ	Active–sterile mixing	$0.5, 0.5, 0$	`dark_portal.u_*`
p	Source spectral index	2.5	`source.p_index`
$f_{e,S}, f_{\mu,S}, f_{\tau,S}$	Source flavor composition	$1/3, 2/3, 0$	`source.f_*_S`
$z_{\max}$	Largest source redshift	4	`source.z_max`
$m_{\rm lightest}$	Lightest neutrino mass	0.01	`cosmology.m_lightest_eV`
$T_{\nu,0}$	Relic-neutrino temperature today	1.676389×10^{-4}	`cosmology.T_nu0_eV`
$n_{n,0}$	Cosmological neutron density	3.091×10^{-8} cm^{-3}	`cosmology.n_n0_cm3`
$\theta_{12}, \theta_{13}, \theta_{23}$	Mixing angles	$33.41°, 8.58°, 49.1°$	`standard_mixing.th*_deg`
$\delta_{\rm CP}$	Dirac phase	$132°$	`standard_mixing.d_cp_deg`
$\Delta m_{21}^2, \Delta m_{31}^2$	Mass splittings [eV^2]	$7.41 \times 10^{-5}, 2.511 \times 10^{-3}$	`standard_mixing.dm*_sq`

2.2 From the equations to the code

The solver (`src/`, entry point `run_solver` in `src/dark_portal_solver.py`) divides the propagation into three phases. Each phase hands its result to the next, and neutrinos and antineutrinos run separately before their fluxes are combined.

Cosmological phase.—From $z_{\max} = 4$ to $z = 0$ the flux is carried as populations in the instantaneous eigenbasis of the Hamiltonian (T4), one per mass state, with no coherences between them. At each step the populations decay at the thermal absorption rate (T7) and receive the regenerated flux of the cascade kernel (T8) and the injected source flux (T18). We project them onto flavor with the mixing probabilities $|U_{\alpha i}(E, z)|^2$ of the full Hamiltonian. The cascade kernel is separable, which reduces the regeneration integral to a suffix sum that costs $O(N_E)$ per step instead of $O(N_E^2)$ (`step_cosmological_propagation` in `src/dp_cosmo.py`).

Transit to the Earth (since 2026-10-13).—Through the Galaxy the populations are carried adiabatically: they become the same populations of the local eigenstates at the Sun, where the potential is the Yukawa integral of the Galaxy's neutrons over the mediator's range. Through the Sun's potential and the Earth's exterior the density matrix is integrated coherently in the instantaneous eigenbasis, along eight directions to the Sun and one radial path for the Earth (`src/dp_transit.py`, `transit_to_earth` in `src/dp_spatial.py`; Appendix A.12). Until 2026-10-13 a Milky Way phase evolved the density matrix from 100 kpc along six directions in 25 steps each.

Earth phase.—The density matrix is then evolved along $n_\oplus$ chords through the Earth (PREM densities, optional standard matter potential) to a detector 1.5 km deep, starting at twice the Earth's radius below $m_{\text{eff}} = 2 \times 10^{-5}$ eV, where the exterior matters. The chords are evenly spaced in the cosine of the zenith angle over the whole sky, and the zenith average integrates their dark-portal phase (Sec. 4.5). The flavor fluxes at the detector are the diagonal of the averaged density matrix.

Antineutrinos.—The antineutrino run uses U^*_{PMNS} and the transformed potential $-V^*$ (T10). With the default `is_antineutrino = "sum"` the two runs are added, weighted by `nubar_to_nu_ratio` (= 1).

2.3 Potential textures

The flavor structure of both coherent potentials can be replaced through the configuration: `dark_portal.U_LRI_struct` for the dark-portal potential (default $u_\alpha u^*_\beta$, as in Eq. (T9)) and `standard_mixing.V_SM_struct` for the standard matter potential (default $\text{diag}(1, 0, 0)$, as in Eq. (T5)). Each is a 3×3 nested list whose entries are real numbers or [Re, Im] pairs. The regression tests `test_structure_override_*` and `test_parse_structure_matrix_3x3_formats_and_validation` check the parsing and the antineutrino convention. In addition, we ran rank-zero, full-rank complex and indefinite textures through both spatial phases and the cosmological mixing table (2026-07-29); any fixed Hermitian texture works.

Limits of the override.—The texture must be Hermitian to 10^{-12} relative precision; the parser rejects anything else, because the propagator reads only one triangle of the Hamiltonian. It must be constant within a run, since the cosmological mixing table assumes a Hamiltonian of the form $M + \sigma S$ with one fixed matrix S. Overrides are given in the neutrino convention; the solver applies the antineutrino transformation itself. The override changes only the coherent potential: the Debye mass and the absorption rates remain set by u_e, u_μ, u_τ. Its overall scale is degenerate with $g_D\epsilon'$.

2.4 Physical inputs

Sources.—A diffuse population following the star-formation history up to $z = 4$, with spectrum $E^{-2.5}$, flavor ratios 1/3 : 2/3 : 0 (pion decay) and equal numbers of neutrinos and antineutrinos. At each redshift the injected flavors split incoherently over the eigenstates of the Hamiltonian there; by default the sources sit in the cosmic mean neutron density, and `source.host_galaxy_scale` places them inside a Milky-Way-like host instead (Sec. 6).

Relic neutrinos.—The absorption rates integrate the Fermi–Dirac distribution of the relic neutrinos at temperature $T_{\nu,0}(1+z)$. Therefore, the relic density in our rates follows from $T_{\nu,0}$ and scales as $T^3_{\nu,0}$. The nominal 56 cm^{-3} per state enters only as a ratio that carries flavor asymmetries. The temperature is $T_{\nu,0} = (4/11)^{1/3}\, T_{\mathrm{CMB}} = 1.9454$ K $= 1.676389 \times 10^{-4}$ eV for $T_{\mathrm{CMB}} = 2.7255$ K. Until 2026-10-12 the code used 1.95×10^{-4} eV (Appendix A.1).

Neutron densities.—The dark-portal potential is proportional to the neutron number density along the path, because the Z-like couplings of protons and electrons cancel in neutral matter. Cosmologically, the neutrons sit almost entirely in helium-4, $Y_p/2$ per baryon, so $n_{n,0} = n_b Y_p/2 = 3.091 \times 10^{-8}$ cm^{-3} for $n_b = 2.515 \times 10^{-7}$ cm^{-3} (PDG 2024) and $Y_p = 0.2458$ (LBT Y_p project, 2026). In the Galaxy we convert the stellar mass density at 0.20 neutrons per nucleon, the electron density of the hot halo at 0.15 neutrons per electron, and the cold HI and H_2 discs of McMillan (2017) at 0.142 neutrons per nucleon (Appendices A.2 and A.4). The Sun, the Moon and the Earth also act as point sources, with 1.929×10^{56}, 2.232×10^{49} and 1.849×10^{51} neutrons. In the Earth we use $1 - Y_e$ neutrons per nucleon, with electron fractions $Y_e = 0.4656$ in the core and 0.4957 in the mantle and crust, and the composition of water for PREM's top layer (Appendix A.3).

The Earth's own potential.—The potential at a point is a Yukawa integral over the neutrons around it, with range $1/m$, $m = (m^2_{Z_D} + m^2_D)^{1/2}$. Distant bodies need only its far limit, a point source; the cosmic mean density, uniform, gives n/m^2, cut at the Hubble radius (Appendix A.12). A local (contact) density holds only where $1/m$ is short against every length over which the density varies: for the Galaxy, below $\sim 10^{-25}$ eV, we integrate its model over the range (Appendix A.12). Inside the Earth the range can match the size of the body, so we evaluate the integral itself. For a spherical profile it is one-dimensional,

$$n_{\mathrm{eff}}(r) = m^2 \left[\frac{e^{-mr}}{r} \int_0^r r'^2 n(r')\, \frac{\sinh mr'}{mr'}\, dr' + \frac{\sinh mr}{mr} \int_r^R r' n(r')\, e^{-mr'} dr' \right], \tag{1}$$

and the potential is the contact expression evaluated with n_{eff} in place of n. This is the form used for the long-range potential in the Sun in Magnus and in the Earth in NuOscProbExact (notebook 20). Appendix A.6 gives our evaluation and its tests.

3 The transport equation in full

This section collects, in one place, the equations the code solves from the sources to the detector: every term, every source, and where each lives in the code (Mauricio, 2026-10-25; to be used for the paper). Later sections derive the individual rates; here they are only defined. The status of each term is that of 2026-10-25.

3.1 What is followed

Kinds and states.—Neutrinos and antineutrinos are followed separately, as numbers per unit energy, F_a and $\bar{F}_a$, in the eigenstates $a = 1, 2, 3$ of the Hamiltonian of each kind,

$$H(E,z) = \frac{U\, \mathrm{diag}(m_1^2, m_2^2, m_3^2)\, U^\dagger}{2E(1+z)} + V(z), \qquad \bar{H}(E,z) = \frac{U^* \mathrm{diag}(m_i^2)\, U^T}{2E(1+z)} - V^*(z), \tag{2}$$

with $V(z) = A(z)\, S$ the long-range potential (texture S; Sec. 2.3). Its eigenvalues are $\lambda_a(\sigma)/2E(1+z)$, λ_a the eigenvalues of $UM^2U^\dagger + \sigma S$ (the effective masses squared), and its eigenvectors $|a\rangle$ depend on E and z only through $\sigma = 2E(1+z)A(z)$. Each state's flavor content is $P_{\alpha a}(\sigma) = |\langle\nu_\alpha|a\rangle|^2$ and its mass content $T_{ia}(\sigma) = |\langle\nu_i|a\rangle|^2$; both are tabulated on nodes in $\log\sigma$ (`src/dp_cosmo.py`). The labels a follow the eigenstates as H changes (sorted eigenvalues): the cosmological potential changes over a Hubble time, against oscillation lengths of at most parsecs, so the evolution is adiabatic and no term moves populations between states except the collisions and decays below. With no potential the eigenstates are the mass states, $T_{ia} = \delta_{ia}$.

Energy and time.—E is the energy observed today; the physical energy at redshift z is $E(1+z)$, so the redshift needs no term of its own. Time is cosmic time, $dt = -dz/[(1+z)H(z)]$. All rates below are physical rates at redshift z.

3.2 The cosmological phase

From $z_{\max} = 4$, where $F_a = \bar{F}_a = 0$, to $z = 0$:

$$
\begin{aligned}
\left.\frac{\partial F_a(E,z)}{\partial t}\right|_E = \underbrace{\mathcal{S}_a}_{\text{sources}} \; & \underbrace{- \Gamma_a^{\text{abs}} F_a + \beta_a \int_E^\infty dE' \Big[\frac{3E^2}{E'^3} \sum_c \Gamma_c^{\text{abs}} F_c + \frac{3(E'-E)^2}{E'^3} \sum_c \bar{\Gamma}_c^{\text{abs}} \bar{F}_c\Big]_{E'}}_{\text{resonant absorption on the relics, and its secondaries}} \\
& \underbrace{+ \sum_{b\neq a} \big(R_{ab}F_b - R_{ba}F_a\big)}_{\text{soft scatterings on the relics}} \quad \underbrace{+ \frac{\partial}{\partial E}\big(b_a F_a\big)}_{\text{relic mass-change loss}} \\
& \underbrace{- \Gamma_a^{\text{dec}} F_a + \sum_{a'} \int_E^\infty dE' \Big[\frac{d\Gamma^{\text{keep}}_{a'\to a}}{dE}(E' \to E)\, F_{a'}(E') + \frac{d\bar{\Gamma}^{\text{flip}}_{a'\to a}}{dE}(E' \to E)\, \bar{F}_{a'}(E')\Big]}_{\text{decay into the mediator, and its daughters}},
\end{aligned}
\tag{3}
$$

and the same for $\bar{F}_a$ with $U \to U^*$, $V \to -V^*$, $G \to G^*$ (barred and unbarred quantities exchanged in the secondaries: an absorbed antineutrino gives an antineutrino at $3E^2/E'^3$ and a neutrino at $3(E'-E)^2/E'^3$, and a flipped daughter of a neutrino is an antineutrino). The terms:

Sources.—Sources of comoving density $R(z)$ (the star-formation rate, Eqs. T19, T20, with the extra evolution $(1+z)^{p_{\text{evol}}}$) that emit $\Phi_0 E_s^{-p}$ per unit source energy, split incoherently over the eigenstates at the source:

$$
\mathcal{S}_a(E,z) = w_\nu\, \Phi_0\, [E(1+z)]^{-p}\, R(z)\, (1+z)\, \Theta_{\text{inj}}(E) \sum_\alpha f_{\alpha,S}\, P_{\alpha a}(\sigma_{\text{src}}), \tag{4}
$$

the $(1+z)$ being dE_s/dE (Appendix A.13); w_ν, $w_{\bar{\nu}}$ the two kinds' weights ($\nu : \bar{\nu} = 1 : 1$ by default); Θ_{inj} the emission window (`source.injection_E_min/max_GeV`; one by default); σ_{src} at the sources' density, the cosmic mean or a host galaxy's (`source.host_galaxy_scale`). The broken power law of the analysis is built from runs at fixed p (spectral-index nodes, interpolated in $\log F$; Sec. 6). Code: `dp_cosmo.step_cosmological_rates`, `dp_sweep.step_full`.

Resonant absorption and its secondaries.—$\nu_a +$ relic $\to Z_D \to \nu + \bar{\nu}$. The rate of state a is a sum over the relic mass states r,

$$
\Gamma_a^{\text{abs}}(E,z) = \sum_r \Gamma_r(E,z)\, Q_{ra}(\sigma), \qquad Q_{ra} = \tfrac{1}{2}\big|(U^\dagger G\, W)_{ra}\big|^2, \tag{5}
$$

W the eigenvectors of H in the flavor basis, Γ_r the Breit–Wigner rate on relic r per unit coupling weight, averaged over its thermal momenta at $T_{\nu,0}(1+z)\eta_r^{1/3}$ (Sec. 4.2; normalization of Appendix A.9; the relics' composition η_r after their own decays, item 2 of Sec. 13.8). Each absorption emits two secondaries (Appendix A.11): one of the absorbed kind with the spectrum $3E^2/E'^3$, one of the other kind with $3(E'-E)^2/E'^3$, each in eigenstate a with the share $\beta_a = \mathrm{Re}(W^\dagger G G^\dagger W)_{aa}/(2\Sigma G)$ ($\sum_a \beta_a$ is the mediator's branching into neutrinos). It matters in the MeV planes, where the resonance energy $m_{Z_D}^2/2m_r$ falls in the band. Code: the same functions, the kernels integrated exactly over the bins (`cascade_kernel_weights`, `cross_kernel_weights`).

Soft scatterings on the relics.—The forward-peaked t channel below 1 keV moves neutrinos between the eigenstates at their own energy (Sec. 13.7):

$$R_{ba} = \Gamma_\mathrm{u}(z)\,|G_{ba}|^2\,\frac{\omega_{ab}^2}{\omega_{ab}^2 + D_{ab}^2}, \quad \omega_{ab} = \frac{\lambda_a - \lambda_b}{2E(1+z)}, \quad D_{ab} = \Gamma_\mathrm{u}\Big[\frac{(G^2)_{aa} + (G^2)_{bb}}{2} - G_{aa}G_{bb}\Big] + \frac{\Gamma_a^\mathrm{dec} + \Gamma_b^\mathrm{dec}}{2}, \tag{6}$$

with $G_{ba} = \langle b|G|a\rangle$ and $\Gamma_\mathrm{u}(z) = 2n_\mathrm{hel}(1+z)^3 \sum_r \eta_r (G'G'^\dagger)_{rr}/[4\pi M_t^2(z)]$, $G' = U^\dagger G U$, $n_\mathrm{hel} = 3\zeta(3) T_{\nu,0}^3/4\pi^2$ the relics per helicity and mass state today, $M_t^2 = m_{Z_D}^2 + m_D^2(z)$ (m_D the relics' Debye mass). The decay widths in D protect a fast-decaying state from conversions into it (Zeno). Kept wherever they move more than 10^{-4} of the flux over the path (`SOFT_JUMP_MIN`). Code: `dp_cosmo._soft_jumps`, `dp_sweep._jump_rates`.

Relic mass-change loss.—Soft scatterings that leave the relic in another mass state cost the beam $\omega_{rs} = (m_s^2 - m_r^2)/2p_r^-$ each (item 1 of Sec. 13.8): a loss of observed energy at the rate

$$b_a(E,z) = \frac{C(z)\,\big[(G+\kappa)(G+\kappa)^\dagger\big]_{aa}}{1+z}, \qquad C(z) = \sum_{r\neq s} \frac{2n_\mathrm{hel}(1+z)^3 \eta_r |G'_{sr}|^2}{4\pi M_t^2}\,\frac{m_s^2 - m_r^2}{2}\Big\langle \frac{1}{E_r}\Big\rangle_{T(z)}, \tag{7}$$

independent of E (κ the Z mixing's universal coupling; zero below 1 keV in the planes). Code: `src/dp_relic_loss.py`.

Decay into the mediator, and its daughters.—$\nu \to \nu' + Z_D$ where the masses allow it (m_{Z_D} below the mass difference). A neutrino in state a' at energy E' gives a daughter in state a at $E < E'$ that keeps its helicity (a neutrino, "keep") or, the Majorana vertex's P_R piece, flips it (an antineutrino, "flip"), and $\Gamma_a^\mathrm{dec} = \sum_b \int dE\,[d\Gamma_{a\to b}^\mathrm{keep} + d\Gamma_{a\to b}^\mathrm{flip}]/dE$. The mediator is taken as invisible (it decays back to neutrinos only for $m_{Z_D} > 2m_1$, into the softest part of the spectrum; decision B of 2026-10-24 checks this). Every decay keeps one neutrino, so these two terms conserve the number. Without a potential the states are the mass states and the rates are the vacuum ones (item 2 of Sec. 13.8, and its kinematics paragraph): $\Gamma_{i\to j} = \Gamma_{ij}^\mathrm{rest}\, m_i/[E'(1+z)]$ and

$$\frac{d\Gamma_{i\to j}^\text{keep, flip}}{dE}(E' \to E) = \frac{\Gamma_{i\to j}}{E'}\,\big[a + b\,x + c/x\big]_{ij}^\text{keep, flip}, \qquad x = E/E' \in [x_-, x_+], \tag{8}$$

with the exact light-front helicity spectra of `dp_decay.daughter_spectra` (its z is this x); the daughter's share x is the same in observed and physical energy. In a potential the decays act between the eigenstates of H (Sec. 13.8, "Decay in a medium"): a parent in state a at E' gives a daughter in state b at $E = xE'$, each with its own eigenvector, at its own σ (the parent's σ, the daughter's $x\sigma$); the effective masses set the window, $k_T^2 = (1-x)\,[x\,\mu_a(\sigma) - \mu'_b(x\sigma)] - x m_{Z_D}^2 \geq 0$ (Eq. 27), and the rate is $d\Gamma_{a\to b}/dx = |\mathcal{A}|^2/[16\pi E'(1+z)]$ with the coherent amplitudes of Eqs. (26) and (28). A state cannot decay into itself, nor into a higher state of its own kind; the lowest state of a kind decays only by flips, and a potential opens $\nu_i \to \bar\nu_i$, closed in vacuum. Without a potential this is Eq. (8) exactly. It holds while each width is small against the splittings of its kind, $\zeta_a = 2E'(1+z)\Gamma_a/\min_b|\mu_a - \mu_b| \ll 1$: Fermi's golden rule between the eigenstates, with corrections of order ζ (at most 1.5×10^{-2} at the first validation points, but up to 0.1–0.5 at couplings $g_D/m_{Z_D} \sim$ 10–100 eV^{-1} just below the unitarity line, where the new limits may fall; being mapped and recorded per run, 2026-10-25). Until 2026-10-25 the code made each state decay at $\sum_i T_{ia}\Gamma_i$ with vacuum kinematics, which let the most stable state decay into itself and slid the flux down in energy. Code: `src/dp_medium.py`, `src/dp_decay.py` (the vacuum spectra), `dp_sweep.step_full`.

The mediator's longitudinal mode in the scatterings (since 2026-10-25).—With massive neutrinos the vector's propagator keeps its $q^\mu q^\nu/m_{Z_D}^2$ part: a beam neutrino scatters on the relics also as through a scalar with the Goldstone couplings $Y_{ij} = i(G'_{ij}m_i + G'_{ji}m_j)/m_{Z_D}$, without interference with the transverse part and not forward-peaked (checked with explicit spinors; Sec. 7, "The longitudinal-mode scatterings on the lepton-number planes"). These are hard scatterings: an absorption term $-\Gamma^{\rm L}_a F_a$ with its secondaries, the scalar channels of the secret interactions with $\lambda = Y$. On the lepton-number planes they enter as a factor per mass state on the eigenstate populations at $z = 0$, from two runs of the mass-basis transport with and without them; exact where the potential leaves the eigenstates mass states where the absorption happens, the bound recorded per run. They grow with the masses: at $g_D/m_{Z_D} = 0.1$ eV^{-1} they leave the flux at 10 TeV at 0.95 of itself for the lightest mass 10^{-6} eV and at 0.03 for 0.1 eV. Left out, with an after-campaign check that the band $g_D/m_{Z_D} \geq 10^{-2}$ eV^{-1} is excluded without them: the dark-portal planes, Dirac neutrinos and the vector secret interactions.

The grid's edges.—The equation is solved on the analysis grid extended internally one decade above its top (`GRID_EXTENSION_DECADES`), since decays, losses and secondaries move neutrinos down in energy; the Galactic emission's grid only up to the templates' end, 10^8 GeV (half a decade). What falls below the grid's bottom (~ 4.6 TeV) leaves the band and is dropped.

3.3 The Galactic emission

Neutrinos emitted in the Galaxy (Sec. 8) reach the Sun along each line of sight $\hat{n}$ from the distance ℓ at which they are emitted, at $z = 0$. With $G_a(E, \ell, \hat{n})$ the number per unit energy travelling toward the Sun,

$$-\frac{\partial G_a}{\partial \ell} = j_a + \sum_{b\neq a} \big(R_{ab}G_b - R_{ba}G_a\big) + \frac{\partial}{\partial E}\big(b_a G_a\big) - \Gamma^{\rm dec}_a G_a + \sum_{a'} \int_E^\infty dE' \Big[\frac{d\Gamma^{\rm keep}_{a'\to a}}{dE}\, G_{a'} + \frac{d\bar{\Gamma}^{\rm flip}_{a'\to a}}{dE}\, \bar{G}_{a'}\Big], \tag{9}$$

with every rate of Eq. (3) at $z = 0$ and in the eigenstates of the local potential, and the emission

$$j_a(E, \ell, \hat{n}) = \tfrac{3}{2}\, N_{\rm G}\, I(E, \hat{n}) \big[w_{\rm gas}(E, \hat{n})\, q_{\rm gas}(\ell, \hat{n}) + (1 - w_{\rm gas})\, q_{\rm src}(\ell, \hat{n})\big] \sum_\alpha f_\alpha\, P_{\alpha a}(\sigma(\ell)), \tag{10}$$

I the template's intensity (CRINGE, per flavor, $\nu + \bar{\nu}$: three flavors' worth, half per kind; `dp_galactic.emitted`), $N_{\rm G}$ its normalization (a fit parameter), $w_{\rm gas}$ its gas share, $q_{\rm gas}$, $q_{\rm src}$ the distributions along the line of sight of the gas emission and of the unresolved sources ($\int q\, d\ell = 1$), f_α the composition $(x, 1-x, 0)$. The resonant absorption is left out: the Galactic relic column is $\sim 10^{-5}$ of the cosmological one. Code: `src/dp_galactic_path.py` (97 distance bins, emission at each bin's centre); without the new processes the populations at emission go straight to the Sun (`dp_galactic.emission_populations`).

3.4 From the Sun to the detector

The cosmological populations at $z = 0$ cross the Galaxy adiabatically: each becomes the same population of the local eigenstate at the Sun, where the potential is the Yukawa integral of the Galaxy's charges over the mediator's range (Appendix A.12). There both components form the density matrix $\rho = \sum_a (F_a + G_a)\, |a\rangle\langle a|$ (and $\bar{\rho}$), which evolves coherently through the Sun's potential, the Earth's exterior and the Earth along each chord,

$$\frac{d\rho}{dx} = -i\,[H(x), \rho], \qquad H(x) = \frac{UM^2U^\dagger}{2E} + V(x) + V_{\rm SM}(x), \tag{11}$$

and the flux of flavor α at the detector is $\langle \nu_\alpha | \rho | \nu_\alpha \rangle$, averaged over the chords with the phase-aware zenith average (Sec. 4.5, App. A.7), per declination band or for the whole sky. No collisions and no decays act from the Sun on. For the decays this was checked with the in-medium widths (Sec. 13.8, “Decays the transport leaves out”): across the Sun and the Earth a state decays by more than 10^{-3} only where the same potential’s coherent effects are already overwhelming ($\sigma/\Delta m^2_{21} \geq 840$ and 3.4×10^5), and the lowest states, which an in-band beam arrives as, never do.

The extragalactic beam’s crossing of the Galaxy (since 2026-10-25).—Where the Galaxy’s potential opens decays that the cosmic one does not (flips between the two kinds’ lowest states, $g_D/m_{Z_D} \gtrsim 10^{-2}$ eV^{-1}), the beam’s populations at $z = 0$ are not carried adiabatically to the Sun but evolved along every line of sight through the Galaxy by Eq. (9) without its emission term ($j_a = 0$), with the same rates, then averaged per declination band and handed to the Sun and the Earth (`EXTRAGALACTIC_CROSSING`; Sec. 13.8, “Decay in a medium”). Measured effect at $g_D/m_{Z_D} = 0.1$ eV^{-1}: 13–41% of the flux at the top of the band. Both kinds of Galactic paths, the emission’s and the crossing’s, run on an $N_{\rm side} = 4$ sky (`GALACTIC_PATH_NSIDE`, `CROSSING_PATH_NSIDE`): against $N_{\rm side} = 8$ the crossing changes by 3×10^{-4} and the Galactic emission by 5.5% (inside its free normalization’s 18% prior), and the fits’ χ^2 by 0.1–0.3 at $\Delta\chi^2 \approx 30$ (the pilot of 2026-10-25), at a quarter of the cost.

3.5 The other transport: dp_si

The same physics in the mass basis, with every $2 \to 2$ channel and the pair production of the secret interactions (Sec. 7.3), carries the cosmological phase where the potential is negligible: the secret-interaction planes (below 1 keV with the decays, the relics after their decays and the vector’s screening and relic loss, Sec. 7.7), the MeV planes’ `'si'` points, and below 1 keV the points with hard scatterings (couplings above 10^{-4}) or decays and no potential (`dark_portal_solver._cosmology_mode`). For one Majorana flux holding both kinds, $F_i = F_i^{\nu} + F_i^{\bar\nu}$,

$$\left.\frac{\partial F_i}{\partial t}\right|_E = \mathcal{S}_i - \Gamma_i F_i + \sum_j \int_E^\infty dE'\, K_{ij}(E, E')\, \Gamma_j(E') F_j(E') - \Gamma_i^{\rm dec} F_i + \sum_j \int_E^\infty dE'\, \frac{d\Gamma_{j\to i}}{dE}(E' \to E)\, F_j(E'), \tag{12}$$

Γ_i the total collision rate on the relics (all channels; their exact thermal averages with `thermal_products`), K_{ij} the secondaries’ spectra (two per $2 \to 2$ collision, four per pair production; the forward peak’s returns as a drift in $\ln E$), and the decays with both daughter halves added. Where the potential matters and the hard scatterings act (`'combined'`), the dark-portal phase’s populations are rescaled per energy by dp_si’s flux for a fixed reference source.

3.6 How the code solves it

The redshift is cut into steps (200 in the analysis, uniform in z, and twice as many where the neutrinos decay in the dark-portal phase, `DECAY_Z_REFINE`: where the decays and the soft scatterings cycle, 200 steps left up to 6×10^{-3} in the spectrum’s shape, 400 at most 1.7×10^{-3}; decided 2026-10-25; dp_si’s steps uniform in $\ln(1+z)$). In each step the energy bins are swept from the highest down: each bin solves its equations exactly over the step with constant rates and sources, the sources being the injection and the secondaries and daughters of the bins above, computed from the numbers those bins actually absorbed and decayed during the step; a bin’s own secondaries and daughters are inside its equations. Where the neutrinos decay (`dp_sweep.step_full`, 2026-10-25) the six populations of a bin (both kinds, three states) are solved together, with the soft scatterings, the decays, the absorption and the injection in one matrix exponential, the in-medium decay rates tabulated once per run on a ladder of σ values with the grid’s bin ratio (`src/dp_medium.py`); elsewhere the absorption alone (`step_cosmological_rates`) and the soft scatterings after it. The soft scatterings’, the decays’ and the loss’s rates are taken at the step's midpoint and their exponents over its exact times ($\int dz/[(1+z)H]$ for the scatterings, $\int dz/[(1+z)^2 H]$ for the decays and the loss); the absorption and its time step at the step's low-redshift end, as since 2026-10-12 (first order; the MeV planes are validated with it); the injection over its exact integral below 1 keV, at the step's end in the MeV planes (a uniform 0.5% at 200 steps, absorbed by the free normalization). The loss shifts each bin's content in two halves around the step. The kernels are integrated exactly over the bins, so every step conserves the number to rounding.

4 Numerical method

4.1 Resolution and defaults

Table 3 compares the solver's production defaults with the settings of the statistical analysis. The defaults follow the convergence study of 2026-07-28 (`tools/convergence_study.py`, results in `output/convergence/`). We remeasured the analysis settings on 2026-10-12 with the corrected solver (Appendices A.8 to A.10): the errors of the study, measured on the runaway cascade, were up to two orders of magnitude too large. The analysis uses coarser energy and redshift grids than the defaults.

Table 3: **Resolution settings.** The convergence column states what each setting is known to achieve.

Setting	Solver default	Analysis	Evidence
Energy bins N_E (10 TeV–30 PeV)	1000	400	$\leq 1.5 \times 10^{-5}$ on flavor fractions at 400 (inband), 2×10^{-4} at $m_{Z_D} = 10^{-2}$ eV
Redshift steps N_z ($z \leq 4$)	1000	200	$\leq 7 \times 10^{-6}$ at 200
Directions to the Sun	8	8	χ^2 within 0.03 of 32 directions (Appendix A.12)
Transit table spacing	0.002	0.002	$\sim 10^{-5}$ against brute force; χ^2 within 0.03 of 0.001
Earth chords $n_\oplus$	12	12	12 agree with 50 to $\leq 1.4 \times 10^{-7}$ (Appendix A.7)
Earth steps N_{Earth}	25	25	Certified, Sec. 4.5

The Earth-chord floor.—Until 2026-10-12, changing the number of chords moved the energy-averaged flavor ratios pseudo-randomly at the 10^{-2} level instead of converging. We had attributed this to energy fringes finer than the grid; it was sampling noise of the dark-portal phase, which does not depend on energy, in the equal-weight chord mean. The phase-aware zenith average removes it (Appendix A.7). With the chord floor gone and the solver corrected, the largest numerical error left at the analysis settings is the energy grid, up to 2×10^{-4} on the fractions at $m_{Z_D} = 10^{-2}$ eV, where fine fringes in energy need ~ 1000 bins; at the benchmark points it is below 1.5×10^{-5}.

4.2 Thermal absorption rates

The rates (T7) require a momentum integral over the relic distribution of a Breit–Wigner resonance whose width is far smaller than any practical grid. We integrate it semi-analytically: the momentum at which the resonance enters the angular range has a closed form, the narrow zone around it is integrated with exact antiderivatives, and the smooth remainder with Gauss–Legendre quadrature (`src/dp_thermal.py`). Against a dense reference integration, the significance-weighted error is 2.5×10^{-7}; the midpoint rule used before July 2026 erred by 6–9% at on-resonance cells (round 7, commits `53d74f2`–`ca816a4`).

Coupling cache.—Within a fit at fixed m_{Z_D} the rate grid depends on the couplings only through the total coupling s, as $U(s) \approx I_A + I_B/s$ to $\sim 7 \times 10^{-8}$ over brackets of up to two decades. The cache fits this form from two exact grids and uses it only after checking it against a third (tolerance 3×10^{-7}). Until 2026-07-29 its minimum bracket width (0.3 decades) exceeded the coupling steps of the minimizer, so it never fired in real scans. We replaced that width with a numerical floor. The cache then served 83% of the evaluations of minimizer-like sequences at three mediator masses, with a worst error of 6.5×10^{-9} against fresh exact grids (commit `5e7877c`).

4.3 Cosmological mixing

The eigenvectors of $M/(2E(1+z)) + A(z)S$ depend on E and z only through $\sigma = 2E(1+z)A(z)$, because eigenvectors are unchanged by an overall factor. Therefore, we tabulate the mixing probabilities on 4096 nodes in $\log\sigma$ and interpolate, instead of computing $\sim 10^6$ eigenvector sets per run. Our table agrees with direct computation to 2.7×10^{-14} at a stress point with a level crossing in range (`README.md`, Sec. 12). Since 2026-10-13 the eigenvectors come from a Jacobi iteration in the texture's eigenbasis, accurate at any σ; the closed form used before lost the vacuum splitting of the rank-one texture's two degenerate eigenvalues once σ exceeded M by $\sim 10^{13}$ (Appendix A.12). The table now stops where the eigenbasis has become the texture's own to 10^{-9}.

4.4 Coherent propagation

The evolution operator of each step, $\exp(-iH\Delta x)$ for a 3×3 Hermitian H, is the spectral sum $\sum_k e^{-i\lambda_k \Delta x} v_k v_k^\dagger$ over an eigenbasis built to stay orthonormal to rounding at any eigenvalue gap (`src/dp_propagator.py`). Closed-form roots of the traceless part order the eigenvalues; the isolated one's eigenvector is a column of the product of the matrices shifted by the other two; the remaining pair comes from a 2×2 block. The operator is therefore unitary to 10^{-15} at any phase (Appendix A.10). A pair whose splitting H does not resolve in double precision evolves as degenerate; under a dominant potential this drops the vacuum splitting inside the potential's degenerate subspace, which changes flavor populations by at most 5×10^{-6} across the Earth's diameter at 10 TeV. When the potential is so large that double precision cannot follow the oscillation phase, the fluxes carry only the decoherence-averaged information. This happens roughly where $m_{Z_D}^2 + m_D^2 \lesssim 3 \times 10^6\, g_D \epsilon'$ eV^2, i.e., below $m_{Z_D} \approx 5$ eV for the default couplings ($g_D = 10^{-4}$, $\epsilon' = 0.1$); the analysis warns when its scan window reaches this region.

4.5 Earth steps

In each Earth step we use the exact average of the PREM density over the step, integrated layer by layer with the layer crossings in closed form. When the mediator's range approaches the thickness of a layer, the density of Eq. (1) relaxes across each jump over a length $1/m$; we integrate these boundary layers on subintervals graded toward them. The accumulated phase then equals the integral of the potential along the chord for any number of steps. Fluxes computed with 8 and with 1000 steps agree to $\leq 4.5 \times 10^{-9}$, and their energy-averaged flavor ratios to $\leq 6.5 \times 10^{-10}$, at the three benchmark points of Sec. 5 that are not in the phase-noise regime (commit `e1b6660`). Since the scheme converges to the exact path-ordered result as the steps shrink, this independence of the step count shows that the result is converged. The earlier left-endpoint scheme accumulated a phase error of thousands of radians, which moved the fluxes by up to 10^{-2} whenever the step count changed. The default is 25 steps since 2026-07-29 (commit `95fef9f`, Table 4).

Zenith average.—The chords are combined by integrating each block of their density matrices against its dark-portal phase over $\cos\theta_z$ (Appendix A.7), not by an equal-weight mean.

The whole sky (2026-10-13).—A detector sees neutrinos from every direction. Since 2026-10-13 the average runs over the whole sky with equal weight per $\cos\theta_z$ (the decided default; up-going only remains an option): down-going neutrinos cross only the ice above the detector, 1.5 km straight down to ~ 140 km at the horizon, against up to 12,740 km through the Earth. The down-going chords mirror the up-going ones, and the phase-aware average reproduces the exact integral over the sky to 10^{-5} (a test against 400,001 directions). How much the sky dilutes the Earth's effect depends on the potential: by half while the ice adds little phase (a dark-portal phase through the core below ~ 50 rad), less as the near-horizon paths start to mix flavors, not at all once the ice mixes them from above too. At the MeV benchmarks the phase is 6×10^5 rad through the core, 680 rad at the horizon and 7 rad straight down, so near-vertical down-going neutrinos keep part of the flavor structure of the flux arriving at the Earth: f_e and f_μ, equal from below, differ by 10^{-3} at "inband" and 3×10^{-3} at "strong". The detector's own acceptance per direction will replace the equal weights in the event-rate analysis (Sec. 13.1).

4.6 Parallelism and cost

Earth chords are independent; they run in a persistent pool of processes started with `spawn`, the start method the analysis requires. The transit to the Earth runs on threads, over energy bins, and only in the bins that a source actually turns: ~ 6 s per evaluation where the Sun matters (400 bins, 8 directions), nothing elsewhere. The cosmological phase and the thermal grids use threads (numba, TBB layer). On the development laptop (Intel i5-1334U, 8 cores), a solver run at the defaults took 0.32 s on 2026-07-29 and takes 0.71 s now (Sec. 14). One likelihood evaluation at the analysis settings (400×200, 6 paths, 12 chords) takes 0.35 s on one core (2026-10-12). Below $m_{\text{eff}} \approx 2 \times 10^{-5}$ eV the Earth's Yukawa columns depend on the couplings through the Debye mass and are recomputed for every new point, adding ~ 0.3 s; they are cached per m_{eff}, so the antineutrino pass reuses those of the neutrino pass. Before the step count fell from 100 to 25, the Earth phase took 66% of a single-core evaluation and the Milky Way phase (replaced on 2026-10-13) 18%.

5 Validation

5.1 The four benchmark points

Every change that can affect the physics is checked against the full solver at production settings ($N_E = N_z = 1000$, 6 paths, 12 chords, 4 cores) at four parameter points, with the reference and the modified code in separate processes:

Point	m_{Z_D} [eV]	g_D	ϵ'
below	3×10^5	10^{-4}	0.1
inband	2×10^6	10^{-4}	0.1
strong	2×10^7	10^{-2}	0.1
cosmoX2	40	10^{-4}	0.1

At "below" all three absorption resonances lie under the energy range; at "inband" they lie inside it; "strong" has a hundred times the coupling. At "cosmoX2" the mediator is light, so the potential dominates the Earth transit. Until 2026-10-12 a change in the last floating-point digit moved its pointwise fluxes by $\sim 10^{-3}$, which we attributed to the large phase; it was the non-unitary propagator (Appendix A.10). It now moves them by 1.8×10^{-12}.

Acceptance rules.—We compare two quantities: the largest relative flux difference over all energies and flavors, and the energy-averaged flavor ratios. A change that should not alter the arithmetic must reproduce "below", "inband" and "strong" to the floor at which two runs of identical code agree, ~ 2–7×10^{-12}. The "strong" point has an additional common-mode floor of $\sim 7.5 \times 10^{-8}$ under any reordering of the last floating-point digit, with its flavor ratios at $\lesssim 10^{-12}$. Until 2026-10-12 we judged "cosmoX2" on the energy-averaged flavor ratios and the energies of the absorption dips only; it now meets the same rules as the other points. Any recompilation of the spatial kernels changes the results at the last digit: two textually identical kernels compiled separately differed by 3.7×10^{-16}.

Absolute checks.—Comparing a modified code with its predecessor cannot reveal an error that both share; three such errors survived from the first version of the solver until 2026-10-12 (Appendices A.8–A.10). We now also check every change against conservation: the energy and number of neutrinos reaching the Earth must not exceed those of the same run without interactions, the spatial phase must conserve every energy bin's total, and vanishing sterile mixing must reproduce standard propagation. At production settings the energy (number) reaching the Earth is 0.99910 (0.99827) of the no-interaction value at "below", 0.94135 (1.09231) at "in-band", 0.99995 (1.00074) at "strong" and 0.99944 (0.99901) at "cosmoX2"; the energy-averaged fractions are $(0.32903, 0.32896, 0.34201)$, $(0.33241, 0.33138, 0.33621)$, $(0.33050, 0.32751, 0.34199)$ and $(0.32899, 0.32899, 0.34201)$ over the whole sky ($f_e = f_\mu$ from below: Sec. 4.5), and change by $\leq 1.2 \times 10^{-7}$ between 1000 and 2000 redshift steps (2026-10-13, Appendix A.11). The number can exceed the no-interaction value: each absorption turns one particle into two.

5.2 Record of validated changes

6 Statistical analysis: current state

6.1 Likelihood and parameters

The analysis (`analysis/`, which replaced FrequentistAnalysis on 2026-10-12; `docs/dev/ANALYSIS_DESIGN.md`) predicts the flavor composition at Earth, $f_\alpha = \int \Phi_\alpha \, dE / \int \Phi \, dE$ over 10 TeV–30 PeV, and compares it with a projected measurement: an asymmetric Gaussian fitted to the expected 2050 sensitivity of IceCube with IceCube-Gen2, KM3NeT, Baikal-GVD, P-ONE and HUNT, centred on full pion decay (surface `pion_2050_case_5` in `analysis/data/flavor_surfaces.json`, one of 35, whose $\ln L$ we reproduce bit for bit). We use $\chi^2 = -2\ln(L/L_{\max})$, zero at the centre. Detector uncertainties enter only through the width of that projection. In the two analyses ported from FrequentistAnalysis four parameters float, m_{Z_D}, g_D, u_e and u_μ (three at $m_{Z_D} = 10^{-30}$ eV); everything else is fixed, including the source composition, the spectral index and the lightest neutrino mass. The mixing parameters are fixed at the NuFIT 6.0 best fit for normal ordering with Super-Kamiokande atmospheric data ($\sin^2\theta_{12} = 0.308$, $\sin^2\theta_{13} = 0.02215$, $\sin^2\theta_{23} = 0.470$, $\delta_{\rm CP} = 212°$), which reproduces the projection's centre to 10^{-6}. FrequentistAnalysis fixed them at the solver's defaults ($\delta_{\rm CP} = 132°$, $\sin^2\theta_{23} = 0.571$), which put standard physics 0.0034 away from the centre in f_μ, 0.7σ of the projection; only new physics could absorb that offset.

Limitations we found.—The flavor fractions are plain integrals of the flux; a detector would weight them by its effective area and cross sections, which rise with energy. For an $E^{-2.5}$ spectrum the plain integral is dominated by energies near 10 TeV. The event-rate calculation of Sec. 13.1 removes this approximation. The priors that FrequentistAnalysis' configurations defined for the mixing parameters were never applied (Appendix A.15); the new package applies Gaussian and half-Gaussian priors, also on derived quantities such as the sum of the neutrino masses.

6.2 Scans and limits

Each fit is a bounded quasi-Newton minimization (L-BFGS-B) from several seeded starts, polished by Powell's derivative-free method; parameters spanning decades are fitted in $\log_{10}$. A profile fixes the parameter of interest on a grid and minimizes over the others; a plane does

Table 4: **Changes checked at the benchmark points.** Differences are the largest relative flux difference and the largest change of an energy-averaged flavor fraction.

Date, commit	Change	Result
2026-07-29 `e1b6660`	Phase-exact Earth stepping	Step-count independent to 4.5×10^{-9} (Sec. 4.5)
2026-07-29 `0d6392c`	Solver split into modules	Identical at inband, strong, cosmoX2; below at the 7×10^{-12} floor
2026-07-29 `95fef9f`	Earth steps 100 → 25	below 1.0×10^{-9} / 1.2×10^{-10}; inband 2.1×10^{-11} / 2.3×10^{-12}; strong 6.2×10^{-12} / 8.3×10^{-13}; cosmoX2 ratios 2.9×10^{-3}, dips unchanged
2026-10-12 `5ecf10e`	Relic temperature corrected	A physics change; effects in Appendix A.1
2026-10-12 `0db8111`	Cosmological and Galactic neutron densities	No change at the four points beyond their floors; effects at light mediators in Appendix A.2
2026-10-12 `9c025e6`	Neutron fractions along Earth chords	Flavor fractions move by up to 0.026, as much as adding one chord (Appendix A.3)
2026-10-12 `2dec771`	Point sources and cold disc gas	No change at the four points beyond their floors; effects at light mediators in Appendix A.4
2026-10-12 `5866227`	Exact length of the Earth chords	Flavor fractions move by 9×10^{-4} (below) and $\leq 5 \times 10^{-5}$ (inband, strong); dips unchanged (Appendix A.5)
2026-10-12 `70ef709`	The Earth's potential as its Yukawa integral	Bit-identical at the four points; effects where the range reaches the Earth's scale in Appendix A.6
2026-10-12 `8f1910b`	Phase-aware zenith average of the Earth chords	Chord-sampling floor removed; flavor fractions move by up to 0.029 (Appendix A.7)
2026-10-12 `7dc95e9`	Cosmological cascade conserves number	A correction: the energy at the Earth was up to 10^{67} times the injected one (Appendix A.8)
2026-10-12 `944360d`	Resonant absorption scales with the couplings	A correction: energy at the Earth $3 \times 10^{-3} \rightarrow 0.999$ of the no-interaction value at "below" (Appendix A.9)
2026-10-12 `2066f3b`	Exactly unitary propagator	below, inband, strong unchanged to five digits; "cosmoX2" fractions move by 3.5×10^{-4} and become reproducible (Appendix A.10)
2026-10-13	Both secondaries of every absorption	inband fractions move by 1.8×10^{-3}; energy at the Earth $0.841 \rightarrow 0.941$ (inband), $0.982 \rightarrow 0.99995$ (strong) (Appendix A.11)
2026-10-13	The whole sky at the detector	$f_e - f_\mu$ from 0 to 10^{-3} (inband), 3×10^{-3} (strong); energies unchanged (Sec. 4.5)
2026-10-13	The transit to the Earth; the cosmological potential at ultra-light masses	below, inband, strong to 3×10^{-12}; cosmoX2 1.4×10^{-3} pointwise, 1.8×10^{-8} on the fractions, the old method's own error (Appendix A.12)

the same for two parameters. The grid points are fitted in parallel, in chains walked outward from the global fit, each point starting from its predecessor's solution and from the global fit. Repair passes then refit points that their neighbours' solutions beat, points that stand out above their neighbours, and the edges of the grid. Every fit is checkpointed, so an interrupted scan resumes. We read intervals off $\Delta\chi^2$, measured from the lowest χ^2 found anywhere, at 1, 2.706 and 3.841 for one parameter (68.27%, 90%, 95%, two-sided) and at 2.296, 4.605 and 5.991 for two; FrequentistAnalysis labelled 4.61 as 2σ.

Exclusion planes (decided 2026-10-13).—We read an exclusion plane per mediator mass: in each column of fixed m_{Z_D}, $\Delta\chi^2$ is measured from that column's own minimum, and the couplings above $\Delta\chi^2 = 3.84$ are excluded at 95% (a raster scan, one parameter per column). The two readings differ most where the effect saturates: on the plane below, the Earth's potential alone gives $\Delta\chi^2 \approx 4.5$ over most light masses, above 3.84 but below the 5.99 of a joint two-parameter region. A column whose own minimum lies above the best fit by more than 3.84 is flagged: its couplings never reached the region without an effect, so its range must be extended before a limit is read. Coverage tests still have to confirm the threshold; for a coupling bounded at zero the asymptotic one-sided value is 2.71.

Planes by regime (decided 2026-10-13).—The observables depend on g_D and the sterile mixing only through products, so a free mixing would always slide to zero; the planes scan the products, and their limits hold for any mixing. MeV planes set $\epsilon' = 0$, which switches off the coherent potential and leaves the resonant absorption, a function of $g_D|U_{\alpha 4}|^2$ alone. Potential planes run from ultralight to GeV mediators in $g_D\epsilon'|U_{\alpha 4}|^2$, to which the potential is proportional; with $\epsilon' = 0.1$ the absorption coupling $\Sigma G \approx g_D|U_{\alpha 4}|^2\, \epsilon' g/c_W$ depends on the same product. At MeV masses the Earth's own potential is far from negligible: $\sim 10^{-8}$ eV at $g_D\epsilon' = 10^{-5}$ and 2 MeV, against a vacuum term of $\sim 10^{-16}$ eV. Toward lighter mediators the potential at the detector comes from the Earth's exterior, then the Sun (10^{-26} to 10^{-18} eV), then the Galaxy smoothed over the mediator's range (Appendix A.12). Each regime is run for the three benchmark textures (T13), (T16), (T17). A test checks that splitting a product differently leaves the fractions unchanged (10^{-12} with $\epsilon' = 0$; 10^{-6} for the potential).

The potential planes (2026-10-13, commit `52f36fc`*, Tycho).*—With the transit of Appendix A.12 the three potential planes (40×55 points each, ~ 3 minutes on 16 cores) follow the source that dominates the potential at the detector. For the $e\mu$ texture the lowest excluded coupling product falls from 3×10^{-48} to 2×10^{-55} between 10^{-24} and 10^{-18} eV (the Sun) and from 3×10^{-47} to 5×10^{-51} around 10^{-16}–10^{-14} eV (the Earth's exterior), and rises from 10^{-60} to 2×10^{-57} at 10^{-30} eV (the Galaxy, no longer overstated); above 10^{-8} eV nothing changes. The $\mu\tau$ texture moves the same way (by up to 15 decades at 10^{-18} eV). For the $e\tau$ texture the potential-dominated composition happens to lie close to the standard one: the excluded region is the transition itself, where the Galactic or solar potential crosses the vacuum term, and the region above $g_D\epsilon'|U|^2/m_{Z_D}^2 \approx 10^4$ eV^{-2}. Every scattered point of the old planes is gone.

A band set by the sources' environment.—All three planes have a narrow band along $g_D\epsilon'|U|^2/m_{Z_D}^2 = 10^4$ eV^{-2}, one grid row wide, where $\Delta\chi^2$ dips (1.3 for $e\mu$, inside the excluded region). There the cosmological potential at the sources ($z \sim 1$) equals the vacuum term, so the injected flavors project halfway between the vacuum states and the potential's, and the composition at the Earth passes close to the standard one. The band existed before above 3×10^{-13} eV; the mass floor of Appendix A.12 flattened it below, into the old planes' horizontal stripe at 10^{-21}. It assumes the sources sit in the cosmic mean density; for ranges shorter than ~Mpc a source inside its host galaxy sees 10^6–10^7 times more neutrons, which would move the band. The source configuration therefore has a knob (decided 2026-10-13), `source.host_galaxy_scale`, off by default: with $s > 0$ the sources sit inside a Milky-Way-like host at a Sun-like position, its neutrons scaled by s, with the same Yukawa-smoothed profile as our own Galaxy (Appendix A.12) added to the cosmic mean and not diluting with $(1+z)^3$. Only the injection changes: the injected flavors split over the host's eigenstates, which leave the host adiabatically. With $s = 1$ the transition moves from $g_D\epsilon'|U|^2/m_{Z_D}^2 \approx 10^4$ to $\sim 10^{-3}$–1 eV^{-2} (a test: deep in it both settings give the same composition). The three potential planes with $s = 1$ (Tycho, commit `971e44b`): for $e\mu$ the lowest excluded coupling is unchanged in 37 of 40 mass columns and one grid row (a decade) lower in the other three (10^{-30}–10^{-26} eV); for $\mu\tau$ it moves by one row at most; for $e\tau$, whose high-coupling exclusion is this transition, the excluded region now starts at $g_D\epsilon'|U|^2/m_{Z_D}^2 \sim 10^{-2}$ instead of 10^3–10^4 eV^{-2}, which lowers the limit by ~ 7 decades between 10^{-25} and 10^{-17} eV. The $e\mu$ and $\mu\tau$ limits therefore do not depend on where the sources sit; the $e\tau$ limit below 10^{-17} eV does.

A first plane (2026-10-13; not for quoting; computed before the regeneration fix of Appendix A.11).—For the $e\mu$ texture at $u = 0.5$ and $\epsilon' = 10^{-4}$ (configuration `pion_2050_case_5_exclusion`, 45×41 points, commit `19f953d`), the 2050 projection excludes at 95% $g_D > 3.7 \times 10^{-9}$ at

$m_{Z_D} = 0.1$ MeV, 3.3×10^{-7} at 1 MeV (with an allowed island at $10^{-5} < g_D < 0.013$, where the resonant absorption offsets the potential), 3.7×10^{-5} at 10 MeV and 3.7×10^{-3} at 100 MeV: the edge follows $g_D \propto m_{Z_D}^2$, where the Earth's potential reaches the vacuum term. Below ~ 30 keV the columns stay 4.3–4.7 above the best fit down to $g_D = 10^{-10}$, the bottom of the scan, so no limit can be read there yet. Three things keep this from being a result: $u = 0.5$ is far above laboratory bounds on the mixing, only up-going directions enter the average, which overstates the Earth's effect, and the light-mass columns need couplings scanned far lower.

Profiles of null data.—The projection is centred on standard physics, and with every coupling free the fit can always switch the new physics off ($u \to 0$, $g_D \to 0$, or a resonance outside the energy range). The profiles of the ported analyses are therefore expected to stay at $\Delta\chi^2 \approx 0$, and an exclusion region needs fixed couplings: the configuration `pion_2050_case_5_exclusion.json` maps $\Delta\chi^2$ over (m_{Z_D}, g_D) for the $e\mu$ texture (T13) at $u = 0.5$.

Degeneracies.—Flavor fractions alone do not identify the parameters. We fitted Asimov data, the model's own prediction at $m_{Z_D} = 2$ MeV, $g_D = 3 \times 10^{-4}$, $u = 0.5$: with m_{Z_D} and g_D free the fit reached $\chi^2 = 7 \times 10^{-7}$ at $m_{Z_D} = 8 \times 10^{-14}$ eV (both the resonant absorption and the coherent potential pull f_e and f_μ together when $u_e = u_\mu$); with the mixing free as well it returned the injected mass to 5×10^{-6} with $g_D = 10^{-4}$ ($\chi^2 = 8 \times 10^{-4}$). Injection-recovery tests must therefore check that the fit reaches $\chi^2 \approx 0$, not that it returns the injected values.

What we have not shown.—Nothing yet checks that the intervals have their stated coverage. Near the boundaries of the parameter space (couplings at zero, the lightest mass at zero) the asymptotic thresholds can fail. For an upper limit on a coupling bounded at zero the one-sided values are 1.64 and 2.71, so limits read at the two-sided values are probably conservative. Section 13.3 describes the tests.

6.3 Thin excluded bands with fitted source compositions (2026-10-21)

What they are.—With the source compositions fitted (the `_fx` planes of the campaign of 2026-10-20), the ultralight $L_e - L_\mu$ and $L_e - L_\tau$ planes exclude, besides their main region, a thin band about 0.5 decade wide in g_D (Fig. 1). It runs along the lower edge of the region excluded with pion-decay compositions: flat near $g_D \approx 3 \times 10^{-28}$ up to $m_{Z_D} \approx 10^{-20}$ eV, climbing between 10^{-19} and 10^{-13} eV, then, for $L_e - L_\tau$ and the future detectors, along $g_D \approx 10^{-1.75} m_{Z_D}$. Its peak $\Delta\chi^2$ is 4.5–13 for the future detectors and about 4–6 for the HESE and DR2 events, whose plane also has a second row along $g_D \approx 10^{-0.5} m_{Z_D}$ that barely reaches 3.84. The band has real gaps, where its peak falls below 3.84: between 10^{-19} and $10^{-17.8}$ eV for $L_e - L_\mu$ and near $10^{-17.4}$ eV for $L_e - L_\tau$ (future detectors).

Why they exist.—Well above the band, the new potential dominates the vacuum term at every measured energy and shifts the flavor composition at Earth by the same amount at every energy; a different source composition produces the same shift, so the fit absorbs it (at $m_{Z_D} = 10^{-22}$ eV, $L_e - L_\tau$, $\Delta\chi^2$ falls from 7.3 on the band to 0.4 a third of a decade above it, and at $g_D = 3 \times 10^{-27}$ the fitted electron fractions are 0.67 and 0.76 for the extragalactic and Galactic sources, against 1/3 for pion decay). On the band, the potential is comparable to the vacuum term inside the measured range: the flavor composition at Earth steps near 30–50 TeV, where the future detectors collect most of their events (Fig. 2), and no energy-independent source composition reproduces a step. Below the band the step moves above ~ 1 PeV, where events are few. The band therefore follows the edge of the pion-decay region, where the effect switches on.

Checks.—The band is not a fitting or numerical artifact. Its points refitted from ten kinds of start (their own solution, their neighbours', the pion-decay plane's, the four corner compositions, random starts) return the same χ^2 to 0.1 (`tools/check_edge_islands.py`). Sampled along the edge with 11 000 fits, the coarse grid's isolated islands join into the continuous band (`tools/refine_edges.py`). With every grid of the flavor calculation doubled (energy and redshift

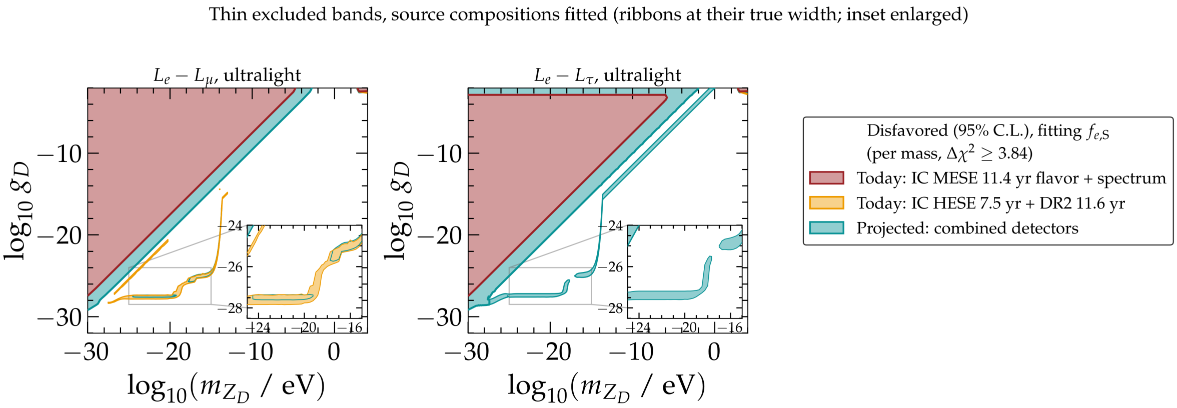


Figure 1: **Thin excluded bands with fitted source compositions** (`tools/plot_campaign_ml.py`): the ultralight planes at 95%, per mass, with the bands drawn at their true width (insets: their flat stretch, enlarged). The exclusion figures leave them out.

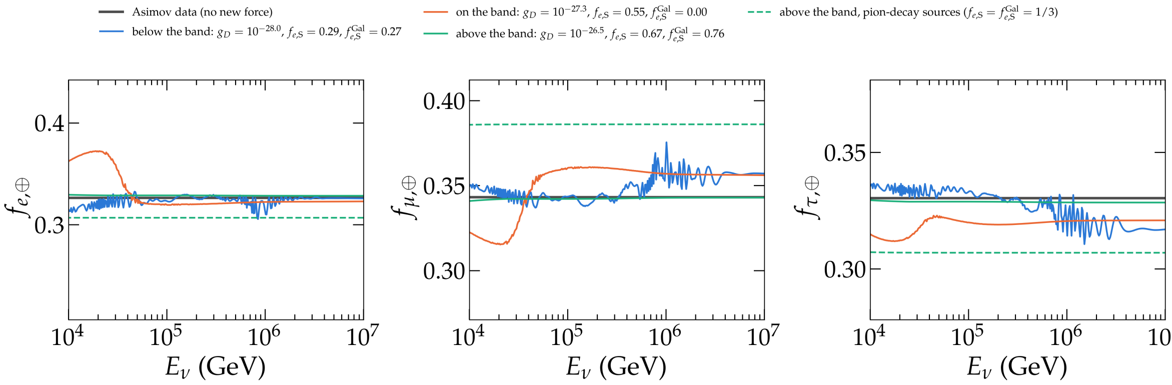


Figure 2: **Why the band exists** (`tools/check_band_mechanism.py`): flavor fractions at Earth against energy for the future detectors, $L_e - L_\tau$, $m_{Z_D} = 10^{-22}$ eV, at couplings below, on and above the band, each with its fitted parameters. Above the band the shift is the same at every energy (dashed: with pion-decay sources) and the fitted sources remove it; on the band the fractions step near 30–50 TeV.

bins, Sun paths, Earth chords and steps), $\Delta\chi^2$ at seven band points moves by at most 0.06 (`tools/check_band_numerics.py`).

Treatment.—The band exists only because the fit takes each source's flavor composition to be the same at every energy (item 9 of `docs/SIMPLIFICATIONS_AND_ASSUMPTIONS.md`). Sources whose composition changes with energy, such as muon-damped sources, could absorb the step as well. We leave the band out of the exclusion figures (`tools/plot_campaign_ml.py`) and will mention it in the paper. [**Decide how the paper describes it**]

6.4 Where the fitted-composition regions end: missing tau neutrinos (2026-10-21)

What the fit sees.—With the source compositions fitted, the MESE flavor + spectrum $L_e - L_\tau$ plane is excluded only just ($\Delta\chi^2 \approx 3.9$), and its region ends below the top of the scan, near $g' = 3 \times 10^{-3}$; with pion-decay sources the same couplings stay excluded ($\Delta\chi^2 = 7.5$–9.9). Splitting χ^2 into its terms (`tools/check_dips.py`, $m_{Z_D} = 10^{-15}$ eV, Table 5) puts every change in the flavor term; the spectrum moves χ^2 by at most 0.33.

Why.—Where the new potential dominates the vacuum term along the whole path, the

$\log_{10} g'$	$\Delta\chi^2$	flavor term	spectrum term	$f_{\tau,\oplus}$
-3.5 to -3	3.89	3.89	0.00	0.000
-2.5	3.64	3.63	0.01	0.011
-2.25	1.92	1.85	0.08	0.100
-2	0.45	0.09	0.33	0.270

Table 5: MESE flavor + spectrum, $L_e - L_\tau$, source compositions fitted, $m_{Z_D} = 10^{-15}$ eV: $\Delta\chi^2$ from the no-effect point, its flavor and spectrum terms, and the fraction of ν_τ at Earth.

flavors stop mixing: each neutrino arrives with the flavor it left with, and since the sources make no ν_τ, none arrives. Fitting the sources reproduces the measured ν_e and ν_μ shares ($f_{e,\oplus} = 0.51$, $f_{\mu,\oplus} = 0.49$) but cannot make ν_τ; MESE measures the τ share only loosely, so the missing ν_τ cost just $\Delta\chi^2 = 3.9$. With pion-decay sources the ν_e and ν_μ shares are off as well (0.33:0.67:0), hence 7.5. Above 3×10^{-3} the ν_τ reappear and the region ends. That they come back through scattering on relic neutrinos, which sets in near these couplings (it makes the HESE + DR2 line near 8×10^{-3}), is our reading, not yet checked. **[Check: the scattering switched off at $g' = 10^{-2}$]** The same missing ν_τ set the main region of every ultralight plane with fitted compositions. Doubling every grid of the flavor calculation changes these values by at most 0.03.

7 Secret neutrino interactions

On 2026-10-13 we widened the project beyond the dark portal (decided that day; the plan is in `docs/dev/WORK_PLAN.md`): the same machinery now tests secret neutrino interactions generally, as in arXiv:2001.04994 (Bustamante, Rosenström, Shalgar & Tamborra), with the full cross sections of arXiv:2107.13568 (Esteban, Pandey, Brdar & Beacom, Appendix A). The switch `secret_interactions.enabled` replaces the dark portal by this model; everything downstream (the analysis, the scanner) is unchanged. This section describes the scalar mediator with Majorana neutrinos (`src/dp_si.py`, commits `781ae43`, `771cd46`), then Dirac neutrinos and a vector mediator (`99a8ccc`, `d82231c`; Sec. 7.2), all implemented and validated on 2026-10-13, and the limits of the four models (Sec. 7.6).

7.1 Model and cross sections

A real scalar ϕ of mass M couples to the neutrinos as $-\frac{1}{2}\sum_{\alpha\beta} G_{\alpha\beta}\,\bar\nu_\alpha \nu_\beta\,\phi$, for any real symmetric flavor texture G (Mauricio, 2026-10-13: non-universal textures must be treated). For Majorana fields, $\nu_\alpha = \sum_i (U_{\alpha i} P_L + U^*_{\alpha i} P_R)\nu_i$, so $\bar\nu_\alpha \nu_\beta = \sum_{ij} \bar\nu_i [(U^T G U)^*_{ij} P_R + (U^T G U)_{ij} P_L]\nu_j$: the mass states couple through the complex symmetric matrix $\lambda = U^T G U$. The paper writes $g_{ij} = \sum U^*_{\alpha i} U_{\beta j} G_{\alpha\beta}$ and takes it real, which agrees only without CP violation; at $\delta_{CP} = 212°$ even a flavor-universal coupling, $g\,U^T U$, has off-diagonal entries of 0.09–0.13 g. The Majorana phases drop out of every rate, and for a single coupled flavor f so does δ_{CP}, $|\lambda_{ij}| = g\,|U_{fi}||U_{fj}|$ (tests). A high-energy neutrino ν_i of energy E scattering on a relic ν_j at rest has $s = 2Em_j$. The cross sections of the appendix cover $\nu_i\nu_j \to \nu_k\nu_l$ through the s, t and u channels and their interference (Eqs. A11, A12), and, above $s = 4M^2$, double-scalar production $\nu_i\nu_j \to \phi\phi$ through neutrino exchange (A11, A13), each scalar decaying at once into two neutrinos with an energy spectrum flat in $[0, E_\phi]$ (A16); the width is $\Gamma = (M/16\pi)\sum_{kl} g^2_{kl}$ (A15). The secondaries' energies follow from $t = (p_i - p_k)^2$: $E_k = E(1 + t/s)$ for a neutrino and $E_A = E(1 + (t - M^2)/s)$ for a scalar.

Three readings of the appendix were settled by checks rather than assumed. The factor 1/2 of Eq. (A12) applies only to identical final states ($k = l$): with it, (A12) integrated over t and summed over the final states reproduces (A11) to 10^{-13}. The kinematic range of the scalar's t is $M^2 - s/2 \mp \sqrt{s(s/4 - M^2)}$, where $st + (t - M^2)^2$ vanishes; the appendix prints it without the factor $\sqrt{s}$ and the shift M^2. And the targets: the cross sections are summed over the relic's spin, and the appendix says to multiply them by half the number of targets, i.e. by the density per helicity state, $n = 3\zeta(3)T_\nu^3/4\pi^2 = 56\ \mathrm{cm}^{-3}$ per mass state today. An independent narrow-width derivation agrees: an incident ν_L on an unpolarized relic has $\sigma = (\pi g^2/2)\,\delta(s - M^2)$ per relic, against 2×56 relics, while the appendix's s channel integrates to πg^2 across the resonance (a test).

Complex couplings.—The appendix assumes real couplings; we generalized it from the helicity structure of the amplitudes. For an incident ν_i of helicity $-\frac{1}{2}$ and the relic at $-\frac{1}{2}$, the s channel ($\lambda_{ij}\lambda_{kl}$), t channel ($\lambda_{ik}\lambda_{jl}$) and u channel ($\lambda_{il}\lambda_{jk}$) lead to the same final state (both neutrinos at $+\frac{1}{2}$) and interfere; the s channel into two neutrinos at $-\frac{1}{2}$ ($\lambda_{ij}\lambda^*_{kl}$) and, with the relic at $+\frac{1}{2}$, the t channel ($\lambda_{ik}\lambda^*_{jl}$) and the u channel ($\lambda_{il}\lambda^*_{jk}$) add incoherently. Equation (A12) becomes the same expression with moduli in the squared terms and real parts in the interference: the s–t term carries $\mathrm{Re}[\lambda_{ij}\lambda_{kl}\lambda^*_{ik}\lambda^*_{jl}(s - M^2 - iM\Gamma)]$ and likewise s–u, the t–u term $\mathrm{Re}[\lambda_{ik}\lambda_{jl}\lambda^*_{il}\lambda^*_{jk}]$. The part with $M\Gamma$ changes sign between neutrinos and antineutrinos (whose couplings are conjugated): a CP asymmetry of relative size $\Gamma/M = \sum|\lambda|^2/16\pi$, which the transport, following both helicities together, averages out. A test integrates the generalized $d\sigma/dt$ for random complex couplings and recovers the closed-form channel sums below to 10^{-10}.

One formula we do not adopt. The amplitudes of double-scalar production, Eqs. (A3) and (A4), sum over the exchanged neutrino ν_k; with the neutrino masses neglected in its propagator the sum is coherent, and the rate carries $|\sum_k \lambda_{ik}\lambda^*_{kj}|^2 = |(\lambda\lambda^\dagger)_{ij}|^2$ (the exchanged neutrino connects λ at one vertex to λ^* at the other, the relic at the opposite helicity). The appendix prints $\sum_k g^2_{ik}g^2_{jk}$ in (A10), (A11) and (A13), the incoherent sum. The two agree for a flavor-universal coupling without CP violation; for a single flavor f the printed form is smaller by $\sum_k |U_{fk}|^4$, about 0.35 for ν_τ, so it fails for non-universal textures. We use the coherent form; the option `phiphi_coupling = "paper"` reproduces their figures (their Fig. 4 is consistent with the printed form).

Summed over the final states, the absorption of ν_a on the relic ν_r is

$$\sigma_{ar} = \frac{|\lambda_{ar}|^2 G_2}{16\pi}\frac{s}{B} + \frac{2H_{aa}H_{rr}N_2 + |H_{ar}|^2 N_5}{16\pi s} + \frac{2\,\mathrm{Re}(\lambda_{ar}Q_{ar})}{16\pi}\frac{(s - M^2)N_4}{B} + \frac{|H_{ar}|^2}{32\pi}I_{\phi\phi}, \tag{13}$$

with $H = \lambda\lambda^\dagger$, $Q = \lambda^*\lambda\lambda^*$, $G_2 = \sum_{kl}|\lambda_{kl}|^2$, $B = (s - M^2)^2 + M^2\Gamma^2$, $N_{2,4,5}$ the integrals over the secondary's energy fraction x of the profiles $(1 - x)^2/(1 - x + \mu)^2$, $(1 - x)/(1 - x + \mu)$ and $x(1 - x)/[(x + \mu)(1 - x + \mu)]$, $\mu = M^2/s$, and $I_{\phi\phi} = (s - 2M^2)\,\Phi(w)/s^2$, $\Phi(w) = 2(3 - w^2)\,\mathrm{artanh}\,w - 6w$, $w = \sqrt{s(s - 4M^2)}/(s - 2M^2)$. The secondaries follow the same couplings: for example the t channel gives ν_c with x-spectrum $|\lambda_{ac}|^2 H_{rr}(1-x)^2/[8\pi s(1-x+\mu)^2]$, and every $2 \to 2$ absorption yields two secondaries and every double-scalar one four (identities that a test checks for random real and complex couplings). The closed forms cancel catastrophically in two places, near the double-scalar threshold ($\Phi \sim w^5$) and far below the resonance ($N_2 \sim s^2/3M^4$); there we use the series and Gauss–Legendre of cancellation-free integrands, and every kinematic integral agrees with 50-digit arithmetic to 4×10^{-15} from threshold to $s/M^2 = 10^{14}$.

7.2 Dirac neutrinos and a vector mediator

On 2026-10-13 Mauricio asked that the physics of both be checked before any code; the checks (`tests/amplitudes_bruteforce.py`) evaluate every tree amplitude with explicit Dirac matrices and helicity spinors, Majorana fermions by the fermion-flow rules of Denner, Eck, Hahn & Küblbeck, independently of every closed form, and the choices below were decided after them.

The brute force reproduces what we took over.—Eq. (A12) for real couplings (10^{-15}), our complex-coupling form of it including the term odd under $\nu \leftrightarrow \bar{\nu}$ (a CP asymmetry of relative size Γ/M, which the transport averages), double-scalar production with the coherent factor (the printed one is off by factors 0.08–2), and the textbook Bhabha and Moller helicity amplitudes as a check of the engine itself.

Scalar, Dirac neutrinos.—With Dirac fields the coupling is $-\phi\, h_{ij}\bar{\nu}_i\nu_j$, $h = U^\dagger G U$ Hermitian: the right-handed neutrinos rotate like the left-handed ones (the paper's convention; nothing observable fixes their basis; decided). The scalar flips chirality, so every vertex joins a state that interacts weakly (ν_L, $\bar{\nu}_R$: visible) to a sterile one (ν_R, $\bar{\nu}_L$). The relics, at rest and unpolarized, present both helicities in the collision, and the scalar couples to both: an incident ν_a meets $\bar{\nu}_r$ of helicity $-$ in the s and t channels (final states: one visible, one sterile), ν_r of helicity $-$ in the t and u channels (both sterile), and the opposite helicities in the t channel alone. Summed, the total is the appendix's Eq. (A8) exactly; the split into visible and sterile secondaries, per mass state and energy, was derived by hand and agrees with the brute force to 10^{-15}. The appendix's Table I, read with its stated factors, overstates the visible production by up to 3.5%; we do not use it. The sterile secondaries scatter too, and by parity an incident ν_R gives the secondaries of a ν_L with visibility swapped (checked), so the visible flux is $(S + D)/2$ from two runs of the transport, S with the kernel visible + sterile and D with visible − sterile (decided: tracked; the option `"drop"` discards them, as 2107.13568 does). Width $M \sum |h|^2/8\pi$.

Vector mediator.—Coupled to left-handed currents (decided), $-V_\mu G_{\alpha\beta}\bar{\nu}_\alpha\gamma^\mu P_L \nu_\beta$, as in gauge models like $L_\mu - L_\tau$; mass basis $U^\dagger G U$ for Majorana and Dirac alike. Only the relic components that behave as ν_L (helicity $-$) and as $\bar{\nu}_R$ ($+$) interact; a Majorana relic offers both (56 + 56 per mass state), a Dirac one half of them, so Dirac rates are half the Majorana ones, and every secondary is visible. The brute force confirms, for complex couplings, $|\mathcal{M}|^2 = 4u^2 |A_s/(s - M^2 + iM\Gamma) + A_t/(t - M^2)|^2$ for $\nu\bar{\nu} \to \nu\bar{\nu}$ ($A_s = G_{ra}G_{kl}$, $A_t = G_{ka}G_{rl}$), $4s^2|A_t/(t - M^2) + A_u/(u - M^2)|^2$ for $\nu\nu \to \nu\nu$, and $4|(GG^\dagger)_{ar}|^2[t/u + u/t + 4M^2 s/tu - M^4(1/t^2 + 1/u^2)]$ for $\nu\bar{\nu} \to VV$ (to 10^{-12}); the s channel gives the Z-burst $\langle\sigma\rangle = 2\sqrt{2}\pi G_F$ for the Standard-Model coupling. Each V decays at once into $\nu\bar{\nu}$; in the relic rest frame, where it is ultrarelativistic, a transverse one gives $3z^2 + 3(1-z)^2$ for the pair and a longitudinal one $6z(1-z)$ per neutrino. Its longitudinal state there is the light-cone polarization $k/M - Mp_2/(k \cdot p_2)$, and the Ward identity removes one diagram: $|\mathcal{M}_0|^2 = 8sM^2(t - M^2)^2/[t^2(u - M^2)^2]$ (derived by fitting the brute force, exact to 3×10^{-14}). The longitudinal fraction reaches 55% near threshold and $\sim 20\%$ for soft vectors at $s = 100M^2$ (decided: exact).

The forward peak.—Above the resonance the vector's t channel is forward-peaked: $\sigma \to |G|^4/4\pi M^2$, constant, where the scalar's falls as $1/s$ (Fig. 3). Nearly every such scattering leaves the neutrino in its own energy bin and gives $\sim M^2/2m_r$ to the recoiling relic, which becomes a neutrino near the resonance energy; with a flavor-dependent coupling it may also change the mass state. These are real, incoherent scatterings (the relic recoils). The transport removes from a bin's equations the returns that change nothing (complementary integrals, without cancellation), keeps every recoil, and solves the bin's three mass states exactly over the step with a matrix exponential, because the quasi-elastic conversions are fast enough to make them stiff (the scalar keeps its scheme, its returns being of order $\Delta \ln E$). Each return that changes nothing still costs the neutrino $\sim M^2/s$ of its energy, and at small M there are enough of them to move it by many bins (at 1 keV and $g = 0.03$, $\sim 10^7$ returns of 10^{-6} each; two thirds of the energy loss is in the returns, a third in the jumps out of the bin). The transport carries their losses as a drift in $\ln E$: an upwind flux to the bin below, in the daughter's state, at the rate of the returns times their mean shift $\int_{1/h}^{1}(-\ln x)P(x)\,dx$ (closed form plus a bounded remainder, 10^{-7} against 40 digits), which the top-down sweep passes on within the step however fast it is. Without it (until 2026-10-13) the energy-averaged fractions of flavor-universal vector couplings at 1 keV and $g = 0.03$ moved by 7×10^{-3} between 401 and 1601 bins, as much as the effects measured; with it, 10^{-4} (3×10^{-5} at 801 bins), and ν_e couplings at 1 and 10 keV agree to 10^{-4} or better at 401 bins; where nearly all the flux is gone (10^{-4} of it left, ν_e at 100 keV and $g = 0.1$) to 10^{-3}, falling with the bin width. The scalar has no forward peak and no drift.

Far above the resonance this is all that happens: at $M = 1$ keV ($s/M^2 > 10^5$ in the band) a scattering moves the neutrino by $\sim M^2/s \sim 10^{-6}$ in energy but projects it onto the coupled

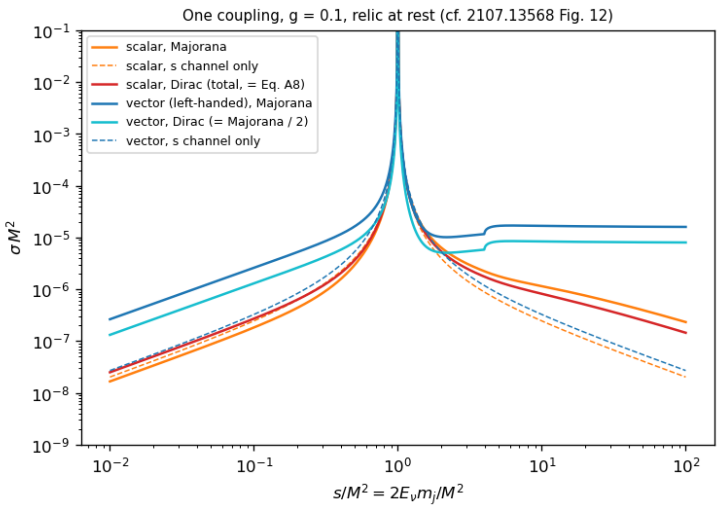


Figure 3: **Total cross sections** for one coupling, $g = 0.1$, relic at rest: scalar (Majorana, Dirac) and vector (Majorana; Dirac is half), with the s channels alone. The scalar Majorana curves reproduce 2107.13568, Fig. 12; all from the closed forms checked by brute force.

flavor ($\sum_k G_{ka}|\nu_k\rangle \propto |\nu_e\rangle$ for ν_e couplings), at the rate $W_{a\to x} = 2\kappa n\,|G_{xa}|^2\,\mathrm{Tr}(G^\dagger G)/4\pi M^2$ from both relic components. The flux at each energy then obeys $dF/dz = (1+z)^2 K_0 F/H(z)$, $K_0 = W - \mathrm{diag}(\sum_x W_{x\cdot})$, solved by one matrix exponential per source redshift; the transport reproduces its flavor fractions to 10^{-6} (Majorana and Dirac, $g = 5\times10^{-4}$ and 10^{-3}; a test). W being symmetric, the conversions drive the flux toward equal mass states, 1/3 of each flavor at Earth: a shift of ~ 0.01 from the pion sources' $0.326 : 0.343 : 0.330$, which the 2050 projection resolves. The rate grows as g^4/M^2, so the limits below the resonant window follow $g \propto \sqrt{M}$ (Sec. 7.6).

7.3 Transport

Over cosmological distances the mass states decohere and there is no coherent potential to speak of (the relic neutrinos' own contribution to the neutrino masses through ϕ is negligible), so the transport follows the number of each mass state per unit observed energy, $F_a(E)$, on the solver's geometric grid; the physical energy at redshift z is $E(1+z)$, so redshift losses need no term of their own, and the sources inject $\Phi_0[E(1+z)]^{-p}R(z)\sum_\alpha |U_{\alpha a}|^2 f_{\alpha,S}$ per unit time, as for the dark portal. Both helicities of a mass state are followed together: with real couplings the interaction is CP-symmetric, so their sum evolves exactly whatever the source's $\bar\nu/\nu$ ratio (the solver returns it split in the injection's proportions; the split of the secondaries is not modelled).

Redshift steps are uniform in $\ln(1+z)$, one per energy bin by default (`z_substeps`), so that μ of bin j at the midpoint of step m, $\mu = (M^2/2E_0 m_r)\,e^{-(jq+m+1/2)\Delta}$, falls on one lattice per relic state: the exact bin integrals of the seven regeneration profiles are tabulated once per mediator and relic masses (not per coupling, so scans in g reuse them; ~ 30 MB at the analysis grid). Each step integrates the optical depth exactly: the Breit–Wigner terms in $\theta = \arctan[(s-M^2)/M\Gamma]$, in which the resonance is smooth however narrow (a crossing contributes $\int A(s)\,ds/B = M\Gamma^{-1}\int A\,d\theta$, with A the slowly varying density and Jacobian), the others by Gauss–Legendre in s. Then the bins are swept from high to low energy, each solving its equations over the step exactly with constant rates and sources, and passing down the number it absorbed: its own secondaries through a 3×3 system over the mass states, those of lower bins through the tabulated profiles. The number at $z = 0$ equals the injected number plus the secondaries kept in the grid minus the absorbed, to 10^{-14}. The grid extends tenfold above $E_{\max}$ so that what is injected above it cascades in. A run takes 0.3–1 s at the analysis grid (400 bins, 81 steps).

7.4 Validation

Table 6: **Checks of the secret-interaction module** (`tests/test_si.py`, `tools/si_validation.py` on Tycho). Deviations of fluxes are relative to the free-streaming flux at the same energy.

Check	Result
Channel sums vs Eq. (A11) as printed; vs quadrature of (A12) + (A13)	10^{-13} (A11 itself loses digits above $s/M^2 \sim 10^3$)
Kinematic integrals vs 50 digits	4×10^{-15}, threshold to $s/M^2 = 10^{14}$
Profile bin integrals vs 30 digits; their sums vs closed-form norms	1.5×10^{-14}; 10^{-14}, $\mu = 10^{-9}$ to 10^8
Double-scalar neutrino spectrum vs 30 digits	2×10^{-10}
No coupling vs free streaming	10^{-14}
Number balance	10^{-14}, every texture, M = 1 keV–1 GeV, g up to 1
Absorption alone vs quadrature of $e^{-\tau}$ along each path	median 10^{-10}–7×10^{-5}; up to 1.3% in the bins at a narrow resonance's sharp edge (one step per bin; 0.3% with four)
Bins and steps vs 1601 bins, four steps per bin (nested grids: no interpolation)	at 401 bins, spectra 1.4×10^{-3} pointwise (median 2.6×10^{-4}), up to 3.4% in the one bin below a very narrow resonance's edge ($g = 0.01$; first order in the bin width); energy-averaged fractions $\leq 1.2 \times 10^{-4}$ in all eight cases (scalar and vector, Majorana and Dirac)
2107.13568 Fig. 5	reproduced (Fig. 4)
2107.13568 Fig. 3 (Majorana, Dirac) vs their code, nuSIprop	0.6% above the first dip, 0.2% at the Dirac peak (with the corrected source weighting; Table 9)
Tree amplitudes by brute force (explicit spinors; Majorana fermions by fermion flow)	(A12), its complex form, the Dirac visible/sterile split 10^{-15}; vector $\lvert\mathcal{M}\rvert^2$ 10^{-12}, longitudinal fraction 3×10^{-14}
Vector: tensors vs $\int d\sigma/dt$; narrow s channel vs the Z-burst normalization	10^{-10}; 2×10^{-4}
Dirac: absorption alone; tracked regeneration over Majorana's	equal to Majorana's (3 digits); 0.499–0.500
Vector far above the resonance vs its closed form (conversions at fixed energy)	flavor fractions 10^{-6}
Vector drift integrals vs 40 digits; fractions at 201 vs 401 bins (1 keV)	10^{-7}; 1.5×10^{-4} (5×10^{-3} without the drift)

The one error above the per-mille level is local and understood: within the step in which a bin crosses a narrow resonance, the particles injected during the step are treated as exposed to the step-averaged absorption, while they are in fact absorbed only if injected before the crossing. It is first order in the step and confined to the one or two bins at the dip's high-energy edge; energy-averaged fractions do not see it (10^{-5}). Splitting the crossing step at the crossing would remove it, if the event-rate analyses need it.

Figure 4 repeats Fig. 5 of 2107.13568 (ν_τ couplings, $M = 5$ MeV, $\Sigma m_\nu = 0.1$ eV in normal ordering, an E^{-2} flux): the two dips at $E_{\rm res} = M^2/2m_j$, the bump of the regenerated neutrinos below them, the slow recovery above the resonance at $g = 0.1$ (where the t and u channels and double-scalar production dominate), and the narrow-width approximation that does not depend on g once $g \gtrsim 10^{-3}$. At $g = 0.1$, with their inputs and without double-scalar production (our `phiphi = "off"`, their default), our transport agrees with their code, nuSIprop, extrapolated

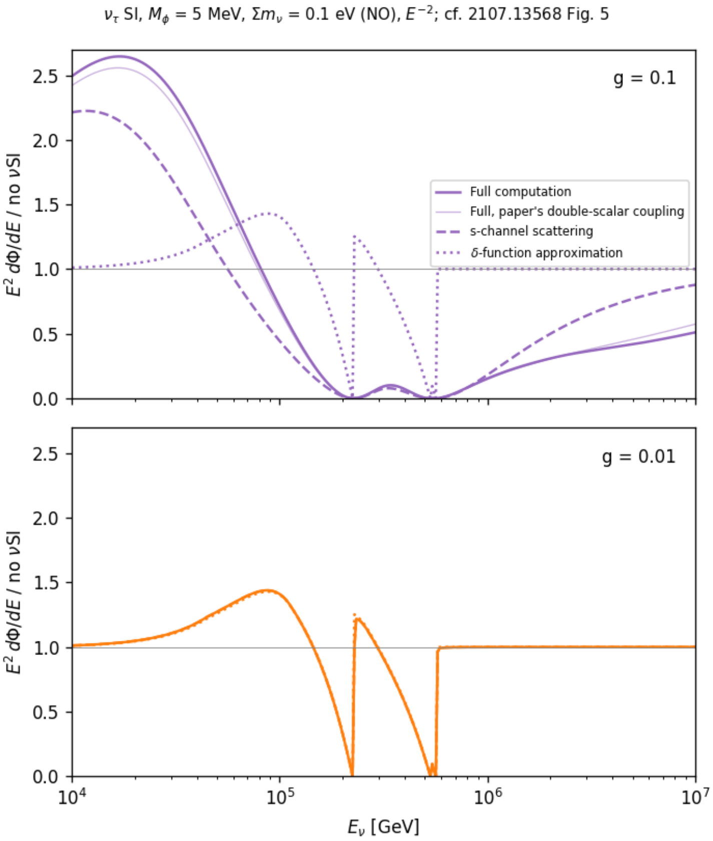


Figure 4: **Fig. 5 of arXiv:2107.13568 recomputed** with `src/dp_si.py` (all-flavor flux over the flux without interactions; ν_τ couplings, $M = 5$ MeV, $\Sigma m_\nu = 0.1$ eV in normal ordering, E^{-2} sources up to $z = 4$; 400 bins). Thin line: the double-scalar coupling as printed in the paper. `tools/si_validation.py fig5`.

in the bin width, to 1% from 10 TeV to 10 PeV; at its default 300 bins nuSIprop is still 6–11% low near the peak (2.09 against 2.35 at 10 TeV), and so is their figure. (Before the correction of App. A.13 our curve differed from theirs by up to 35%, at 100 TeV.)

Figure 5 repeats the bottom panel of their Fig. 3, the same model at $g = 0.01$ with Majorana and with Dirac neutrinos. With their inputs ($z_{\max} = 5$, $H_0 = 70$, NuFIT 5.0, their relic density) our transport agrees with their code, nuSIprop, which reproduces their figure: to 0.6% just above the first dip and 0.2% at the Dirac peak, Majorana and Dirac alike (Table 9). The 9% difference first found just above the first dip for Dirac neutrinos came from our sources' redshift weighting, not from the Dirac physics (App. A.13). Our curves also obey two exact relations: absorption alone is the same for both kinds of neutrino (to three digits at every energy), and with the sterile secondaries tracked the Dirac regeneration is exactly half the Majorana one (0.499–0.500 of it at every energy), as the parity mirror of Sec. 7.2 requires; dropping them, as the paper does, keeps 0.42–0.47 of it. The source composition plays no role here: with $1:1:1$ (equal mass states, as theirs) or $1:2:0$ the curves agree to three digits, every state being absorbed in the dips.

7.5 The flavor-triangle planes

We ran the flavor-triangle analysis of Sec. 6 with the secret interactions in place of the dark portal: the projected 2050 measurement for pion sources, the projection's mixing, $m_{\text{lightest}} = 0.01$ eV, $E^{-2.5}$ sources up to $z = 4$, the whole sky (no Earth effect here), and planes in (M, g)

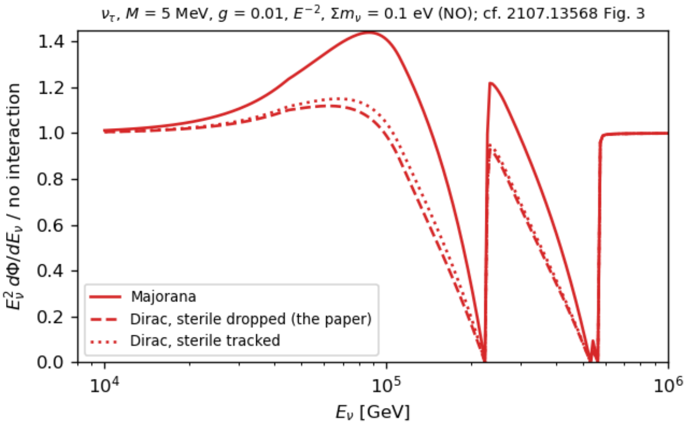


Figure 5: **Fig. 3 (bottom) of arXiv:2107.13568 recomputed**: Majorana neutrinos, and Dirac neutrinos with the sterile secondaries dropped (as the paper does) or tracked; ν_τ couplings, $M = 5$ MeV, $g = 0.01$, otherwise as Fig. 4. `tools/si_validation.py fig3`, `tools/plot_si_summary.py fig3`.

for four textures, ν_e, ν_μ or ν_τ couplings alone and flavor-universal ones, and three non-universal textures of two flavors, g on ν_e and ν_μ, ν_e and ν_τ, or ν_μ and ν_τ (49 × 49 points from 1 keV to 1 GeV and $g = 10^{-6}$ to 1, read per mass at $\Delta\chi^2 = 3.84$; Tycho, commits `fe8e2dc` for the single flavors and `a6570a6`, with the couplings $U^T G U$, for the others; 4–9 minutes each on 16 cores; configurations `pion_2050_case_5_si_*.json`). Every mass column reaches the region without an effect: $\Delta\chi^2 = 0.00$ at $g = 10^{-6}$ (a first run starting at $g = 10^{-4}$ did not, near 1 MeV). Figure 6 shows the planes and Fig. 7 how the composition moves.

The limits have three regimes. Below $\sim$ 200 keV every resonance lies below the band and the non-resonant channels act alone, with $\sigma \sim g^4/8\pi s$ independent of M: the limit is flat, $g < 5.8 \times 10^{-3}$–8.2×10^{-3} (ν_e), 6.9×10^{-3}–1.1×10^{-2} (ν_μ, ν_τ), 4.5×10^{-3}–8×10^{-3} (universal), 5×10^{-3}–10^{-2} for the pairs with ν_e. Between $\sim$ 0.5 and 3 MeV the resonances $E_{\rm res} = M^2/2m_j$ fall at 10–50 TeV, at the bottom of the band, where the $E^{-2.5}$ flux dominates the energy-averaged fractions, and the limits drop by two to three decades: $g < 1.5 \times 10^{-5}$ (ν_e, at 0.56 MeV), 2.9×10^{-5} (ν_μ, 0.56 MeV), 8.3×10^{-5} (ν_τ, 0.75 MeV), 8.0×10^{-6} (universal, 0.56 MeV), and 1.3×10^{-5}, 1.0×10^{-5}, 1.7×10^{-5} for the pairs $e\mu$, $e\tau$, $\mu\tau$. Above, the resonances move above most of the flux and the limits weaken: at 10 MeV $g < 0.13$, 0.27, 0.30, 0.064 and 0.086, 0.073, 0.12; above 30 MeV only the universal coupling and the pairs with ν_e are still limited ($g < 0.75$, 0.98 and 0.81 at 100 MeV).

Two features follow from flavor fractions being the only observable. Flavor-blind absorption does not change the composition: universal couplings act on it through the mass-dependent resonance energies and through the off-diagonal couplings of $g\,U^T U$. With $g\mathbb{1}$, as the paper would have it, the effects of the m_3 and the $m_{1,2}$ resonances compensated near 1 MeV and left an allowed gap ($g < 10^{-2}$ there, as the couplings diagonal in the mass basis of Sec. 7.6 still show); the off-diagonal couplings close most of it (1.7×10^{-4} at 1 MeV). And as g grows the composition can pass back through the projection's centre: thin allowed islands and channels run through the excluded regions (the widest, for the $\mu\tau$ pair near 100 keV, raises the limit read there to 0.09; Fig. 7 shows the paths, nearly straight lines through the standard point). Spectral data will remove both; they will also exclude much of the plane outright, since at $g \gtrsim 10^{-2}$ below 1 MeV more than 99% of the flux below 10^7 GeV is absorbed, which normalized fractions do not see. The resonant window overlaps the region below $\sim$ 2 MeV that Big-Bang nucleosynthesis already disfavors (2107.13568, Fig. 1).

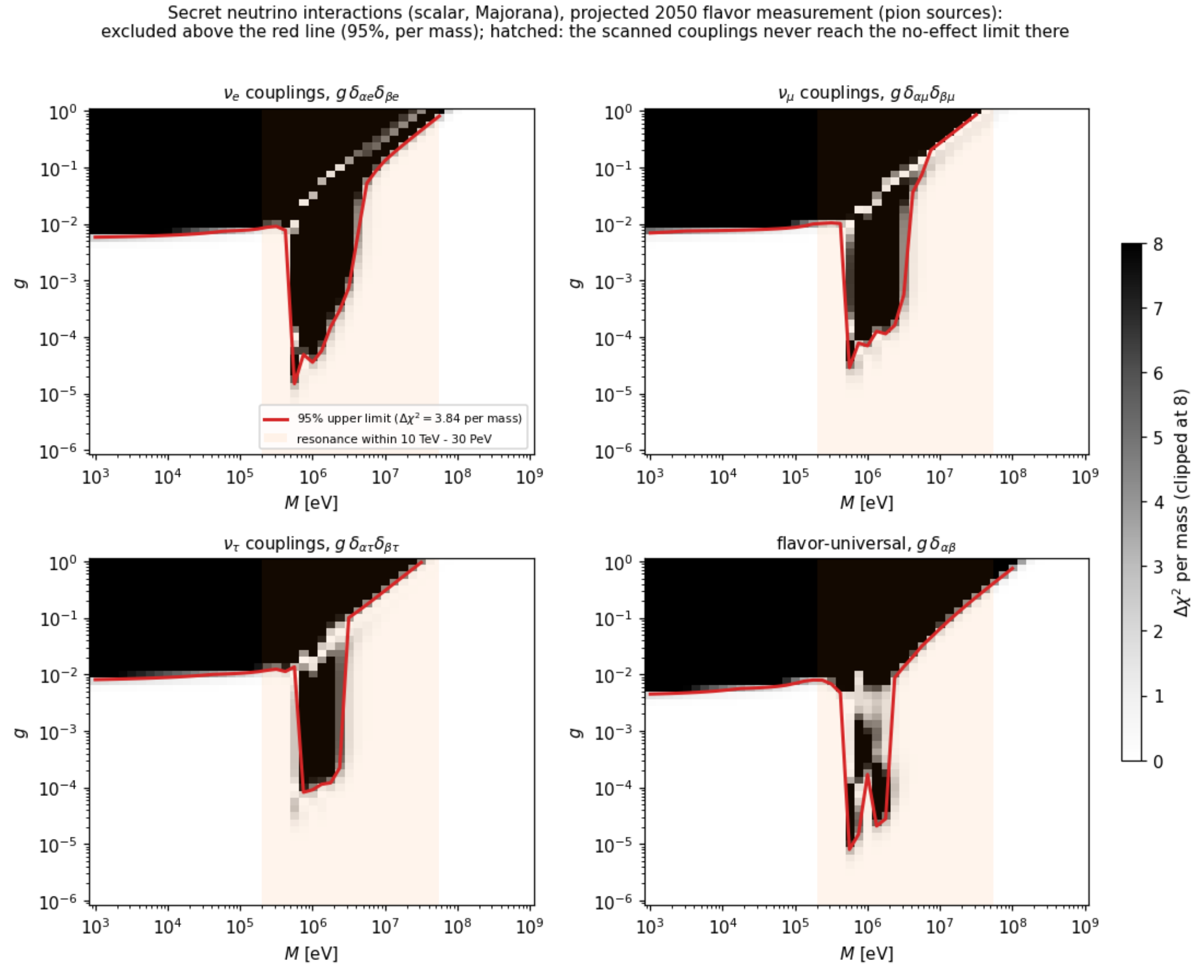


Figure 6: **Secret-interaction exclusion planes** of the projected 2050 flavor measurement (pion sources; scalar mediator, Majorana neutrinos): $\Delta\chi^2$ per mediator mass and the 95% upper limit (red; the top of each column's lowest allowed interval). Shaded: masses whose resonance falls in 10 TeV–30 PeV for some $z \leq 4$. `tools/plot_si_summary.py planes`.

7.6 Four models

We repeated the four planes of Fig. 6 for the scalar with Dirac neutrinos and for the vector with Majorana and with Dirac neutrinos (configurations `pion_2050_case_5_si_{dirac,vector, vector_dirac}_*.json`, the same grid; Tycho jobs 66307194–97 (commit `124fd03`, with the drift and the corrected source weighting of App. A.13); 7–13 minutes per plane on 16 cores, the Dirac scalar the slowest, with two transport runs per point; the vector Dirac planes are in `output/figures/si_planes_vector_dirac.png`). Figure 10 shows one benchmark under the four models, Fig. 11 and Table 7 the limits, and Figs. 12 and 13 two of the models' planes.

The resonant window (0.5–3 MeV) gives the strongest limits in all four, within a factor ~ 1.5 of each other at its best mass (absorption alone being the same for Majorana and Dirac neutrinos, Sec. 7.4). For the Dirac scalar the excluded region inside the window breaks into islands (Fig. 13), where the composition's path passes back through the projection's centre, as it does more rarely for Majorana neutrinos (Sec. 7.5); the per-mass reading then jumps with the channels: for ν_τ at 0.56 MeV it is 5.6×10^{-5} for Dirac neutrinos, while an allowed channel lifts the Majorana reading to 1.4×10^{-2}; at 0.75 MeV, 1.8×10^{-4} against 8.3×10^{-5}. Fine scans of these columns (121 couplings, with the planes' own objective; Fig. 9) reproduce the planes' points exactly and their readings within one grid step, so the channels are resolved and real:

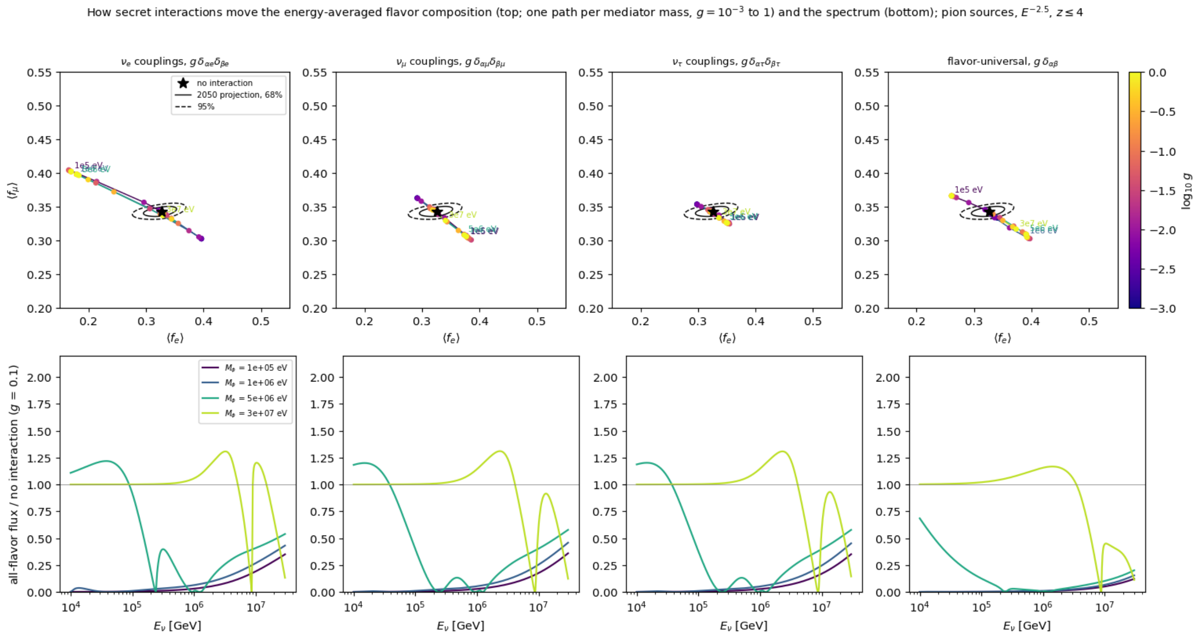


Figure 7: **How the composition moves.** Top: energy-averaged (f_e, f_μ) at the detector as g goes from 10^{-3} to 1 at four mediator masses, against the projection's 68% and 95% contours. Bottom: the all-flavor flux at $g = 0.1$ over the flux without interactions. `tools/si_flavor_paths.py` (Tycho), `tools/plot_si_summary.py paths`.

along a column $\Delta\chi^2$ falls back to $\simeq 0$ wherever the composition's path crosses the projection's centre. They also show what the single line hides: at 0.56 MeV the Dirac column crosses 3.84 only in a narrow peak near $g \simeq 6 \times 10^{-5}$ and stays allowed ($\Delta\chi^2 \simeq 0.5$) from 10^{-4} to 10^{-2}; the planes' shading shows it, the per-mass reading of the lowest allowed interval does not.

Below the window the vector's quasi-elastic conversions (Sec. 7.2) set the limit for the single-flavor textures, $g \propto \sqrt{M}$: 4.4×10^{-4} for ν_e at 1 keV, thirteen times below the scalar's 5.8×10^{-3}, with the flux intact. Their effect saturates at equal mass states, and where that shift is no longer enough (20–50 keV for ν_e) the limit comes from hard scatterings, as for the scalar. Flavor-universal couplings stay diagonal in the mass basis for the vector and for the Dirac scalar, $U^\dagger(g\mathbb{1})U = g\mathbb{1}$: no conversions, and the 1-MeV gap of $g\mathbb{1}$ that U^TU closes for the Majorana scalar (Sec. 7.5) remains for all three ($g < 0.010$–0.011 there, against 1.7×10^{-4}). Below 400 keV universal vector couplings are not limited at all: the energy loss of their forward-peaked scatterings can remove the whole flux from the band (to 2×10^{-4} of its number at 1 keV and $g = 0.03$) while hardly moving its composition (f_e by -0.016, $\Delta\chi^2 < 3.84$), which normalized fractions cannot see. Spectral data will exclude that region outright. (Before the drift of Sec. 7.2, it appeared excluded above $g \simeq 4 \times 10^{-3}$: an artifact of the missing energy loss.)

Above the window the vector is limited further, because its cross section stays constant above the resonance where the scalar's falls as $1/s$: $g < 0.071$ for ν_e at 10 MeV (scalar 0.13), and for ν_e a limit up to 100 MeV (0.74). Dirac rates of the vector are half the Majorana ones, and its limits are weaker by $2^{1/4} \simeq 1.2$ wherever one process dominates (1 keV: 5.2×10^{-4} against 4.4×10^{-4} for ν_e).

Not yet included.—The relic neutrinos' thermal momenta (relics are at rest here, as in both papers; this matters for a nearly massless lightest state, whose relics are relativistic), the timing within the crossing step (Sec. 7.4), and spectral data, which would exclude outright the regions where the flux is absorbed without changing its composition.

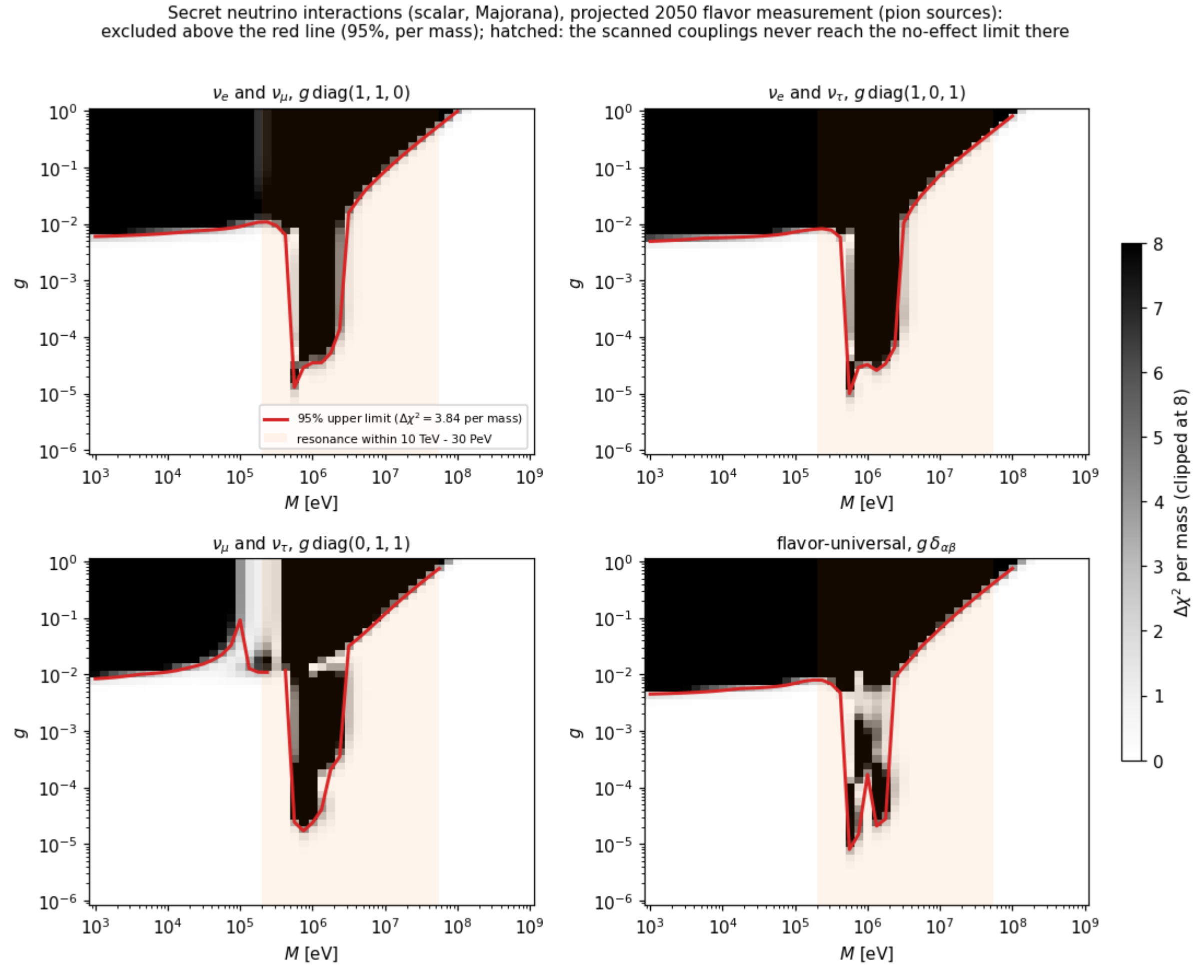


Figure 8: **Non-universal textures of two flavors**, as Fig. 6: g on ν_e and ν_μ, ν_e and ν_τ, ν_μ and ν_τ (complex mass-basis couplings $U^T G U$ at $\delta_{CP} = 212°$), and the universal coupling again. `tools/plot_si_summary.py planes2`.

7.7 Secret interactions below 1 keV (2026-10-25)

Mauricio asked (2026-10-25) that the secret-interaction campaign run once over the mass range of the dark-portal and lepton-number planes, $M = 10^{-30}$ eV–1 GeV, instead of 1 keV–1 GeV. Below 1 keV the resonance is far below the band ($s = 2E\sigma \le M^2$ needs $E \le M^2/2\sigma \lesssim 10^9$ eV for every relic momentum), and the light mediator opens processes that the planes above 1 keV do not have. This subsection lists them for the four models (scalar or vector, Majorana or Dirac neutrinos), says where each acts in the planes, and how the code carries it (`src/dp_si_light.py`, `src/dp_si.py`). Nothing changes at 1 keV and above (checked bit for bit, below). Numbers are for the μ texture and $m_{\text{lightest}} = 10^{-6}$ eV unless stated; the couplings in the mass basis are $\lambda = U^T G U$ (scalar, Majorana), $h = U^\dagger G U$ (scalar, Dirac) and $G' = U^\dagger G U$ (vector).

What acts, and where.—(i) *Hard 2 → 2 and pair channels* (dp_si's): below 1 keV every collision has $s \gg M^2$, and the cross sections depend on M only through the pair channels' collinear logarithm $\ln(s/M^2)$; the scalar's limit stays near $g \approx 6 \times 10^{-3}$ and the vector's energy loss acts from $g \sim 10^{-3}$. (ii) *The vector's soft t channel*, at a rate $\propto g^4/M_t^2$, converts the beam between mass states (Sec. 7.2); its limit continues the one at 1 keV, $g \simeq 1.4 \times 10^{-5}\sqrt{M/\mathrm{eV}}$ (ν_e), down to the mass splittings. The scalar has no forward peak: its vertex flips the helicity, so $|\mathcal{M}|^2 \propto |t|(4m_r^2 - t)/(t - M^2)^2$ vanishes at $t \to 0$. (iii) *The relic mass-change loss* (vector;

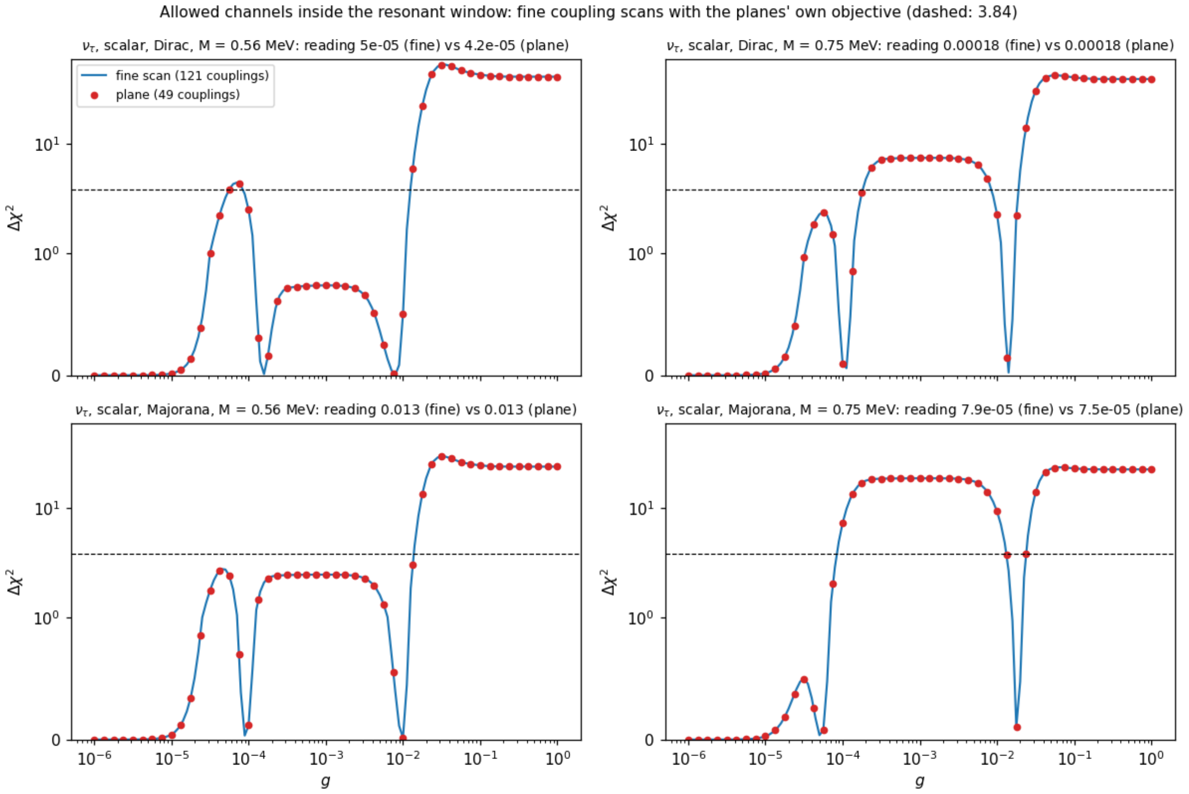


Figure 9: **Allowed channels inside the resonant window**: $\Delta\chi^2$ along four mass columns of the ν_τ planes (scalar, Majorana and Dirac, $M = 0.56$ and 0.75 MeV), fine scans of 121 couplings with the planes' objective (line) against the planes' 49 points (dots); dashed: 3.84.

Sec. 13.8, item 1) acts only within about a decade of the unitarity line. (iv) *The relics' Debye screening* of the vector: $m_D \approx 10^{-5} g$ eV (Majorana relics, whose charge is v-suppressed) to $4 \times 10^{-4} g$ eV (Dirac, $|G'|^2 mT \ln 2/4\pi^2$); where the models hold $M \geq 50\, m_D$, so $M_t^2 = M^2 + m_D^2$ differs from M^2 by at most 4×10^{-4}. (v) *Decay*, $\nu_i \to \nu_j + X$ for $M < m_i - m_j \leq 0.11$ eV. For the vector the rate grows as $(g/M)^2$ (the longitudinal mode), and at 10 TeV the beam decays over the path for $g/M \gtrsim 1.1 \times 10^{-6}$ eV^{-1} (μ, τ; 3.6×10^{-6} for e; 2.5–8 times larger at $m_{\text{lightest}} = 0.1$ eV), that is $g \sim 10^{-36}$ at 10^{-30} eV. The scalar has no longitudinal mode: its rest rate is $|\lambda_{ji}|^2 m_i/16\pi$ and the beam decays for $g \gtrsim 5 \times 10^{-8}$ (μ), 6×10^{-8} (τ), 2×10^{-7} (e, universal), whatever M. A flavor-universal coupling is diagonal in the mass basis for the vector and the Dirac scalar (no decays), but not for the Majorana scalar ($U^T U$ at $\delta_{CP} \neq 0$). (vi) *The relics' own potential*: the next paragraphs. (vii) *Where the models hold*: the vector's longitudinal mode (unitarity) and the scalar's relic field, below.

The potential the relics put on the beam.—For the vector it vanishes at first order. Its source is the relics' number current, neutrinos minus antineutrinos, which cancels without a lepton asymmetry (for Majorana relics the left-handed current's charge is $-G'hv$ per helicity h, and the two helicities are equally populated). The second order, the dispersive part of the s- and u-channel forward amplitudes, cancels at $O(1/s)$ between neutrinos and antineutrinos and is bounded by $g^2 n/\langle E_r\rangle \leq 1.2 \times 10^{-7} g^2$ eV^2 (all relics relativistic, $z = 4$), $1.6 \times 10^{-3} g^2$ of Δm^2_{21}, independent of M below the MeV: negligible here and above 1 keV. (With a relic lepton asymmetry as large as the baryon asymmetry, a Dirac vector's first-order potential $g^2 \eta n_\gamma/M^2$ would reach Δm^2_{21} only for $g/M \gtrsim 0.1$–1 eV^{-1}, five decades inside the region the decays exclude; we take no asymmetry.)

The scalar is different, and only the scalar (decided, Mauricio, 2026-10-25; `docs/`

Table 7: **95% upper limits on g of the four models** (projected 2050 flavor measurement, pion sources; per mass at $\Delta\chi^2 = 3.84$, the top of each column's lowest allowed interval). S/V: scalar/vector; M/D: Majorana/Dirac; best: the strongest limit and its mass in MeV; –: no limit up to $g = 1$ (or the column never reaches the no-effect limit). `output/analysis/`.

Texture	Model	Best (MeV)	1 keV	100 keV	10 MeV	100 MeV
ν_e	S, M	1.5×10^{-5} (0.56)	5.8×10^{-3}	7.7×10^{-3}	0.13	–
	S, D	1.8×10^{-5} (0.56)	5.7×10^{-3}	7.0×10^{-3}	0.2	–
	V, M	1.2×10^{-5} (0.56)	4.4×10^{-4}	5.7×10^{-3}	0.071	0.74
	V, D	1.7×10^{-5} (0.56)	5.2×10^{-4}	6.6×10^{-3}	0.084	0.85
ν_μ	S, M	2.9×10^{-5} (0.56)	6.9×10^{-3}	8.7×10^{-3}	0.27	–
	S, D	3.2×10^{-5} (0.75)	7.0×10^{-3}	8.4×10^{-3}	0.15	–
	V, M	2.2×10^{-5} (0.56)	8.1×10^{-4}	4.7×10^{-3}	0.14	–
	V, D	3.1×10^{-5} (0.56)	9.5×10^{-4}	5.7×10^{-3}	0.17	–
ν_τ	S, M	8.3×10^{-5} (0.75)	8.1×10^{-3}	0.011	0.3	–
	S, D	5.6×10^{-5} (0.56)	8.0×10^{-3}	0.01	0.18	–
	V, M	5.8×10^{-5} (0.75)	7.2×10^{-4}	5.3×10^{-3}	0.15	–
	V, D	8.1×10^{-5} (0.75)	8.7×10^{-4}	6.4×10^{-3}	0.18	–
universal	S, M	8.0×10^{-6} (0.56)	4.5×10^{-3}	7.0×10^{-3}	0.064	0.75
	S, D	1.1×10^{-5} (0.56)	5.2×10^{-3}	7.3×10^{-3}	0.065	0.75
	V, M	5.7×10^{-6} (0.56)	–	–	–	–
	V, D	7.9×10^{-6} (0.56)	–	–	–	–

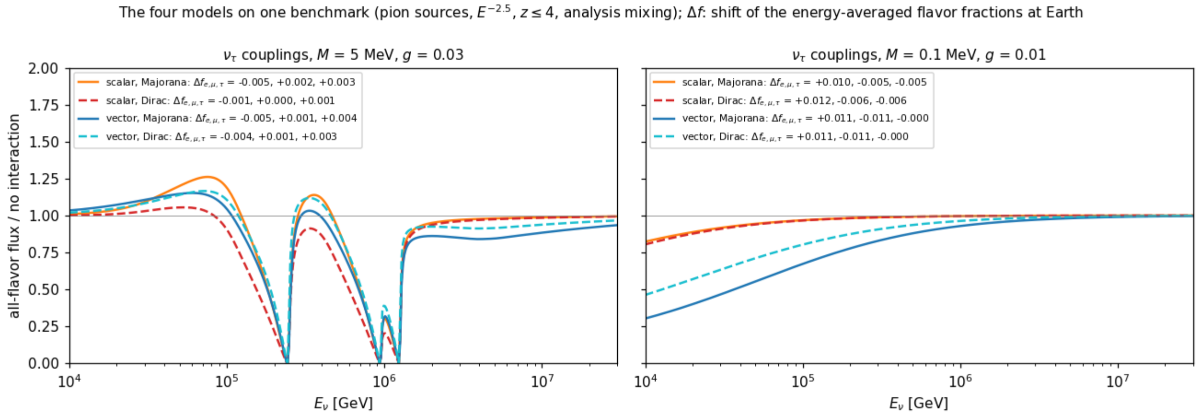


Figure 10: **One benchmark under the four models**: all-flavor flux over the flux without interactions, ν_τ couplings, and the shift of the energy-averaged flavor fractions at Earth. Left: $M = 5$ MeV, $g = 0.03$ (the resonant dips; above them the vector's constant cross section keeps acting). Right: $M = 0.1$ MeV, $g = 0.01$ (resonances below the band: the scalar's non-resonant channels against the vector's forward peak). `tools/si_validation.py models`, `tools/plot_si_summary.py models`.

`SIMPLIFICATIONS_AND_ASSUMPTIONS.md` item 23). It couples to the neutrinos' scalar density, to which relic neutrinos and antineutrinos contribute with the same sign, so the relic background sources a steady field; the dark-portal and lepton-number vectors, in contrast, get their potential from ordinary matter, already transported. Minimizing $V(\phi) + \rho_{\text{relics}}(m_{\text{eff}}(\phi))$ gives $M^2\phi = -\sum_r \operatorname{Re} c_{rr}\, n_{s,r}$, with $c = \lambda$ or h and $n_{s,r} = 2n_{\text{hel}}\eta_r(1+z)^3\langle m_r/E_r\rangle$ the scalar density of relic state r, and every neutrino, beam and relic alike, then has the mass matrix $D + c\phi$: a shift $g^2 n_s/M^2$ that grows as the mediator gets lighter. It changes the masses squared by more than $10^{-3}\Delta m^2_{21}$ (at $z = 4$, where the density is largest) above the line $g/M = 10^{c_s}$:

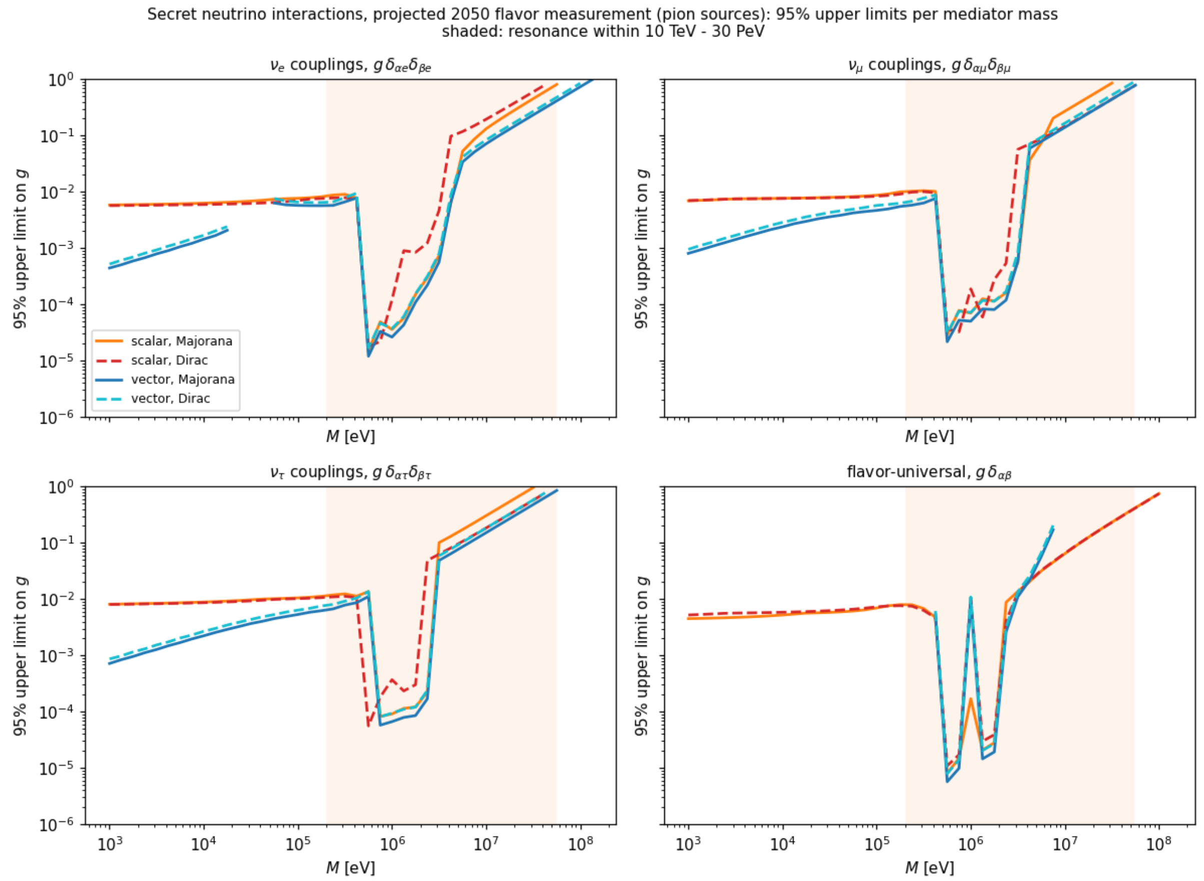


Figure 11: **95% upper limits of the four models** per mediator mass and texture (as Fig. 6). Gaps: masses with no limit up to $g = 1$, or whose lowest allowed interval is open. `tools/plot_si_summary.py limits`.

texture	relics as produced	relics decayed, $m_{\text{lightest}} = 10^{-6}$ eV	$m_{\text{lightest}} = 0.1$ eV
e	2.65 (Dirac 2.63)	3.94	1.6–1.8
μ	2.06	3.99	1.76–2.01
τ	2.07	3.83	1.76–1.86
universal	1.78	3.40 (Dirac: no decays, 1.78)	1.51

(c_s, the decimal logarithm of g/M in eV^{-1}, `dp_si_light.scalar_field_measure`). The line is highest where the relics have decayed into a light ν_1, which carries almost no scalar density; the relics decay wherever the decay is open and $g \gtrsim 4 \times 10^{-15}$. Above $g/M \sim 10^3$ eV^{-1} (0.9–7×10^3 for the non-relativistic relic states, $z \leq 4$) the scalar's attraction between relics also beats their pressure, $g^2 n_s/M^2 \gtrsim \langle p^2 \rangle / m$, far stronger than gravity: the uniform relic gas is unstable and clumps. The region is shaded rather than modelled. In the clumping regime every rate the transport uses (absorption, scattering, decays, screening) assumes a uniform relic gas of known density, temperature and masses; with clumps these are unknown without a structure-formation calculation, so neither an exclusion nor an allowed region computed there would be valid. Modelling the smooth shift below the clumping onset (a self-consistent field per redshift, fed into the transport; estimated at 1–1.5 days, not measured) would recover only the band between the two lines, one to two decades of mediator mass near the decay limit. At the scalar's limits the region covers the decay limit below $M \sim 10^{-11}$–10^{-9} eV and the hard-scattering limit below 10^{-6}–10^{-4} eV. The fits skip it, capped to the lightest masses where the model holds (as decision A for unitarity: `exact_profile.lightest_cap` through `exclusion.not_modelled_measure`, the measure growing with the lightest mass), and the fig-

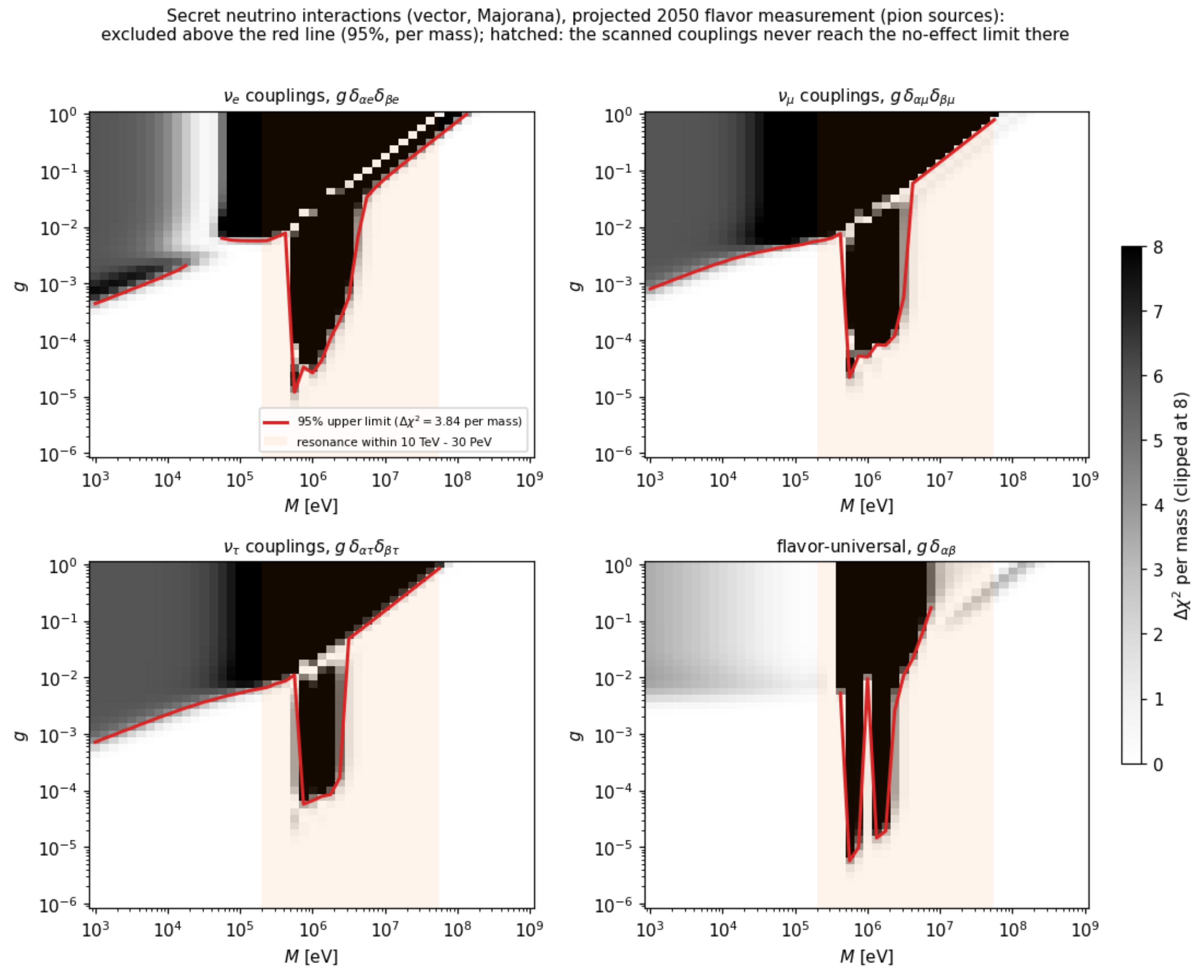


Figure 12: **Exclusion planes of the vector mediator** (Majorana neutrinos), as Fig. 6. Below the window, the quasi-elastic conversions ($g \propto \sqrt{M}$) for single flavors. `tools/plot_si_summary.py planes`, tag `vector_`.

ures shade it (`exclusion.not_modelled_mask`, `draw_not_modelled`).

Decay, the four models.—The vertices of $\nu_i \to \nu_j + X$ (amplitude $\bar{u}_j \Gamma u_i$) are $\lambda_{ji} P_L + \lambda^*_{ji} P_R$ (scalar, Majorana), h_{ji} (scalar, Dirac: both chiralities, a visible ν_L into a sterile ν_R), $\gamma^\mu (G'_{ji} P_L - G'^*_{ji} P_R)$ (vector, Majorana; `src/dp_decay.py`, unchanged) and $\gamma^\mu G'_{ji} P_L$ (vector, Dirac). The rest rates, spin-averaged, with p^* the daughter's momentum:

$$\Gamma^{\rm S,M}_{ij} = \frac{p^*}{8\pi m_i^2}\Big[|\lambda_{ji}|^2 (m_i^2 + m_j^2 - M^2) + 2 m_i m_j \,\mathrm{Re}\,\lambda^2_{ji}\Big], \qquad \Gamma^{\rm S,D}_{ij} = \frac{p^*}{8\pi m_i^2}|h_{ji}|^2\big[(m_i + m_j)^2 - M^2\big], \tag{14}$$

and the vector's of Sec. 13.8 (Dirac: the P_L piece alone). The daughters' spectra in their share z of the parent's energy are the light-front traces of `dp_decay.daughter_spectra` with the general vertex and polarization sum: on the window $[z_-, z_+]$ of two-body kinematics $d\Gamma/dz = a + bz + c/z$ exactly (the tests: residual 10^{-10} of the mean at three further angles), each parent helicity's kept and flipped spectra adding up to the closed-form rate to 10^{-12} (the total does not depend on the spin). Light daughter and mediator: the scalar keeps the helicity with $2z$ and flips it with $2(1-z)$, Majorana and Dirac alike (as Lindner, Ohlsson and Winter); the Dirac vector's visible parent keeps it with the longitudinal $2z$ and does not flip, and a sterile parent (helicity +) flips into a visible daughter with $2(1-z)$. With massive daughters the shares move: the Dirac vector flips 2.6% of $\nu_3 \to \nu_2$ at $m_{\rm lightest} = 10^{-6}$ eV and 30–50% at 0.1 eV. (The fit's $1/z$

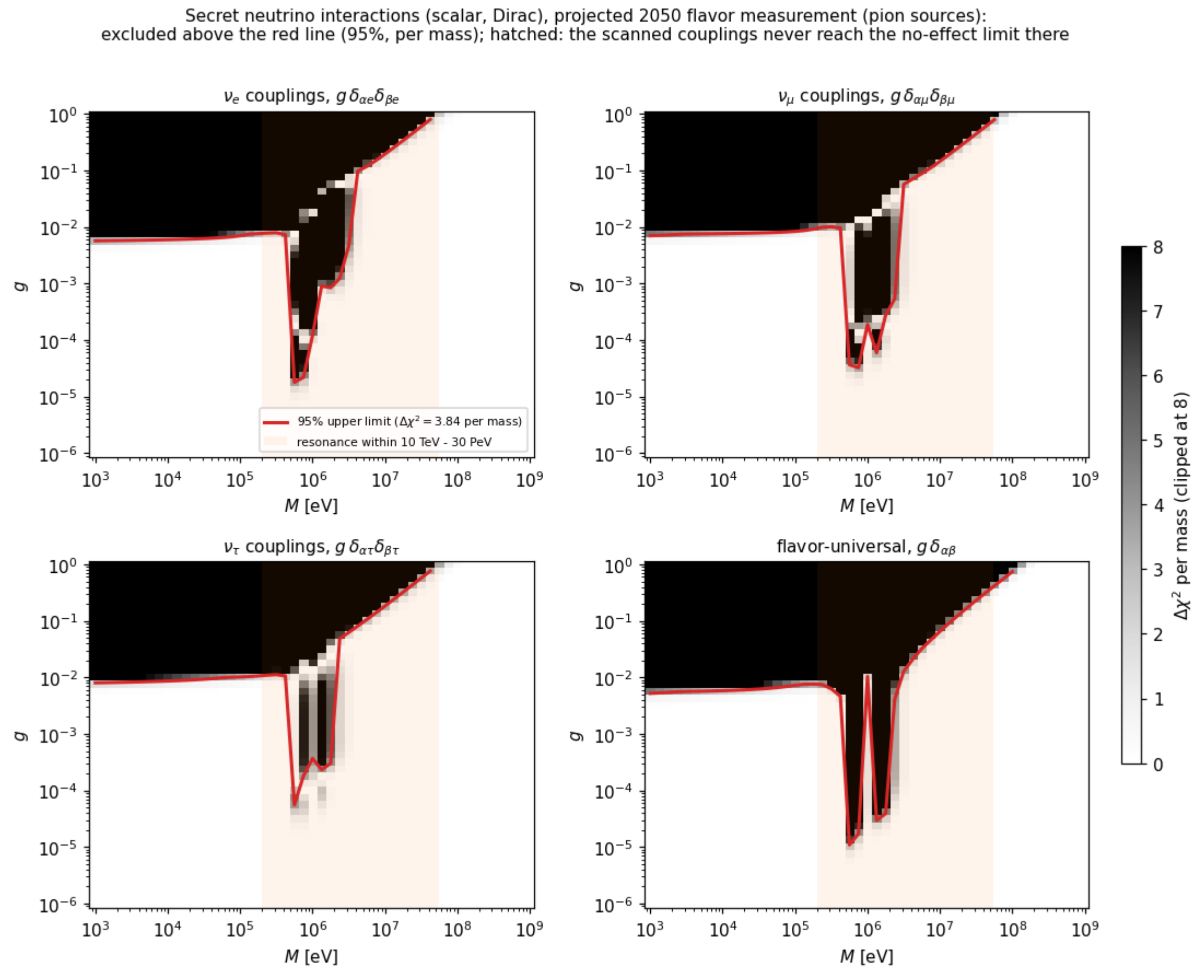


Figure 13: **Exclusion planes of the scalar with Dirac neutrinos** (sterile secondaries tracked), as Fig. 6: inside the resonant window the excluded region breaks into islands. `tools/plot_si_summary.py planes`, tag `dirac_`.

column is scaled by z_-: unscaled, at $z_- \sim 10^{-16}$ it reached 10^{16} and the least squares dropped the other columns; dp_decay's own fit, unchanged, loses accuracy below $m_{\text{lightest}} \sim 10^{-8}$ eV, outside the analysis' range.)

Dirac neutrinos: six states.—A Dirac neutrino's flipped daughter has the sterile helicity, and a sterile neutrino decays as fast as a visible one, its flipped daughters visible again. dp_si therefore follows six states, visible (0–2) and sterile (3–5), whenever a Dirac model decays (`model_spec(sterile_states=True)`): for the scalar the sterile states scatter with visibility swapped (the parity mirror behind the sum and difference runs), for the vector they do not scatter. The decay channel's arrays (`dp_si_light.transport_arrays`) give each (parent, daughter) pair its spectrum and window; for Majorana neutrinos they are dp_si's earlier ones bit for bit. The relics decay too (`dp_decay.relic_composition` with each model's rates): ν_1 only, wherever the relics' decay is open; a Dirac relic's sterile daughters are counted as relics of their mass state (an approximation that matters only where the decays already dominate).

The collision terms at the smallest masses.—Two changes, both exact where the planes need them. (a) The vector pair's spectrum (`dp_si._vv_nu_bins`) underflowed for $M^2/s < 10^{-48}$ ($M \lesssim 10^{-17}$ eV): its integration grid reached energy fractions $x < 10^{-100}$, where x^3 is zero in double precision; the moments there enter only the cumulative integrals far below every bin edge used, and are now set to zero (bit for bit wherever the routine ran before). (b) The scalar's collision terms take $\max(M, m_{\text{heaviest}})$: they depend on M only through M^2/s ($\leq 10^{-12}$ at the floor) except the pair channel's collinear logarithm $\ln(s/M^2)$, which with massive neutrinos is cut by the exchanged neutrino's mass, not by M; the region where the floor acts lies four or more decades in g inside the one the decays exclude. The vector keeps its own M_t (its forward peak is the mediator's propagator); its pair channel's logarithm is likewise too large below the neutrino masses, where its hard scatterings act only far beyond the unitarity line.

Where the vector holds.—The longitudinal mode couples, through the current's divergence, as a Goldstone boson (Sec. 13.8, item 3). For Dirac neutrinos $\partial \cdot J = i \sum G'_{ij}(m_i \bar{\nu}_i P_L \nu_j - m_j \bar{\nu}_i P_R \nu_j)$, so $-\mathcal{L} = \phi N_i Y_{ij} \nu_{Lj} + \text{h.c.}$ with $Y_{ij} = i G'_{ij} m_i/M$ and $N = (\nu_R)^c$: a Yukawa between the six Weyl fields, whose partial waves (`dp_si_light.goldstone_measure`) reproduce the bound $8\pi/3$ of one Dirac pair (Allwicher et al. Eq. 4.7). The lines $\log_{10} g = \log_{10} M + c$:

m_{lightest}	e	μ	τ	universal
10^{-6} eV, Majorana / Dirac	2.46 / 2.53	1.76 / 1.93	1.74 / 1.90	1.55 / 1.76
0.1 eV, Majorana / Dirac	1.25 / 1.46	1.23 / 1.44	1.22 / 1.44	1.08 / 1.26

(`exclusion.not_allowed_boundary` now reads the secret interactions' couplings). The scalar's Yukawa is $g \leq 1$, within its bounds; its limit is the relic field above.

Left out, with reasons.—(a) The longitudinal mode's own scatterings. With masses the vector's propagator's $q^\mu q^\nu/M^2$ part does not decouple: the beam scatters on the relics through Goldstone exchange and pair production with the Yukawa $Y \sim gm/M$, scalar-like and additive to dp_si's (massless) channels (different final helicities: no interference). Run as dp_si's scalar channels with the Goldstone Yukawa (μ texture, $M = 10^{-6}$ eV, the relics after their decays) they leave the flux at 10 TeV unchanged at $g/M = 0.01$ eV^{-1}, reduce it to 0.62 (0.27 at $m_{\text{lightest}} = 0.1$ eV) at $g/M = 0.1$, and remove it from $g/M = 1$ on. That is inside the region the decays exclude: from $g/M \sim 10^{-4}$ eV^{-1} on the beam arrives decayed and the pilot's fits give $\Delta\chi^2 \approx 10$ (now_ic26, pion sources, vector with Majorana and Dirac neutrinos); the scatterings would only strengthen that. Left out for the secret-interaction vector and the dark portal's ultralight planes, with a check on the campaign's results: every vector point with g/M between 10^{-2} eV^{-1} and the unitarity line must come out excluded, in every data set and composition. On the lepton-number planes they move the limit (at $g/M \sim 0.1$–1 eV^{-1} in the pilot) and are included (Mauricio, 2026-10-25; next paragraph). (b) The Zeno factor in dp_si's quasi-elastic conversions (the dark-portal phase has it): where the decays cycle with the conversions the widths damp the conversions into a decaying state by $1/(1 + \zeta^2/4)$, at most 2% near the unitarity line ($\zeta \leq 0.3$). (c) The hard scatterings on the Galactic paths (as for the secret interactions above 1 keV; the Galaxy's relic column is $\sim 10^{-5}$ of the cosmological one).

The longitudinal-mode scatterings on the lepton-number planes (2026-10-25).—The check by explicit spinors (massive neutrinos, the full propagator $-g^{\mu\nu} + q^\mu q^\nu/M^2$, s and t channels; `tests/test_goldstone.py`): between currents the masses do not conserve, the $q^\mu q^\nu/M^2$ part is exactly the exchange of a scalar with the Goldstone couplings Y ($q \cdot J = m_a \bar{u} P_R u - m_k \bar{u} P_L u$, to 10^{-12}); it adds to the $-g^{\mu\nu}$ part without interference (the beam's helicity flips; $< 10^{-12}$), and that part with massive spinors is dp_si's massless cross section to 10^{-9}: nothing is counted twice (the jumps and the relic loss come from the helicity-keeping forward peak, the decays are another process). With Majorana neutrinos they are dp_si's scalar channels with $\lambda = Y$, whose decay is the vector's longitudinal one (to 10^{-5}). The planes of $L_e - L_\mu$ and $L_e - L_\tau$ have a potential, so dp_si cannot carry them whole; they enter the dark-portal phase as a factor per mass state, $R_i(E) = F_i/F_i^{(0)}$ from two dp_si runs with and without the Goldstone channels (the decays and the relics after their decays in both; the relics thermal; a fixed reference source, equal flavors over the whole spectrum, so that the runs the analysis adds up still add up), each eigenstate's population at $z = 0$ multiplied by $\sum_i T_{ia} R_i$ (`dp_si_light.goldstone_factors`; the solver for those two models below 1 keV wherever the largest $|Y| \geq 10^{-4}$). Exact where the potential leaves the eigenstates mass states at the energies where they absorb: the bound $\max_E \theta(E)(1 - R_{\text{tot}}(E))$, θ the potential's $\sigma/\Delta m^2_{21}$ averaged over redshift with the absorption's weight $(1 +$

$z)/H$, is recorded per run (`meta['zeta']['goldstone']`). Left out: Dirac neutrinos, the Galactic paths ($\sim 10^{-5}$ of the cosmological depth), the conversion–decay cycles' interplay with them (where the jumps cycle the flux is already drained). Validated on Tycho (2026-10-25, job 66919204, `tools/validate_goldstone.py`): bit for bit where they do not act (lepton number below the gate, the dark portal, the secret-interaction vector at 10 keV); at $L_e - L_\mu$, $m_{Z_D} = 10^{-12}$ eV, the flux at 10^{13}, 10^{14}, 10^{15} eV with them over without: 0.955, 0.995, 0.999 ($g/m_{Z_D} = 0.1$ eV^{-1}, $m_l = 10^{-6}$ eV), 0.9996, 0.99996, 1.0000 (0.03), 0.034, 0.24, 0.75 (0.1, $m_l = 0.1$ eV), 0.81, 0.975, 0.997 (0.03, $m_l = 0.1$ eV); the exactness bound at most 2×10^{-2} (where 3% of the flux is left) and $\leq 4 \times 10^{-5}$ elsewhere; flavor fractions moved by $\leq 1.2 \times 10^{-4}$; a run's cost +5–8%. The decayed beam is mostly ν_1, whose Goldstone coupling grows with m_1: hence the strong dependence on the lightest mass.

The Galactic lines of sight.—Neutrinos emitted in the Galaxy decay, convert and lose energy over their 0.05–30 kpc too, at $z = 0$ (the extragalactic flux, by the time those matter, has long decayed). Without a potential every line of sight evolves alike but for where along it the emission sits, so the emission's spectra (per output pixel and distance bin; each pixel its children's, intensity-weighted, exact by linearity), scaled per energy by the sky mean, are split into 57 shapes by an SVD (kept to 10^{-10} of the largest singular value), and each shape in each mass state is evolved once from the Sun outward as a constant source: the populations at consecutive distance edges differ by what each bin's emission sends to the Sun (`dp_si_light.galactic_transfer`). Each distance segment solves every energy bin exactly with its conversions (Zeno-damped, as the dark-portal phase's Galactic paths), decays and sources (`_segment`), the relic loss in two halves around it. The pixels' flavor fluxes per unit emission enter the analysis' bands and sites as the averaged transfer did. Checks: the number kept to 10^{-12} over a segment, decay chains and the Dirac sterile states included (an in-place update in the first version of `_segment` created sterile daughters twice; the test found it); against each pixel's own line of sight swept inward with its actual emission, 10^{-6} (the daughters of the bins above held over different segments: first order in them); 0.4–0.8 s per point on the 150-bin test grid.

The transport and the routing.—Below 1 keV the secret interactions' cosmological phase is dp_si with the inputs of `dp_si_light.subkev_inputs`: the decays (six states for Dirac neutrinos), the relic loss (vector), M_t (vector) or the scalar's floor, the relics after their decays, and two redshift substeps per energy bin (`secret_interactions.z_substeps` 2 in the sub-keV configurations: where the decays and the conversions cycle, as for the dark-portal model's runs through dp_si). The Galactic component goes through `galactic_transfer` wherever one of the processes acts over 30 kpc (conversions above 10^{-4}, decays and loss above 10^{-6} at the grid's lowest energy), on its grid extended to the template's end.

Validation (2026-10-25, Tycho; `tools/validate_si_light.py`, `tests/test_si_light.py`, 16 tests).—(i) *1 keV and above unchanged*: `src/dp_si.py` with this subsection's edits undone (a scratch copy) against the current one, the four models at $M = 1$ keV, 100 keV, 0.5 MeV, 10 MeV: bit for bit, 16 of 16; the solver's secret-interaction branch calls dp_si exactly as before at 1 keV and above. (ii) *Limits*: without coupling the transport is the free-streaming closed form exactly (0.0); the decays alone keep the number of neutrinos, visible and sterile together, to 4.5×10^{-7} (vector, Majorana), 8×10^{-10} (vector, Dirac), 8.6×10^{-8} (scalars), the source above 10^{14} eV on a grid down to 1 GeV (the Dirac models' visible share: 0.995 and 0.979). (iii) *Redshift steps and energy bins* at 18 points of the four models (decays at $g/M = 10^{-6}$ and 10^{-5} eV^{-1}, degenerate masses, decays with conversions and loss, conversions above the splittings, hard scatterings, the relics decayed, the universal Majorana scalar), against 8 substeps per bin: the spectrum's shape within 2.4×10^{-5} at 1 substep, 5.9×10^{-6} at 2 (the configurations'), 1.2×10^{-6} at 4, a flavor fraction within 1.3×10^{-5}, 3.2×10^{-6}, 6.4×10^{-7} (first order in the step); 877 bins against 439 (nested), 6.1×10^{-5} in shape and 5.8×10^{-6} in flavor. At $g/M = 30$ eV^{-1} (decays cycling with conversions) the flux is below

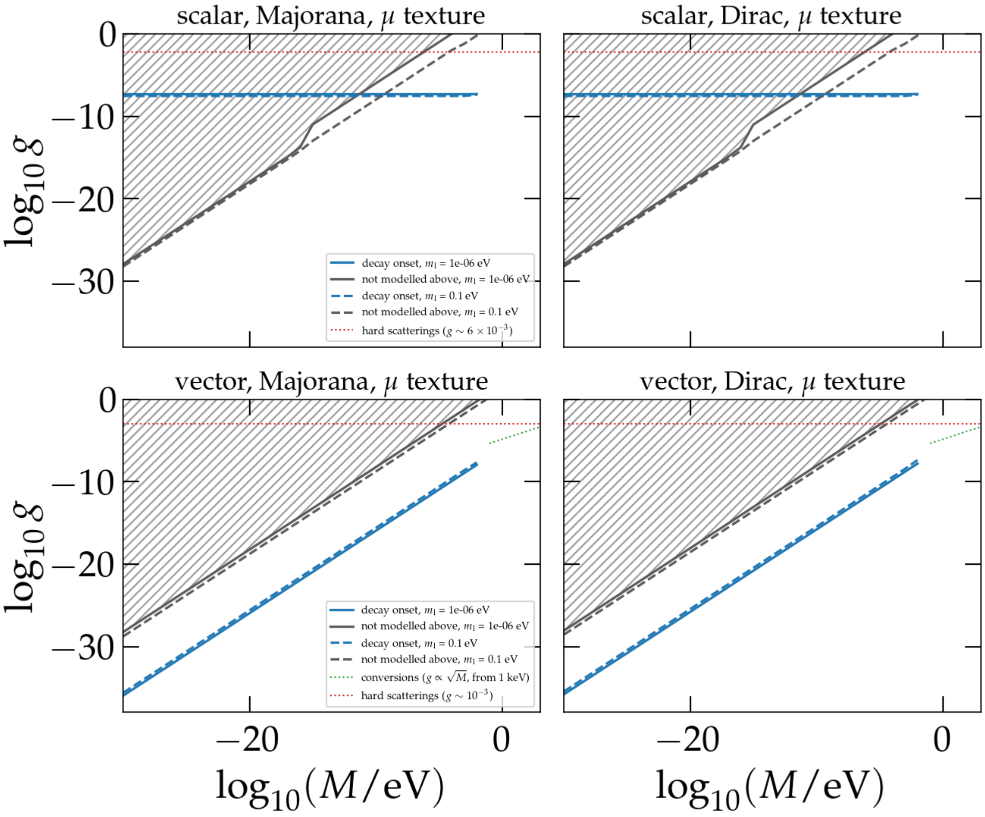


Figure 14: **The secret interactions below 1 keV** (μ texture): where the decay acts (its depth over the path at 10 TeV reaches 1; blue), where the model is not modelled (gray, hatched at $m_{\text{lightest}} = 10^{-6}$ eV; dashed at 0.1 eV): the vector's longitudinal mode beyond perturbative unitarity, the scalar's relic background field (its line rises where the relics have decayed into a light ν_1), and, for orientation, the scales of the hard scatterings and of the vector's conversions carried down from 1 keV. `tools/plot_si_subkev_regions.py`.

10^{-3} of the standard one at every energy. (iv) *Against the dark-portal phase*: the vector with Majorana neutrinos and the μ texture is the dark-portal model at $\epsilon' = 0$ with $g_D \sum |U|^2 = g$ (its eigenstates the mass states, its jumps with their Zeno factor, its decays through dp_medium, its Galactic crossing), an independent transport: at $(M, g) = (10^{-6}, 10^{-12})$, $(10^{-9}, 10^{-14})$, $(10^{-3}, 3 \times 10^{-8})$ at $m_{\text{lightest}} = 0.1$ eV, $(1, 2 \times 10^{-5})$ the shapes agree to 2.8×10^{-6} and the flavor fractions to 6.1×10^{-5}; at $(100, 10^{-3})$, routed through dp_si in both, to 4×10^{-8}. Along g/M at $M = 10^{-6}$ eV, where the decays and the conversions cycle: the two agree to $\leq 1 \times 10^{-5}$ while the flux keeps 0.2 or more of the standard one ($g/M \leq 0.03$ eV^{-1}), to 3×10^{-4} at 3×10^{-3} of it ($g/M = 0.1$), and part below $\sim 10^{-4}$ of it: at $g/M = 0.3$, 1 and 3 eV^{-1} dp_si leaves 3.6×10^{-5}, 3.2×10^{-7} and 5×10^{-9} of the standard flux, the dark-portal phase 0.93, 5×10^{-4} and 10^{-90} times that. dp_si's falls as the steady state of the cycles requires: each neutrino passes through a decade of energy in a few cycle times, so the flux left is $\sim H/\Gamma_{\text{cycle}}$ ($\sim 3 \times 10^{-9}$ at $g/M = 3$, scaling as $(g/M)^{-4}$ at fixed M, as it does); the dark-portal phase drains faster than that where the cycles outrun its redshift steps by $\gtrsim 10^4$. Either way those points are excluded by many units of χ^2; noted for the dark-portal and lepton-number planes. (v) *The Galactic lines of sight*: halving every distance segment moves the flux by $\leq 4.8 \times 10^{-7}$, the SVD's tolerance 10^{-13} against 10^{-10} by 1.2×10^{-8} (57 shapes); 1.3–4.7 s per point at 470 bins.

The planes.—`tools/make_si_campaign_configs.py` writes, per data set (now_ic26, now_events, proj_future), model, texture (e, μ, τ, universal) and composition (pion, fitted), the plane over 1 keV–1 GeV (the first planes' grid, 49 × 49, $g = 10^{-6}$–1; it reproduces those four configurations exactly) and the sub-keV plane, 10^{-30} eV–1 keV with one mass per decade (34) and $g = 10^{-38}$–1 with two per decade (77): the vector's decays act down to $g/M \sim 10^{-6}$ eV^{-1}, and every column must reach the region without an effect. 192 planes in all.

The pilot (2026-10-25, Tycho, now_ic26, pion sources, μ texture, the sub-keV planes; one core per fit, `tools/pilot_fits.py` with `--points`, `slurm/si_pilot2.sh`).—Sixteen points per model in the modelled region, four fits per worker (the first paying the worker's numba compilation): median 40 s per fit (scalar, Majorana), 48 s (vector, Majorana), 73 s (vector, Dirac; one fit 1165 s), 98 s (scalar, Dirac). Where the beam arrives decayed the fits give $\chi^2 \approx 19$–20 against 9.04–9.28 without an effect ($\Delta\chi^2 \approx 10$), and where the decays cycle with the conversions up to 167. Per sub-keV plane, from where the processes switch on (an estimate, not measured): the vector ~ 600 points with an effect, ~ 840 not modelled (not fitted), ~ 1200 without an effect (inherited); the scalar ~ 150, ~ 770, ~ 1700; the universal vector ~ 60 with an effect. With the medians, the full fits of the four models' sub-keV planes of one texture take ~ 26 core-hours for now_ic26 with pion sources; the other data sets and the fitted compositions are not measured.

8 Arrival directions and Galactic sources

On 2026-10-13 Mauricio asked for per-direction output and a Galactic-source component, with an implementation plan backed by published work before any code. The plan (`docs/dev/DIRECTIONS_PLAN.md`, decisions D1–D6) was written from surveys of the solver's direction handling, the literature on Galactic diffuse neutrinos and on new physics with directions, and the sibling projects' data. This section collects its results as they land.

Sky and frames.—`src/dp_sky.py`: HEALPix pixels in RING ordering (Gorski et al. 2005), equatorial J2000 with the IAU 1958 Galactic frame in its FK5 realization (agreeing with astropy to 7×10^{-10} rad), the ecliptic, and the zenith at the South Pole, $\cos\theta_z = -\sin\delta$ at every time, so that the Earth's part of the flux depends on declination only. Output in 20 declination bands of equal solid angle (0.1 in $\sin\delta$) and HEALPix maps at $N_{\rm side} = 16$ (decision D1).

The Earth per band.—The phase-aware zenith average restricted to each band (App. A.7, last paragraph): the bands weighted by solid angle return the whole sky to rounding, and each band matches the exact integral over its range. `propagate_sky` returns the bands for neutrinos and antineutrinos separately; secret interactions are isotropic.

The Sun per direction (decision D4: measure first).—Over a year a direction at ecliptic latitude β sees the Sun at $\cos\psi = \cos\beta\cos(\lambda - \lambda_\odot(t))$ (Kepler orbit); its flux is $\sum_p a_p F_p$ over the transit's Gauss–Legendre paths, with a_p the year-average of the Lagrange basis on the path cosines, which averages over the sky to the Gauss–Legendre weights (isotropy recovered exactly; the path runs recombine to the isotropic one to rounding). At twelve points across the Sun's window ($m_{Z_D} = 10^{-24}$ to 10^{-18} eV; the Sun's potential at the Earth 0.3, 1 and 3 times $\Delta m^2_{31}/(2 \times 100$ TeV), $e\mu$ texture) the yearly per-direction weighting moves the energy-averaged flavor fractions of a pixel by at most 1.2×10^{-3}, and by 10^{-5}–5×10^{-4} typically, against 6.4×10^{-3} for single paths: every direction sweeps through angles to the Sun over the year (Fig. 15). The effect falls as the Sun's potential grows, where its crossing becomes adiabatic and depends only on the endpoints (as found for supernova neutrinos, arXiv:2607.24918). The Sun is kept isotropic; `tools/sun_anisotropy.py`.

The Galactic templates (decision D2).—Two templates, each in IceCube's convention (per flavor, $\nu + \bar\nu$; the release README of doi:10.21234/8v5d-rn16) and reduced on Tycho to equatorial $N_{\rm side} = 16$ maps from 72 GeV to 10^8 GeV (`tools/prepare_galactic.py`): CRINGE (Schwefer, Mertsch & Wiebusch 2023, Zenodo 7859442) and the Fermi-LAT π^0 template with $E^{-2.7}$, whose model flux is $E^2\Phi(100\ \text{TeV}) = 4.4 \times 10^{-12}$ TeV cm^{-2} s^{-1}, sky-integrated per flavor (arXiv:2607.25966, supplement). The frames are read off the maps (CRINGE is Galactic,

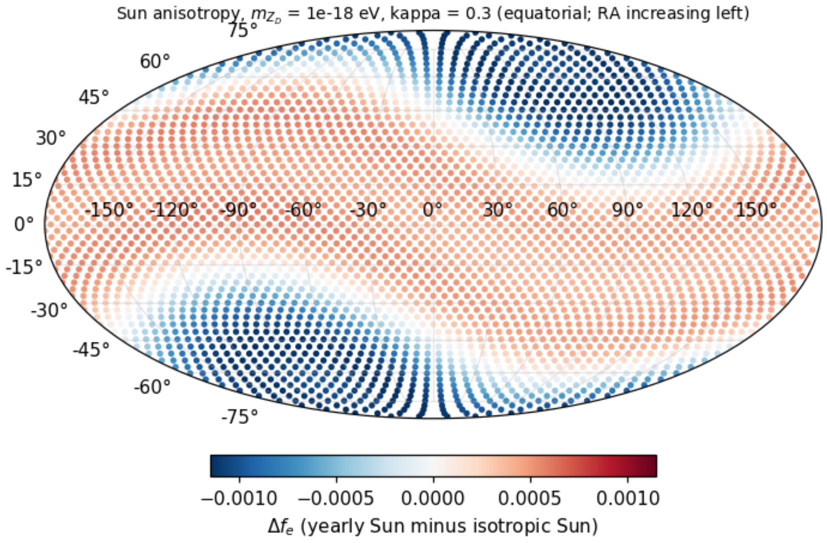

Sun anisotropy, m_{Z_D} = 1e-18 eV, kappa = 0.3 (equatorial; RA increasing left)
75°
60°
45°
30°
15°
0°
-15°
-30°
-45°
-60°
-75°
-150°
-120°
-90°
-60°
-30°
0°
30°
60°
90°
120°
150°
−0.0010
−0.0005
0.0000
0.0005
0.0010
Δf_e (yearly Sun minus isotropic Sun)

Figure 15: **The Sun's anisotropy** at its largest among the points scanned ($m_{Z_D} = 10^{-18}$ eV, the Sun's potential 0.3 times the vacuum term at 100 TeV): the energy-averaged ν_e fraction with the yearly per-direction weighting minus the isotropic Sun average, per HEALPix pixel ($N_{\rm side} = 16$, equatorial). The pattern follows the ecliptic.

IceCube's π^0 map equatorial; toise's Galactic copy of the latter, rotated by our code, correlates with it at 0.997). The test is IceCube's own fit: times its 2026 normalizations (0.91 for CRINGE, 4.99 for π^0) our inner-Galaxy spectra ($|b| < 15°$, $|l| < 20°$) lie on the curves of arXiv:2607.25966 Fig. 4, read from the figure's SVG path coordinates, to 0.25% and 0.47% (Fig. 16). That test exposed what IceCube calls CRINGE: the diffuse map *plus* CRINGE's unresolved-source map. The diffuse map alone reaches only 28–54% of the fitted curve, and is softer; the unresolved sources carry 75% of the sky-integrated flux at 100 TeV and dominate the inner Galaxy above $\sim$30 TeV. For the dark portal's emission points (plan step 6) this makes the source distribution, not the gas, the main term above 30 TeV.

The Galactic component.—We keep each template's all-flavor total and apply our own physics (`src/dp_galactic.py`): sources of composition $(x, 1-x, 0)$ (decision D6, x floated on its own; 1/3 for pion decay), $\nu : \bar{\nu} = 1 : 1$. Under secret interactions the Galactic flux is unattenuated (the Galactic relic column is $\sim 10^{-5}$ of the cosmological one) with oscillations averaged over kiloparsecs. For the dark portal, emission at the local potential (option `local`; the 3-D emission points below are the default): the flavors decohere into the local eigenstates, populations $\sum_\alpha f_\alpha |W_{\alpha k}|^2$, and from there cross the Sun and the Earth as the extragalactic flux does, per declination band. `propagate_sky` returns the component in physical units next to the extragalactic one. Checks (`tests/test_galactic.py`): the transfer conserves the flux to 3×10^{-14} and is linear in x; without the dark potential it is the averaged oscillation to 10^{-12}; with the Earth's standard matter potential on, the bands through the Earth move by the MSW effect on the arriving mass-state mixture, 2.4×10^{-5} at 10 TeV through the core and falling as $1/E$.

Emission points in three dimensions (decision D3).—For the dark portal, a Galactic neutrino is born in the eigenstates of the potential at its emission point and, the Galaxy's potential changing over $\gtrsim 10$ pc against oscillation lengths $\lesssim 10^{-3}$ pc, keeps their populations on its way to the Sun. The potential there is the Yukawa-smoothed neutron density $n_{\rm eff}(\vec{x}; m)$, now tabulated at any point of the Galaxy (`tools/mw_neff_tables.py`): sphere averages of our Galactic model around 48×22 nodes in (R, z), integrated on the solver's own distance nodes, for 20 masses per decade from 10^{-34} to 10^{-12} eV. At the Sun's node it equals the solver's local

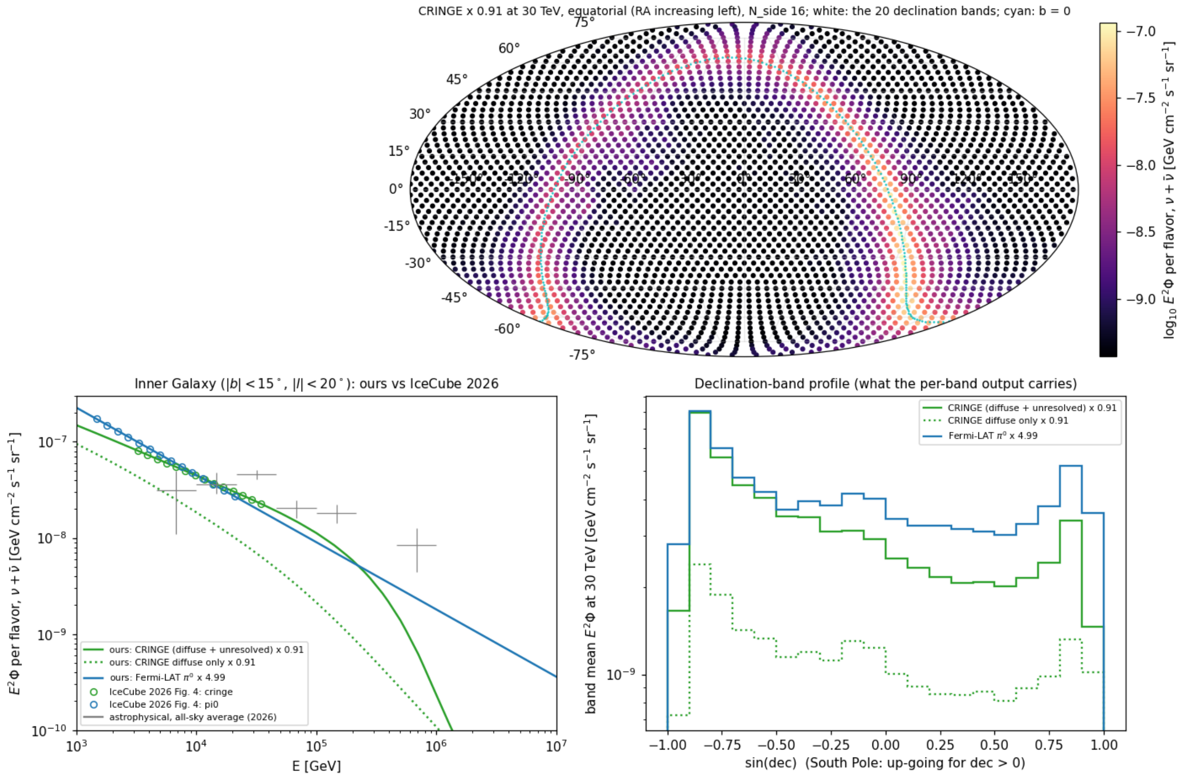


Figure 16: **The Galactic templates.** Top: CRINGE (diffuse plus unresolved sources) times 0.91 at 30 TeV, equatorial, with the 20 declination bands. Bottom left: inner-Galaxy spectra, ours (lines) against IceCube's 2026 best fits (circles, arXiv:2607.25966 Fig. 4) and its all-sky astrophysical flux (grey); the dotted line is CRINGE's diffuse map alone. Bottom right: the band means the per-band output carries.

potential to 10^{-9} (test T7). Far beyond the Galaxy's size $n_{\rm eff}$ is *not* uniform: the Galaxy's share is m^2 times its neutrons' "Coulomb" potential at the point, about six times deeper at the Centre than at the Sun, and it exceeds the cosmic floor down to $\sim 4 \times 10^{-31}$ eV; only beyond the Hubble radius is the pattern frozen. Where along each line of sight the neutrinos are born follows the two components of CRINGE: the gas (hydrogen columns of Söding et al. 2025) times CRINGE's cosmic-ray flux at 20 times the neutrino energy for the diffuse part, and the Lorimer et al. (2006) source distribution with a 0.2-kpc scale height, weighted by distance, for the unresolved sources (the nearby ones are resolved and removed: with $dN/dL \propto L^{-1.5}$ and a flux threshold the unresolved luminosity per source grows as the distance). Integrated along each line of sight, these reproduce the morphology of CRINGE's maps (correlations 0.991 and 0.985). Toward the Centre the neutrinos are born on average 5.7 kpc (gas) and 9.8 kpc (sources) away, toward the anticentre 0.9 and 3.7 kpc (Fig. 17). Per pixel, the emission densities are deposited on nodes 20 per decade apart, each node's populations come from the transit's own frame table, and three runs per kind give each declination band's response to a unit population of each local eigenstate. Checks: the transit from the emission points is adiabatic everywhere we looked (Landau–Zener parameter $\geq 1.5\times10^5$ from 10^{-28} to 10^{-16} eV and 10 TeV to 30 PeV; test T9); beyond the Hubble radius three-dimensional and local emission agree; a Galactic source at the Sun's position and an extragalactic source in a Milky-Way-like host at $z \to 0$ give the same flavors to 10^{-5} (test T10). That test caught an error in the local emission of step 5, which had included the Moon's potential at birth (13% of the Galaxy's at a 6-pc range); Galactic neutrinos are born far from the Earth–Moon system. With emission in 3-D, the Galactic flavor composition becomes anisotropic where the emission potentials straddle the vacuum term: at

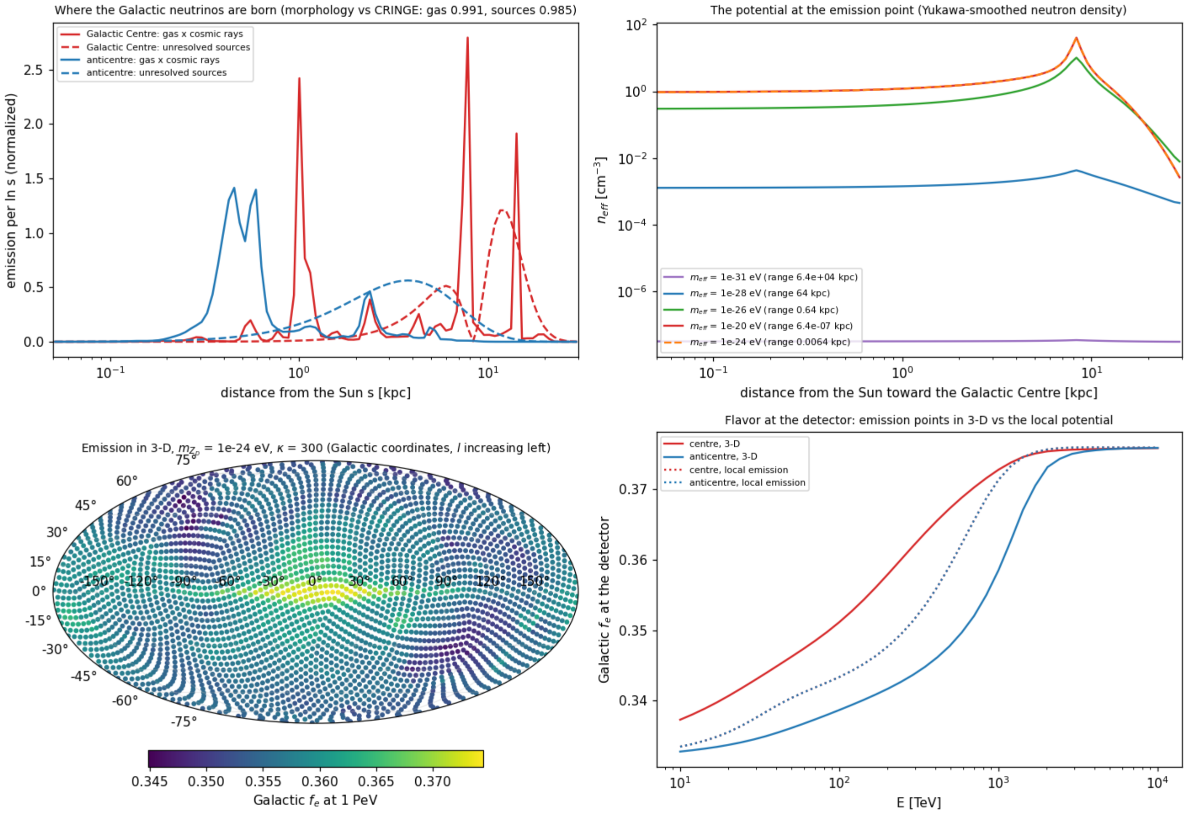


Figure 17: **Galactic emission points.** Top left: where along the lines of sight toward the Galactic Centre and the anticentre the neutrinos are born, for the diffuse emission (gas times cosmic rays) and the unresolved sources. Top right: the potential at the emission point along the Centre's line of sight for several mediator ranges. Bottom: the Galactic ν_e fraction at the detector for emission points in 3-D at $m_{Z_D} = 10^{-24}$ eV, per pixel at 1 PeV (left) and against energy for the Centre and the anticentre (right; dotted: emission at the local potential).

$m_{Z_D} = 10^{-24}$ eV and a coupling that puts the local Galactic potential near the vacuum term at 1 PeV, f_e at 1 PeV ranges over 0.345–0.374 across the sky, highest along the inner Galactic plane.

The pieces together (plan step 7).—To add the Galactic flux to the extragalactic one, the latter can now be put in physical units (`sky.extragalactic_norm`): its per-flavor intensity at 100 TeV without new physics, from the closed form of the free-streaming flux, which the solver reproduces to its redshift grid's first-order error (4×10^{-3} at 400 steps). With IceCube's HESE 7.5-year fit (2.12×10^{-18} GeV^{-1} cm^{-2} s^{-1} sr^{-1} per flavor at 100 TeV, $\gamma = 2.87$) the Galactic share of the flux (CRINGE × 0.91) is 4%, 6% and 1% of the sky-averaged flux at 10 TeV, 100 TeV and 1 PeV, and 22%, 35% and 7% in the inner Galaxy (Fig. 18). An independent check of the template's normalization: at 20 TeV the sky-averaged CRINGE flux is 11% of IceCube's 2026 astrophysical flux in that energy bin; the 2026 paper states "about 10%". (HESE's soft fit lies a factor 4 above the 2026 flux at 10 TeV, the known tension between it and fits with a spectral break, so the Galactic share depends on that choice.) At the dark-portal point of Fig. 17 the anisotropy of f_e at 100 TeV comes from the Galactic component (a spread of 0.018 over the pixels; 0.014 in the total flux), against 10^{-6} from the Earth there and at most 1.2×10^{-3} from the Sun's yearly path (decision D4).

A consumer: HESE (plan step 8).—IceCube's HESE 7.5-year release gives each simulated event its weight over flux and livetime, so its expected rate under a flux is the livetime times the sum over events of that weight times the flux of the event's particle type at its true energy

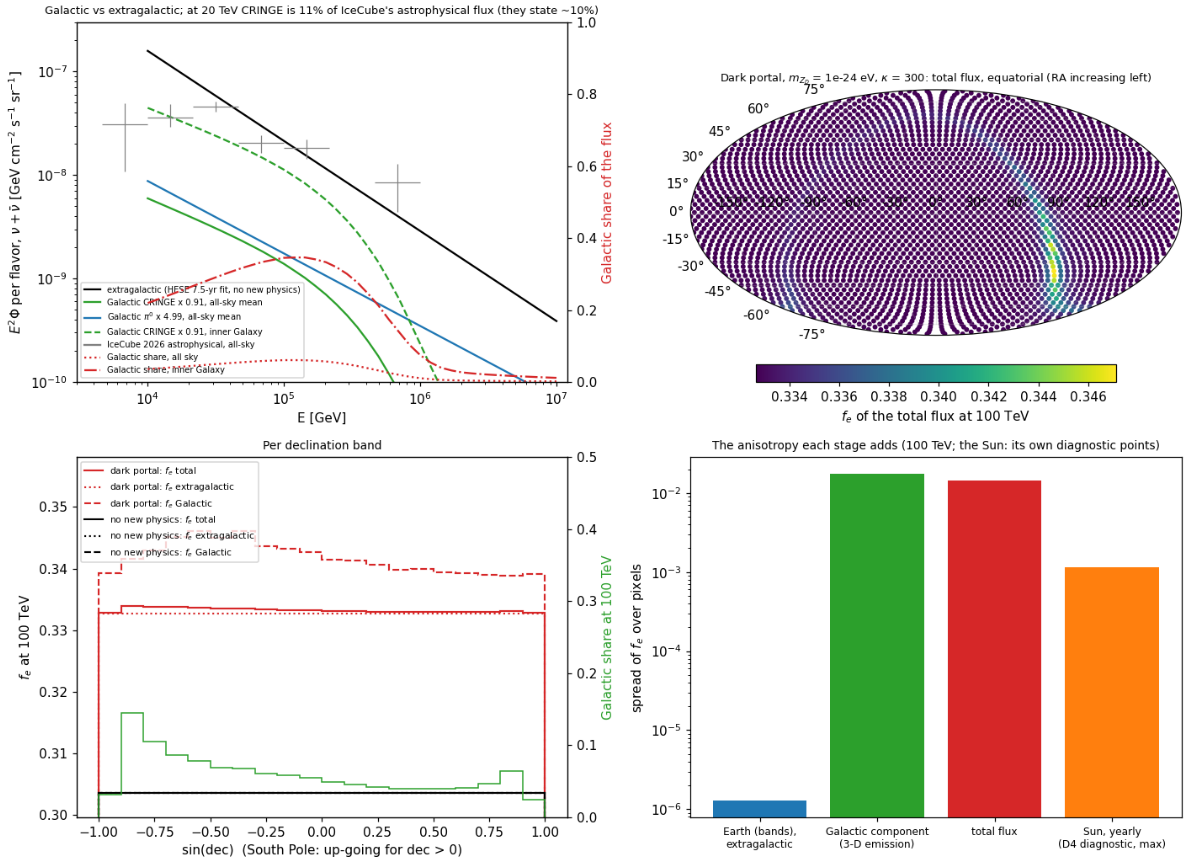


Figure 18: **The sky with the Galactic component** (`tools/sky_validation.py`). Top left: sky-averaged per-flavor spectra of the extragalactic flux (HESE 7.5-year fit), the Galactic templates and IceCube's 2026 astrophysical flux, with the Galactic share (right axis). Top right: f_e of the total flux at 100 TeV for the dark-portal point of Fig. 17. Bottom left: per declination band, f_e of the total, extragalactic and Galactic fluxes with and without the dark portal, and the Galactic share. Bottom right: the spread of f_e over the pixels that each stage adds.

and direction (`tools/hese_check.py`; the release on Tycho). With the release's best-fit flux (6.37 for the six species at 100 TeV, $\gamma = 2.87$) our weighting and the release's own code (run in its own environment, with photospline for its detector corrections) agree event by event to 2×10^{-14}: 50.16 astrophysical events above 60 TeV in 2635 days (39.5 cascades, 8.4 tracks, 2.2 double cascades); the detector corrections at their nominal values are exactly 1. The units test: `propagate_sky` with sources $(1/3, 1/3, 1/3)$, which stay 1:1:1 at the Earth for any mixing, and the extragalactic flux normalized to the same fit reproduces the release's flux for every one of the 736 781 neutrino events to the same factor, 1.0036, the redshift grid's offset of the normalization: the normalization, the neutrino–antineutrino split, the flavors, the declination ($\sin\delta = -\cos\theta_z$) and the energy interpolation are all consistent. Pion sources instead give the flavors $(0.30, 0.36, 0.34)$ at the Earth: 3.7% fewer cascades and 3.8% more tracks. The Galactic component (CRINGE × 0.91) adds 2.8 events (5.6%), mostly cascades below 200 TeV (Fig. 19). Cascades resolve 10–20°, so HESE can use maps of $N_{\text{side}} = 8$ (Mauricio); the release's MC has no azimuth, so the Galactic flux enters averaged over right ascension per declination band.

The planes with the Galactic component (Mauricio's task 2; finished 2026-10-14).—Every secret-interaction plane and the pre-audit dark-portal planes were rerun with the Galactic component and IceCube's 2026 spectrum as data (the extragalactic normalization and index floating; `tools/make_galactic_configs.py`), with the sources' compositions at pion decay and, as a

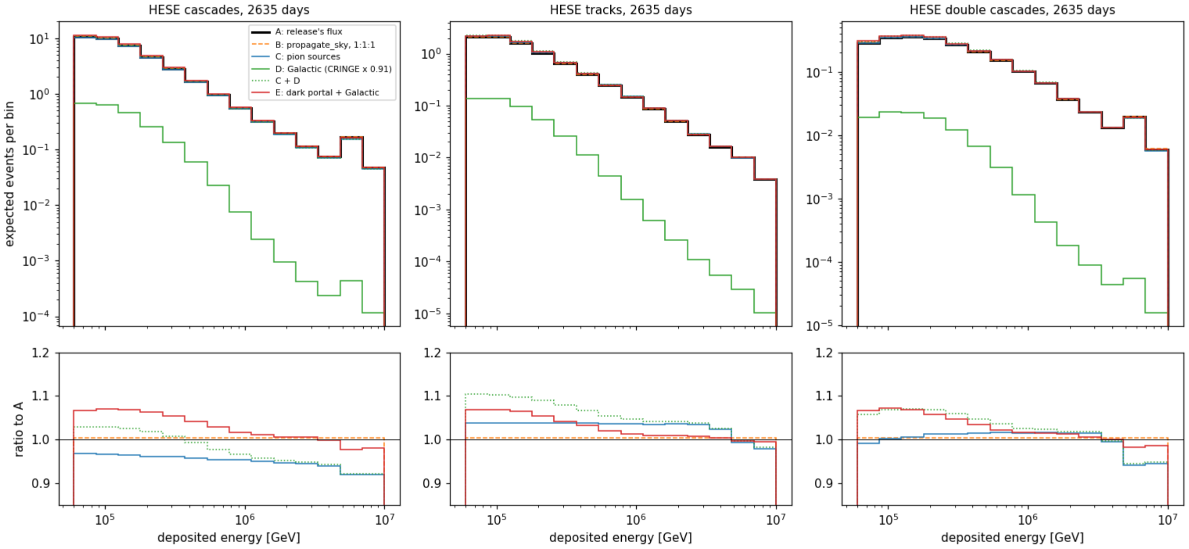


Figure 19: **Expected HESE events** (2635 days, deposited energy 60 TeV–10 PeV) per morphology, from the release's MC: A, the release's best-fit flux; B, our per-direction output with sources 1:1:1 normalized to it (the units test: 1.0036 for every event); C, pion sources; D, the Galactic component; E, a dark-portal point with the Galactic component. Bottom: ratios to A.

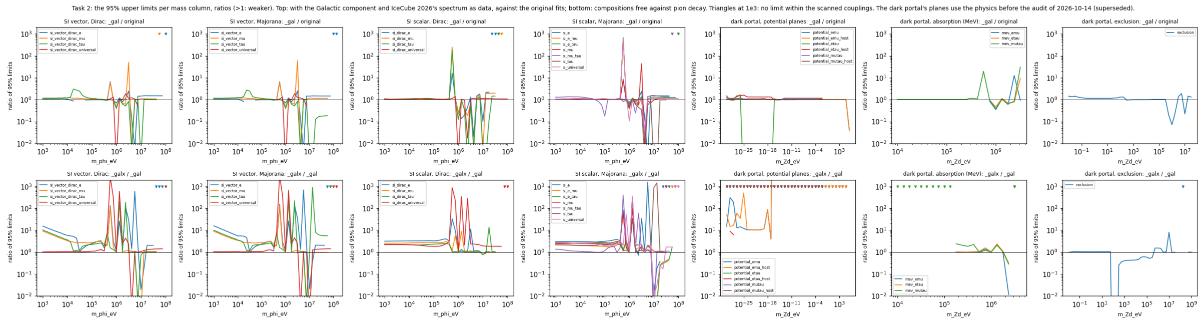


Figure 20: **Task 2: the Galactic component and IceCube's 2026 spectrum.** The 95% upper limits per mass, as ratios (above 1: weaker). Top: with the Galactic component and the 2026 spectrum as data, against the original flavor-only fits. Bottom: the sources' compositions fitted, against pion decay. The dark portal's panels use the physics before the audit (superseded).

variant, fitted (Fig. 20, `tools/plot_galactic_comparison.py`). Read per mass as all planes are, the secret-interaction limits barely move: the median ratio to the original limits is 1.02–1.23 per plane. Two findings. First, with the 2026 spectrum every secret-interaction plane prefers new physics somewhere: $\Delta\chi^2 = 3.6$–10 below the fit without it, always for a resonance at 0.5–7.5 MeV (couplings 10^{-4}–10^{-1}), the dip it carves fitting the measured spectrum better than a power law ($\sim 2\sigma$ for the two parameters); the same will show in the MeV planes of the campaign. (The comparison tool had first dropped every mass column whose minimum lies above the plane's, which this preference makes all of them; fixed.) Second, fitting the sources' compositions instead of fixing them at pion decay weakens the limits below $\sim$1 MeV by 1.5–10 times. The dark portal's planes of this task predate the physics audit (Sec. 12) and are superseded by the campaign; its last four dark-portal analyses (the FrequentistAnalysis ports, with the mixing free) were stopped, with Mauricio's agreement.

9 Track rates with IceTracks-DR2

Expected through-going muon tracks for any flux the solver computes (`IceCubeRates/`; plan in `docs/dev/DR2_TRACKS_PLAN.md`). The response is IceCube's IceTracks-DR2 release (v3.1, 2008–2022, doi:10.7910/DVN/MMIIZA), in four detector configurations: IC40, IC59, IC79 and IC86 (eleven seasons), 4963 days of good runs in all. For each it gives the muon-neutrino effective area $A(E_\nu, \delta)$, in 0.2-dex bins of true energy and 50 bins of true declination (uniform in $\sin\delta$), with absorption in the Earth included, and a five-dimensional smearing table: for each bin of true energy (0.5 dex) and declination (30–52 bins), the fraction of simulated events in each cell of reconstructed muon energy, point-spread angle ψ (between the true neutrino direction and the reconstructed muon direction) and estimated angular error. We sum over the angular error and move the cells, which the release chooses per bin, onto common grids: 0.1 dex in reconstructed energy and 0.25 dex in ψ up to 180°, in proportion to their overlaps (`tools/dr2_reduce.py`). Every filled slice still sums to 1, to 4×10^{-13}.

The fold.—For configuration c, with livetime T_c, the expected number of tracks in a bin of reconstructed energy E_r and reconstructed declination m is

$$N_c(E_r, m) = T_c \sum_{i,k} A_c(i,k) \int_{E_i} dE \int_{\delta_k} d\Omega\ \Phi_{\nu_\mu+\bar{\nu}_\mu}(E,\delta) \sum_q P_c(E_r, \psi_q \,|\, i,k)\, G(k,q,m)\ , \tag{15}$$

with i, k the effective area's bins (the smearing's bins hold their centres) and G the geometry: the probability that a track from true bin k, deflected by ψ in bin q at an azimuth uniform around the true direction, is reconstructed in bin m. On the sphere, $\sin\delta_r = \sin\delta\cos\psi + \cos\delta\sin\psi\cos\phi$; for an edge δ_e the probability $F = P(\delta_r < \delta_e) = 1 - \arccos(c)/\pi$, with $c = (\sin\delta_e - \sin\delta\cos\psi)/(\cos\delta\sin\psi)$ clipped to $[-1,1]$, is exact in the azimuth. As a function of the true declination, F is flat (0 or 1) or smooth between four breakpoints, $\delta_e \mp \psi$ and $\pm(180^\circ - \psi) - \delta_e$, where the circle of radius ψ reaches the edge directly or over a pole, with square-root ends there; we split each true bin at them and integrate each piece by Gauss–Legendre with nodes clustered at its ends, and integrate $\log\psi$ within each bin with nodes at most 0.5° apart. Against Monte Carlo (tracks uniform in $\sin\delta$ within a bin, ψ log-uniform, azimuth uniform) the kernel agrees within the statistics everywhere: the worst of 1584 comparisons is 2.8σ. A first version, plain Gauss–Legendre in $\sin\delta$, was wrong by 6–9% wherever an edge cut a true bin, and then by 1.5% for ψ near 180° at the poles; the Monte Carlo test caught both. The point-spread angle moves 0.1–0.3% of an isotropic flux's tracks across $\delta = -5^\circ$, and conserves the total.

The atmospheric background.—As in NuFlavorSingleSource (Mauricio, 2026-10-14): MCEq with SIBYLL-2.3d and the H3a cosmic rays gives the nominal flux, conventional plus prompt, and its derivatives with respect to the twelve hadron-production parameters of Barr et al. that matter above 30 GeV (pions in regions G, H, I and kaons in W, Y, Z, each charge separately; priors 0.30, 0.15, 0.122 and 0.40, 0.30, 0.122) and to a tilt of the cosmic-ray spectral index (prior 0.05); the prompt normalization varies with prior 0.5. The flux is linear in all of them, $\Phi = \Phi_0\,(1 + \sum_p \theta_p\,\partial_p \ln\Phi) + \theta_{\rm pr}\Phi_{\rm pr}$, so each enters a fit as one folded template. The regions, priors, yield modification and cosmic-ray tilt are NuFlavorSingleSource's code, copied unchanged (`tools/atmospheric_background.py`); for IceCube we use the South Pole's atmosphere (NRLMSISE-00, January and July averaged) instead of the US standard one, and 20 production zenith angles dense near the horizon instead of 8. A two-angle test reproduces NuFlavorSingleSource's relative derivatives to 10^{-3}; only the prompt fraction differs (0.185 against 0.197 at 100 TeV, vertical), from the atmosphere. Up-going directions take the flux produced at the mirror zenith on the far side. daemonflux (MCEq calibrated to muon data) is kept as a cross-check of the conventional flux.

Against the data.—With the nominal atmospheric flux and IceCube's 9.5-year fit of the astrophysical flux ($\nu_\mu + \bar{\nu}_\mu$: 1.44×10^{-18} GeV^{-1} cm^{-2} s^{-1} sr^{-1} at 100 TeV, $\gamma = 2.37$), the tracks reconstructed above $\delta = -5^\circ$ number 1.057, 1.155, 1.067 and 1.123 times our expectation

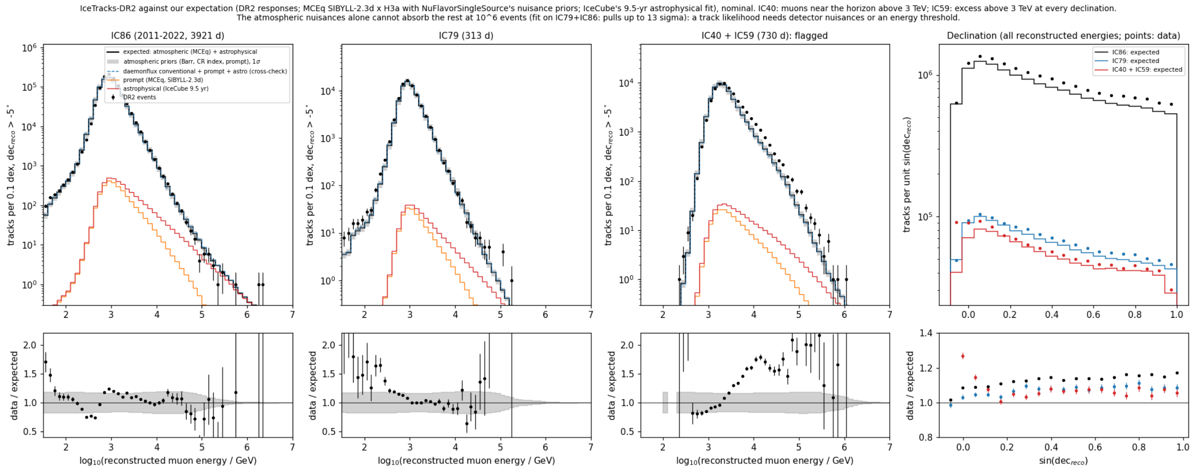


Figure 21: **IceTracks-DR2 against our expectation** (`tools/dr2_validation.py`): tracks reconstructed above $\delta = -5°$ in reconstructed muon energy for IC86, IC79 and IC40 + IC59, and in declination; the expectation is the DR2 responses folded with our atmospheric flux (MCEq SIBYLL-2.3d × H3a at the South Pole; band: the Barr, cosmic-ray and prompt priors) and IceCube's 9.5-year astrophysical fit, at nominal values; dashed: daemonflux's conventional flux instead.

in IC40, IC59, IC79 and IC86 (Fig. 21): the units are right, and the declination distribution is flat in the ratio. Split by declination and energy, IC79 and IC86 agree above 3 TeV (IC86 is 15% high below, the overall normalization). The early configurations do not: IC40 has twice the expected events above 3 TeV, but only within 10° of the horizon (atmospheric muons reconstructed as up-going, absent from our model), and IC59 has 1.2–1.8 times the expected events above 3 TeV at every declination, even above +40°, where muons cannot reach: its published responses and its events disagree at high energy. A fit of the atmospheric nuisances (and of a normalization per configuration and the astrophysical flux) to IC79 and IC86 cannot absorb the rest: with 10^6 events every bin is measured to better than 1%, and the fit pulls Barr parameters by up to 13σ to mimic what are detector effects, for which DR2 publishes no systematics. A track likelihood will therefore need detector nuisances (a normalization and an energy scale per configuration) or an energy threshold above which the astrophysical flux matters and the data and our expectation agree; IC40 and IC59 are left out until then.

New physics in the tracks.—The solver's flux per declination band (the effective area's 50 bands, the Galactic component from $N_{\text{side}} = 64$ data), with the extragalactic $\nu_\mu + \bar{\nu}_\mu$ normalized to the 9.5-year fit without new physics (pion sources put 35.5% of the flux in ν_μ at the Earth: the flavor-averaged normalization times 0.938; the solver's standard tracks then equal the fit's power law in every bin, to 5×10^{-15}), gives 3835 extragalactic and 527 Galactic tracks above $\delta = -5°$ in 2008–2022, 340 and 23 of them above 10 TeV reconstructed (Fig. 22). The dark-portal point of Fig. 17 removes 6% of the extragalactic tracks at every energy (its flavor composition), the secret-interaction point (ν_τ, 5 MeV, $g = 0.05$) 43% at 200 TeV reconstructed (absorption near the resonance) while adding 5–10% below 10 TeV (the regenerated neutrinos). The Galactic share of the tracks is 12% in all, 6% above 10 TeV.

10 Starting events with HESE

Expected HESE events (Mauricio's task 1; plan in `docs/dev/HESE_RATES_PLAN.md`) from IceCube's 7.5-year release: its simulated events weighted per event by any flux, exactly (Section 8's consumer check), in the release's binning (cascades and tracks in deposited energy and zenith,

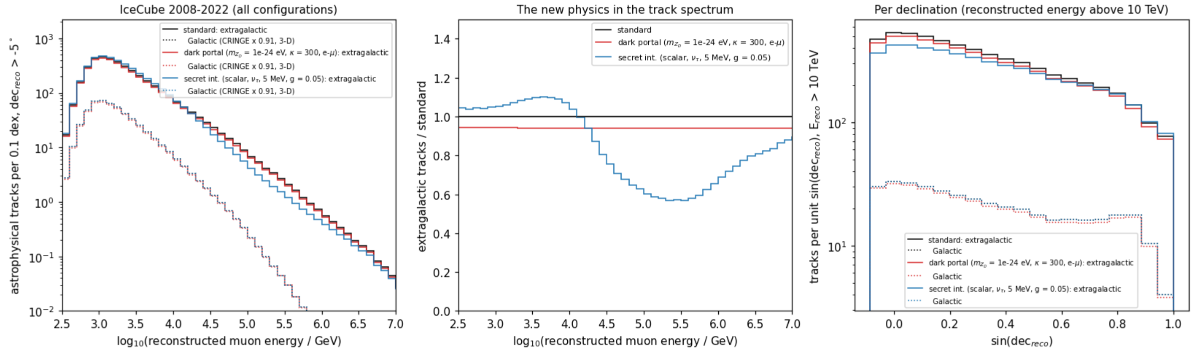


Figure 22: **Astrophysical tracks with new physics** (IceCube 2008–2022, all configurations, reconstructed $\delta > -5°$): extragalactic (solid) and Galactic (dotted) for standard physics, a dark-portal point and a secret-interaction point; their ratio to standard; per declination above 10 TeV.

double cascades in deposited energy and length), with the release's cuts (`tools/hese_reduce.py`: 736 814 simulated events; the 60 events above 60 TeV: 41 cascades, 17 tracks, 2 double cascades). The release's best-fit astrophysical flux gives 50.16 events (39.51, 8.43, 2.21), as before.

Backgrounds.—The atmospheric neutrinos are our MCEq flux (Section 9) per event, the conventional part times the release's conventional self-veto correction and the prompt part times its prompt one (Mauricio, 2026-10-14). The conventional part agrees with the release's own model (Honda-based, its veto) to 5% in every morphology: 7.91 events against 8.01. The prompt part, from SIBYLL-2.3d, is 3.04 events against the release's 1.63 (BERSS): the known difference of the two charm models, within its 50% prior. The atmospheric muons are the release's own simulation, 0.79 events (tracks).

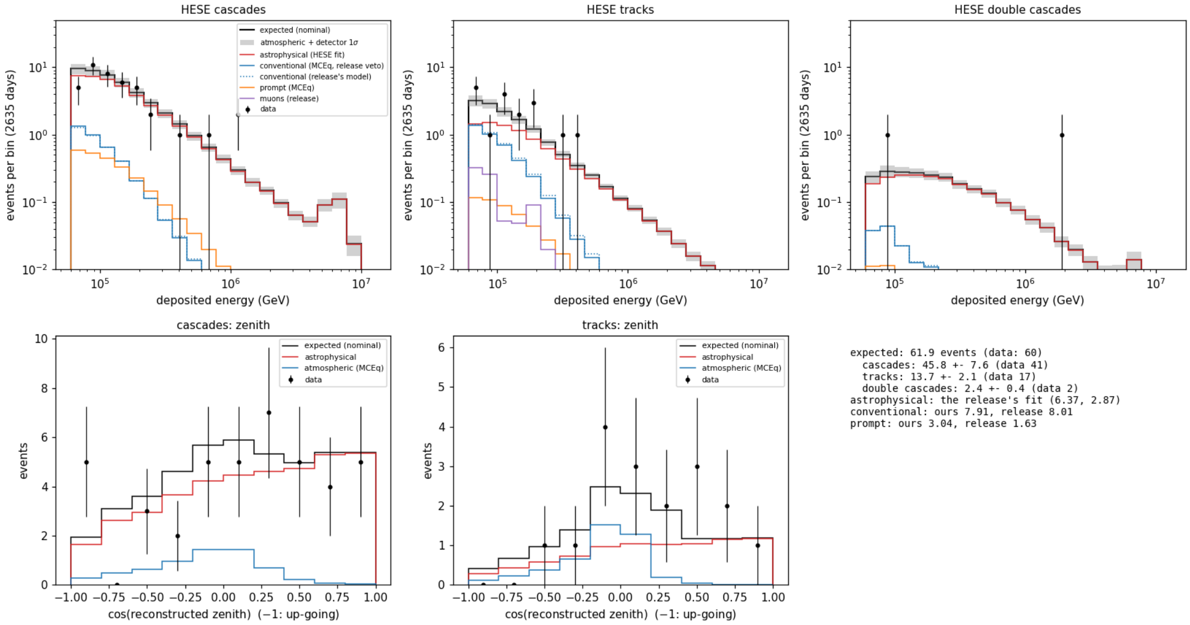
Detector systematics.—The release corrects each event's weight for the DOM efficiency (prior 0.99 ± 0.1), the head-on efficiency (0 ± 0.5) and the ice anisotropy (1 ± 0.2), with splines per flux component. We export the exact weights from the release's own code, run in its own environment, on a grid of each systematic's values per event (`tools/hese_detector_systematics.py`) and interpolate between them. At $+1\sigma$ the DOM efficiency moves the expectation by 7.3 cascades, 1.6 tracks and 0.4 double cascades; the other two by 0.1 events or less.

Expectation against the data.—With the release's astrophysical fit and our backgrounds, we expect 45.8 ± 7.6 cascades, 13.7 ± 2.1 tracks and 2.4 ± 0.4 double cascades (detector and atmospheric 1σ, correlated across bins; Fig. 23), against 41, 17 and 2.

New physics in HESE.—With the extragalactic flux normalized to the release's fit and pion sources, standard physics gives 38.1 cascades, 8.75 tracks and 2.23 double cascades, and the Galactic component 2.25, 0.46 and 0.11 (Fig. 24). The dark-portal point moves 4% of the events from tracks to cascades. The secret-interaction point removes 35% of the cascades and half of the tracks and double cascades: HESE's energies, 60 TeV–10 PeV deposited, sit at the resonance, where the northern tracks (mostly below 100 TeV) lose 14% above 10 TeV.

11 Matter charges and textures: the scope of the dark-portal paper

Decisions (Mauricio, 2026-10-14).—The dark-portal paper widens the model in two ways and claims, in its abstract, the most comprehensive calculation of these effects in high-energy astrophysical neutrinos, with several kinds of current data. Working title: "New forces on high-energy cosmic neutrinos, from the Hubble scale to the Earth". First, a general matter charge: the po-



```
expected: 61.9 events (data: 60)
  cascades: 45.8 +- 7.6 (data 41)
  tracks: 13.7 +- 2.1 (data 17)
  double cascades: 2.4 +- 0.4 (data 2)
astrophysical: the release's fit (6.37, 2.87)
conventional: ours 7.91, release 8.01
prompt: ours 3.04, release 1.63
```

Figure 23: **HESE events against our expectation** (`tools/hese_rates.py`): per morphology in deposited energy, and in zenith for cascades and tracks; the astrophysical flux is the release's fit, the atmospheric neutrinos our MCEq flux with the release's self-veto (dotted: the release's own conventional model), the muons the release's; band: the atmospheric and detector 1σ.

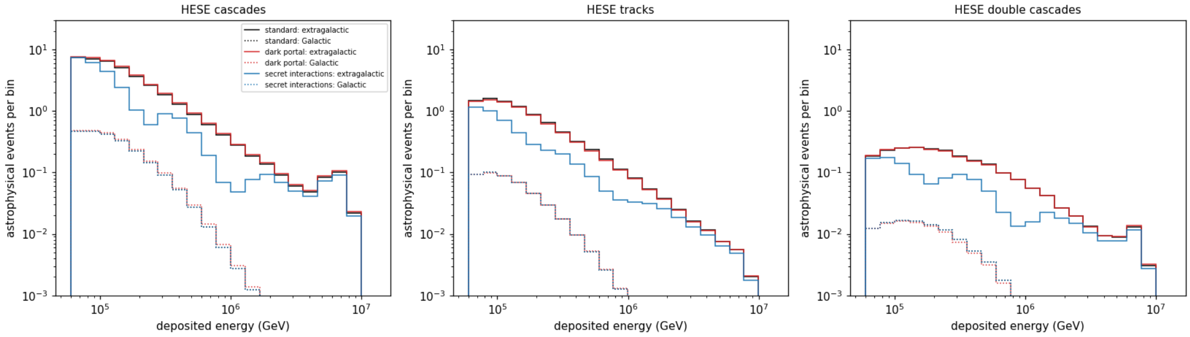


Figure 24: **HESE events with new physics**: extragalactic (solid) and Galactic (dotted) per morphology for standard physics, a dark-portal point and a secret-interaction point.

tential is sourced by $q_n n_n + q_p n_p + q_e n_e$, not by neutrons alone. Second, a family of neutrino textures. Eight benchmark models, each with the matter charge and the texture its gauge structure fixes (Table 8). The dark portal's mixing ratios are free, with full planes for named textures (pure e, μ, τ; $e\mu$, $e\tau$, $\mu\tau$; democratic) and maps of the bound over the ratio triangle at a few masses. The plan is `docs/dev/SCOPE_WIDENING_PLAN.md`.

Conventions.—The theory notes' ϵ' (T9, T12) is half the dark-Z mixing ϵ_Z of Davoudiasl, Lee & Marciano, whose neutron coupling is $\epsilon_Z g/(4c_W)$; the notes' neutron coupling $\epsilon' g/(2c_W)$ and neutrino coupling $\epsilon' g/c_W$ (T12) follow one normalization, so the model is self-consistent. For $L_\mu - L_\tau$, whose Z' reaches the neutrons by the same mixing, the 2023 paper's ξ is $2\epsilon'$.

The matter charge (step 1, done).—In neutral matter $n_p = n_e$, so only q_n and $q_p + q_e$ enter. Every density the solver uses is linear in its source, so each is computed for neutrons and for electrons and combined (`src/dp_matter.py`): the cosmic mean ($n_{e,0} = n_b(1 - Y_p/2) = 2.2059 \times 10^{-7}$ cm^{-3}), the Galaxy at the Sun and at its emission points (electrons per nucleon

Table 8: **The eight benchmark models.** Matter charges in neutral matter ($n_p = n_e$); neutrino charges (e, μ, τ). The first three gauge models are those of Agarwalla, Bustamante, Das & Narang (JHEP 08 (2023) 113).

Model	Matter charge	Neutrino texture
Dark portal	neutrons (Z–Z_D mixing)	$uu^\dagger$, ratios free
$L_e - L_\mu$, $L_e - L_\tau$	electrons	$(1, -1, 0)$, $(1, 0, -1)$
$L_\mu - L_\tau$	neutrons (Z–Z' mixing)	$(0, 1, -1)$
$B - 3L_e$	$n - 2e$	$(-3, 0, 0)$
$B - 3L_\mu$, $B - 3L_\tau$	nucleons	$(0, -3, 0)$, $(0, 0, -3)$
$B - \frac{3}{2}(L_\mu + L_\tau)$	nucleons	$(0, -\frac{3}{2}, -\frac{3}{2})$

0.80 in stars, 0.858 in cold gas; the hot halo's profile is already the electron density), the Sun, the Moon and the Earth's Yukawa profile (Y_e by layer, water on top). The electrons per body are 1.0046×10^{57} (Sun), 1.7480×10^{51} (Earth, PREM integrated) and 2.1895×10^{49} (Moon). The solver carries the potential's sign in its coefficient, so a charge negative in every body ($B - 3L_e$) has its sign moved there, and one that changes sign between bodies is refused. Fig. 25 shows what changes: the distant sources are rich in hydrogen, with four to seven electrons per neutron, while the Earth and the Moon have about one. An electron-sourced force therefore weighs the potential at IceCube 7.1 times the neutrons' at cosmic ranges, 4.2 at Galactic ones and 4.4 where the Sun dominates (from an AU to a kiloparsec), but 0.95 at the Earth's. Checks (`tests/test_matter_charge.py`, 16 tests): linearity of every density to 10^{-12}; the electron counts against the bodies' masses; the general path with the neutron charge equal to the historical one to 10^{-12}; a negative charge equal to a flipped texture. The four validation points are bit for bit those of the previous code, compiled fresh on both sides (a cached and a fresh build of the same code already differ by 7×10^{-15}).

The absorption counted every relic state in full (found 2026-10-14).—The resonant absorption (T7, T11, T12; `src/dp_thermal.py`) gives an incident ν_α the same coupling $(\Sigma G)_\alpha$ with each of the three relic mass states. A relic state offers only its own flavor components: the s-channel amplitude of $\nu_\alpha + \bar{\nu}_i$ is $(CU^*)_{\alpha i}$, with C the coupling matrix in the flavor basis, and summed over the three states this gives $(CC^\dagger)_{\alpha\alpha} = (\Sigma G)_\alpha$, one species' worth of targets where the kernel counts three. The width is the same in both. With the kernel itself, a flavor-diagonal coupling absorbs ν_e as strongly on the heaviest state as on the lightest, although $|U_{e3}|^2 = 0.02$; integrated over energy, the absorption is 2.9 times too strong for the dark portal's $e\mu$ texture and 2.2 (ν_e) and 4.2 (ν_μ) for $L_e - L_\mu$ (Fig. 26). Mauricio decided (2026-10-14) to weigh each state by its own flavor content, $|(CU^*)_{\alpha i}|^2$, for all eight models, and the Debye mass likewise; the dark portal's MeV planes will change, by up to $\sqrt{3}$ in $g_D|U|^2$. This departs from the theory notes.

12 The physics audit of 2026-10-14

Mauricio asked, before any further runs, for every part of the physics code to be checked (`docs/dev/PHYSICS_AUDIT.md` lists each item with its derivation or check). The propagation chain holds: the cosmological evolution and injection, the adiabatic Galaxy, the coherent Sun, the Earth's Yukawa profile and chords, the matter models, the flux assembly and the event folds. A new absolute check: with a potential far above the vacuum term everywhere and no absorption, the composition at the Earth is $\sum_a |W_{\alpha a}|^2 \sum_f |W_{fa}|^2 f_f$ with W the texture's eigenvectors (its null space split by the projected vacuum term), and the solver reproduces it to 10^{-11} for three textures, neutrinos and antineutrinos. The errors were all in the resonant absorption on the relic neutrinos, which followed the theory notes' Eqs. (T7), (T11), (T12).

The absorption, from the Lagrangian.—With the mediator coupled to the neutrinos as

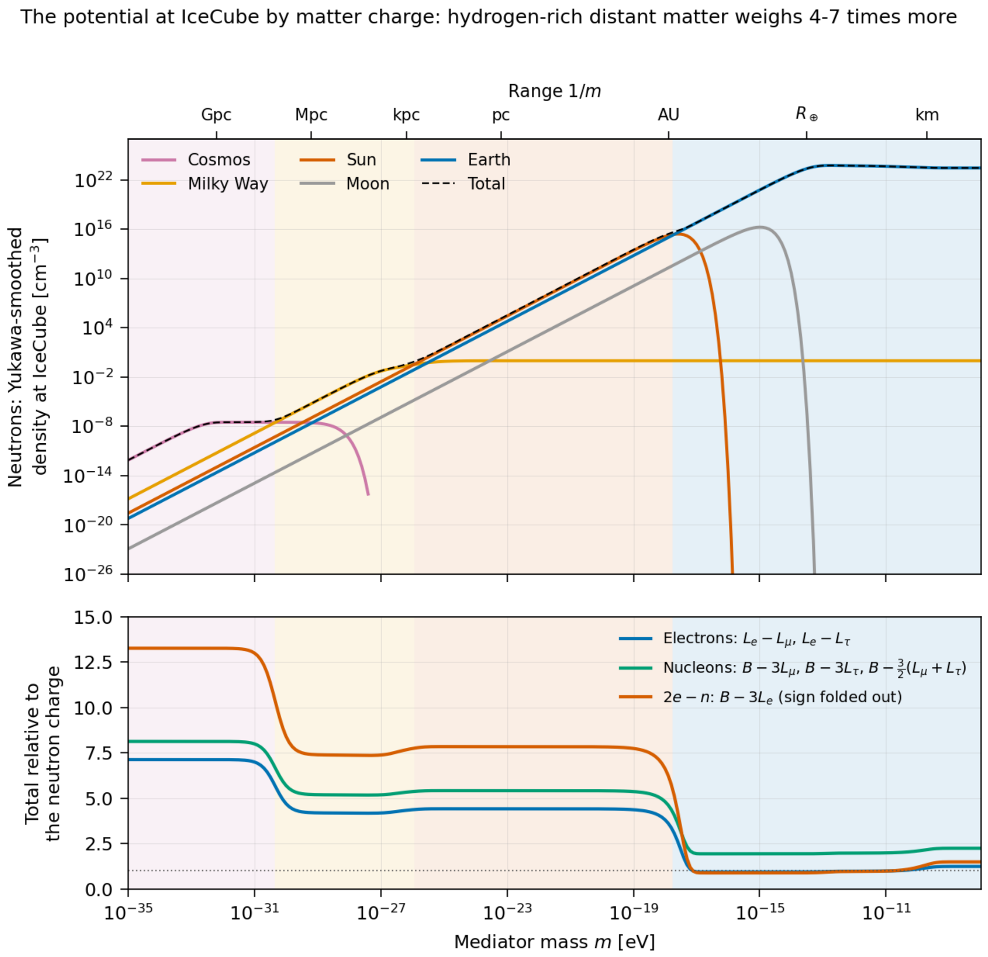


Figure 25: **Where the potential at IceCube comes from.** Top: the Yukawa-smoothed neutron density at the detector per body versus the mediator mass (shading: the dominant body). Bottom: each benchmark charge's total relative to the neutrons'. `tools/plot_matter_sources.py`.

$V_\mu \sum G_{\alpha\beta}\bar{\nu}_\alpha\gamma^\mu P_L \nu_\beta$ ($G = g_D u u^\dagger$ for the dark portal, in the coupling of Eq. (T9)), an incident eigenstate a (flavor amplitudes W_{fa}) meets the relic mass state r with amplitude $(U^\dagger G W)_{ra}$, and the Breit–Wigner of a vector formed by $\nu_L + \bar{\nu}_R$ integrates to $2\pi|(U^\dagger G W)_{ra}|^2$ per target, the normalization of the Z-burst. The code gave every relic state the full flavor coupling (three times the targets), counted each neutrino pair with the width of a Dirac pair with vector couplings, $M/12\pi$, instead of the left-handed one, $M/24\pi$ (twice the rate per target), and projected eigenstates and secondaries on flavor without interference. A flavor-universal coupling, diagonal in the mass basis, must absorb mass state j only on relic state j; the old form spread it over all three. Mauricio decided (2026-10-14) on the Lagrangian form throughout: each relic state by its own flavor content, coherently per eigenstate, the left-handed width, the decays' neutrinos in the eigenstates by $(W^\dagger G G^\dagger W)_{aa}$ (`src/dp_absorption.py`). At 3 MeV and $g_D|U|^2 = 2.5 \times 10^{-4}$ the deepest dip goes from 0.003 to 0.021 of the flux, and much of the old flavor shift disappears. I wrote the 2026-10-12 normalization from Eq. (T11) as it stands; its Z-burst check held only with the Z''s coupling divided by $\sqrt{2}$.

The Z mixing in the absorption.—Eq. (T12)'s term $c\,\epsilon' g/c_W$ is the cross term of $|c+\kappa|^2$ with $\kappa = \epsilon' g/c_W$, the coupling the Z–Z_D mixing gives every neutrino, but it drops κ^2 and puts the cross term on every flavor pair. Mauricio decided to keep all three pieces: $G = g_D u u^\dagger + \kappa I$,

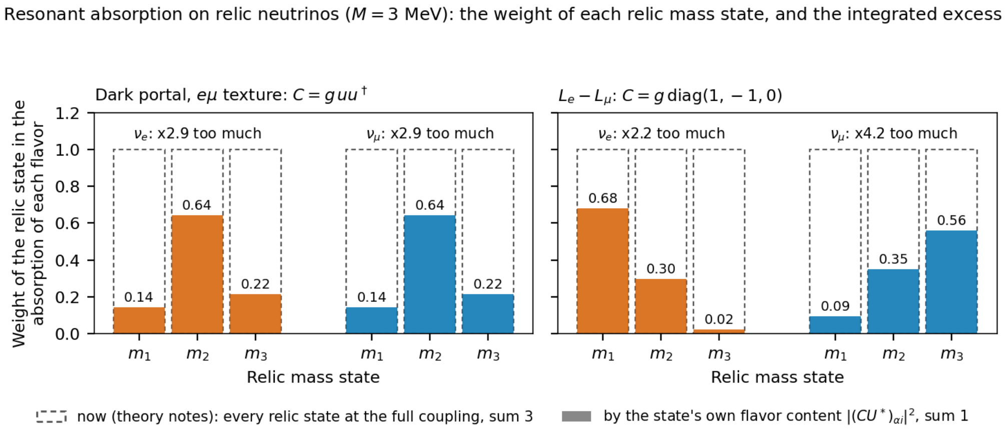


Figure 26: **How the absorption weighs the relic mass states** at $M = 3$ MeV: as computed until 2026-10-14 (dashed, every state at the full coupling) and by each state's flavor content (bars). `tools/absorption_relic_weights.py`.

with the e^+e^- and $\mu^+\mu^-$ decays of the mixing in the width. Then $\epsilon' = 0.1$ alone absorbs every flavor near MeV masses, and the potential planes no longer depend on $P = g_D\epsilon'|U|^2$ alone: they will profile ϵ' (decided).

Relics in motion.—The kernel used the massless relic's flux factor $1-\cos\theta$; the Moller factor of a relic at speed v is $1 - v\cos\theta$, which changes the integrated absorption by +1.7%, +1.3% and +0.3% for the three states at $m_{\text{lightest}} = 0.01$ eV and +9% for a 10^{-3}-eV lightest state.

Screening (a departure from Eq. T9).—The relics' Debye mass, Eq. (T23), is the relativistic one, $m_D^2 = \sum_r |G'_{rr}|^2 T^2/6$ ($G' = U^\dagger G U$). Today's relics are slow, and the mediator couples to the left-handed current, so a relic's charge depends on its speed v. The vector current of a Majorana field vanishes, and its left-handed current is $-\frac{1}{2}\bar\nu\gamma^\mu\gamma_5\nu$, whose charge on a state of helicity h is $-G'_{rr}hv$: a slow Majorana neutrino carries almost no charge. The static polarization $\sum\int d^3p/(2\pi)^3\, q^2(-\partial f/\partial E)$ then gives $9\zeta(3)|G'_{rr}|^2T^3/(2\pi^2 m_r)$ per state, smaller than Eq. (T23) by $3.3\,T/m_r$ (a Dirac relic's charge is $(1+v)/2$, giving $|G'_{rr}|^2 m_r T\ln 2/(4\pi^2)$). The code takes Majorana relics, as for the absorption's targets; the Z mixing's piece is $m_D \approx 2\times10^{-5}$ eV$\times\kappa$ today. Screening needs the relics to move: around the Earth, the Moon, the Sun and the Galaxy they drift in the potential's gradient (they cross the Galaxy in $\sim 10^7$ yr), but the uniform cosmological potential pushes on none of them and could only be screened if the relic background acquired net dark charge everywhere at once, which charge conservation forbids (the uniform background of the "jellium" model is not screened either). We therefore screen the localized bodies only, and the cosmological potential keeps the bare mediator mass; Eq. (T9) screens it too. At $\epsilon' = 0.1$ the screening mass is $\sim 2\times10^{-6}$ eV (a range of ~ 12 cm) and cuts off every ultralight local potential; at $m_{Z_D} = 10^{-20}$ eV, ϵ' and $g_D|U|^2$ must both stay below $\sim 5\times10^{-16}$ for those potentials to act. Since the potential fixes only $P = g_D\epsilon'|U|^2$, a split with a large $g_D|U|^2$ and a tiny ϵ' switches off the Earth, the Sun and the Galaxy (today, for a mediator lighter than the inverse distance: $g_D|U|^2 > 2\times10^{-9}$, 7×10^{-14}, 4×10^{-23}), and no published limit forbids it: MICROSCOPE bounds $\epsilon' < 3\times10^{-25}$ from above, and cosmology bounds a neutrino-only coupling only at $\sim 10^{-7}$. Mauricio decided that each potential plane shows two curves: the bound on P where screening is negligible, which does not depend on the split, with its condition stated, and the bound profiled over all splits, which at ultralight masses comes from the cosmological potential alone (at $m_{Z_D} = 10^{-20}$ eV about 19 orders of magnitude weaker).

The MeV regime.—There the cosmological potential is negligible and the dark-portal absorption is a vector secret interaction with coupling G. Mauricio decided to compute it with `src/dp_si.py`, which has every collision with the relics (s-channel resonance, t and u exchange, pair production), mass state by mass state; its resonance is now averaged over the relics' momenta and it takes a visible width. The two codes' resonance agree to 0.1% (median) at 3 MeV; relics at rest would overstate the dip twofold. Both signs of the dark portal's potential will be scanned (decided): the sign of $g_D\epsilon'$ is fixed only by a model's Higgs sector.

13 Planned work

13.1 Event rates at IceCube

We will compute the event distributions in reconstructed energy for two event classes, from public IceCube responses, in a new package (`IceCubeRates/`).

Northern-Hemisphere muons.—Through-going muon tracks from below the horizon, where the Earth removes the atmospheric muon background. The response is IceCube's IceTracks-DR2 release (2008–2022, doi:10.7910/DVN/MMIIZA), which provides per-season muon-neutrino effective areas and smearing tables. For each bin of true energy and declination, a smearing table gives the fraction of events in each bin of reconstructed muon energy, angular distance from the true direction (point-spread function) and estimated angular error. Its 41 declination bins replace the 3 of the 2018 release, which matters near the horizon. Done with our own fold (Section 9); the cross-check against PLEnuM's fold of the 2018 release was dropped (Mauricio, 2026-10-14).

Starting events.—Cascades, starting tracks and double cascades, from IceCube's HESE 7.5-year release (github.com/icecube/HESE-7-year-data-release). It is the only public simulation of individual starting events that resolves all three morphologies. Each simulated event carries its flavor (neutrino or antineutrino), its interaction (charged current, neutral current or Glashow resonance), its reconstructed deposited energy and its morphology. It also carries a weight per unit flux, together with the atmospheric fluxes and self-veto probabilities at its energy and direction. We reweight each simulated event by our flux. Done (Section 10). For IceCube-Gen2 we scale these weights by the ratio of the Gen2 and IceCube effective areas published for the Gen2 design (doi:10.7910/DVN/XANWAT), per morphology, flavor, energy and zenith angle. Those tables carry no energy resolution, so the IceCube response supplies it.

Earth absorption and direction.—Both responses already contain absorption in the Earth, computed by IceCube with its nominal cross sections. We checked this in the DR2 tables: at 1 PeV the effective area for straight up-going neutrinos is 20 times smaller than at the horizon, and at 10 PeV 120 times smaller, while at 10 TeV it is nearly uniform. Our solver computes the other effect of the Earth, the flavor change caused by the dark-portal potential, which involves no cross section. Because collisions remove all flavors nearly equally, the two act as independent factors along each path; the Glashow resonance, ν_τ regeneration and neutral-current scatterings that continue at lower energy break this slightly. Since both factors depend on direction, the solver must return the flux per Earth chord, and must add down-going directions, instead of the average it returns now.

13.2 Systematic uncertainties

The fits will vary the following, agreed on 2026-10-12. *Astrophysical flux:* its normalization and spectral index. *Source composition:* $f_{e,S}$, with $f_{\tau,S} = 0$. *Atmospheric fluxes:* the conventional normalization, a tilt of the cosmic-ray spectrum and the twelve hadron-production parameters of Barr et al. above 30 GeV (pion regions G, H, I and kaon regions W, Y, Z, each for both charges). We compute their effects with MCEq (SIBYLL-2.3d and the H3a cosmic-ray model), exactly as in the NuFlavorSingleSource project. The prompt and atmospheric-muon normalizations and the self-veto vary as well. *Mixing parameters:* all six, constrained by the joint $\Delta\chi^2$ of NuFIT 6.1 with atmospheric data, per ordering. We use the three-dimensional table over $(\sin^2\theta_{23}, \Delta m^2_{31}, \delta_{\mathrm{CP}})$, the two-dimensional table over $(\sin^2\theta_{12}, \Delta m^2_{21})$ and the one-dimensional table over $\sin^2\theta_{13}$, so that each parameter appears in exactly one table. *Lightest neutrino mass:* free, with a prior on the sum of the masses.

The prior on the mass sum.—As in the GeoneutrinoDecay project, our prior is a half-Gaussian on the sum, which we derive from the lightest mass and the splittings. It adds $(\Sigma m_\nu/\sigma)^2$ to the χ^2, with $\sigma = L/\Phi^{-1}(0.95)$ for a published 95% upper limit L. We use $L = 0.055$ eV, from the cosmic microwave background (Planck, ACT and SPT-3G) with the baryon acoustic oscillations of DESI DR2 in ΛCDM, with a prior on Σm_ν that starts at zero (SPT-3G D1, arXiv:2506.20707, Eq. 96). Then $\sigma = 0.0334$ eV. The smallest sum allowed in normal ordering, 0.0587 eV, costs $\Delta\chi^2 = 3.08$; in inverted ordering, 0.0995 eV costs 8.85. Both orderings are fitted. As a check, this prior excludes the two minimal sums at 96.0% and 99.9%, against 96.6% and 99.9% stated in the paper.

The spectral shape (2026-10-15).—IceCube's 2026 combined fit of tracks and cascades (PRD 113, 062002, Table 3, "CF BPL") bends below ∼25 TeV: 1.77×10^{-18} GeV^{-1} cm^{-2} s^{-1} sr^{-1} per flavor at 100 TeV, $E^{-2.735}$ above $E_b = 10^{4.39}$ GeV and $E^{-1.31}$ below. A single power law fitted to the same data and extended down carries 3 times its flux at 10 TeV and 60 times at 1 TeV, which DR2 (from 3 TeV), IceCube's own spectrum segments (from 4.6 TeV) and the future detectors (from 100 GeV) all see. So the extragalactic flux is now IceCube's broken power law, in every data set: two source populations split at the energy the neutrinos have today, E_b, each a power law, with the upper index p and the lower index $p_{\rm low}$ free and E_b fixed, normalized so that without new physics the flux at Earth is exactly the broken power law (`docs/dev/BROKEN_POWER_LAW.md`). Each population is its own solver run, with the sources emitting only on their side of E_b; the transport is linear in its sources, so everything downstream (conversion, absorption, and the secondaries that an absorption above E_b sends below it) is each population's own. We expect the bend to absorb part of the preference for an MeV resonance that the 2026 spectrum gave every secret-interaction plane (Sec. 8); the rerun campaign will tell.

Not varied.—The cross sections are fixed by the public responses; varying them would require generating our own effective areas. The detector-systematic corrections of the HESE release (optical efficiency, hole ice, ice anisotropy) are deferred, because they need the PHOTOSPLINE package.

13.3 Statistical soundness

We will establish that the analysis is sound in the order used in the GeoneutrinoDecay project. *Closure.*—Each response must reproduce a published IceCube fit before it is used for new physics; the HESE release includes its own power-law fit. *Injection and recovery.*—On Asimov data with known dark-portal parameters, and without a signal, the fit must recover the truth without bias. *Fits.*—Then the fits with the current profile-likelihood procedure. *Coverage.*—We generate pseudo-experiments with Poisson-fluctuated counts and fit each exactly like the data, 200 per point (which gives ±2% on a 10% rate). The points are the no-signal hypothesis, where we compare the test statistic with its asymptotic distribution, and the expected limit and half a decade beyond it. We add the parameter boundaries, the region where the mediator mass and the lightest mass trade off in setting the resonance energy, and the low-count corners (double cascades, highest-energy tracks). The pseudo-experiments change only the data, so the solver predictions on the scan grid are computed once and reused. *Decision.*—If no point under-covers, we keep the profile likelihood; otherwise we construct Feldman–Cousins intervals with PyFC (github.com/mbustama/FeldmanCousins), whose likelihood also accounts for the finite size of the HESE simulation.

The projected flavor analysis.—It has no counts to fluctuate; its pseudo-experiments must shift the centre of the projected measurement by its own width, which is an approximation we will state with its results.

13.4 Computation

Our heavy runs go to Tycho, the cluster of the Niels Bohr Institute, set up as in the GRBsim and NuFlavorSingleSource projects. The repository and a conda environment live on the shared file system, and the job scripts read their core count from SLURM. Jobs take 16 cores by default and up to 32. Before the first run, the solver must size its process pools from the cores the job was given; it now uses the core count of the whole node, so a 16-core job would start 127 processes. The first job will repeat the benchmark validation of Sec. 5 against the laptop's results.

13.5 Relic motion in the collision transport (2026-10-18)

In the MeV regime, the transport splits each collision of a high-energy neutrino with a relic into ten profiles q (the s channel, the t channel, their interferences, pair production). Each is a kinematic factor $K_q(s)$ times a shape $G_q(\mu)$, $\mu = M^2/s$: the absorption takes the shape's integral, the regeneration its integrals over the lower energy bins. Since 2026-10-14 the factors of the resonance, s/BW and $(s - M^2)/\mathrm{BW}$, are averaged over the relics' thermal momenta; the shapes, and the other two factors ($1/s$ and the pair cross section), take the relic at rest, $s = 2E(1 + z)m$. Since 2026-10-18 a switch averages every product (below).

Kinematics.—A neutrino of energy E' meeting a relic of energy E_p at angle θ has $s = m^2 + 2E'E_p(1 - v\cos\theta)$. Every product is boosted along the incoming neutrino by $\gamma \approx E'/\sqrt{s} \sim 10^8$, so its energy fraction in the cosmological frame equals its value in the relic's rest frame, $(k \cdot p_\mathrm{r})/(p \cdot p_\mathrm{r})$, to $\sim 1/\gamma$. The shapes are therefore functions of s alone, and so are the spin-averaged cross sections for Majorana relics, which are unpolarized at rest and in motion. Relic motion enters only through the distribution of s.

The distribution.—With $\sigma \equiv s/2E'$ ($\sigma = m$ at rest), Fermi–Dirac momenta at $T_\mathrm{r} = T_{\nu 0}(1 + z)\eta_\mathrm{r}^{1/3}$, isotropic directions and the flux factor $1 - v\cos\theta$ give

$$\rho(\sigma) = \frac{\sigma}{3\zeta(3)T_\mathrm{r}^3} \int_{p_X(\sigma)}^{\infty} \frac{p\, f(p/T_\mathrm{r})}{E_p}\, dp, \qquad p_X = \frac{X^2 - m^2}{2X}, \quad X = \max(\sigma, m^2/\sigma), \tag{16}$$

normalized to one, with $\langle 1/\sigma \rangle = \langle 1/E_p \rangle$ and a kink at $\sigma = m$; it becomes $\delta(\sigma - m)$ for slow relics.

The rule.—Every rate the transport uses is a product of K_q and an integral of G_q, so every product must be averaged, $\langle K_q G_q \rangle = \int d\sigma\, \rho(\sigma)\, K_q(2E'\sigma)\, G_q(M^2/2E'\sigma)$. Averaging K_q alone while G_q stays at rest is inconsistent wherever G_q depends on s: the totals of seven of the ten vector profiles and the spectra of eight. A first change of 2026-10-18 did exactly that for $1/s$ and the pair cross section; it produced negative fluxes (at 31.6 keV, $g = 7.8 \times 10^{-3}$, $m_\mathrm{lightest} = 10^{-6}$ eV) and was reverted. The resonance averaging of 2026-10-14 has the same inconsistency for the two s–t interference profiles, and it switches between the thermal and the at-rest treatment abruptly, so that χ^2 jumps as m_lightest varies.

Where it matters.—Taking the relic at rest understates its mean energy, and so s, by factors of 500, 1000 and 2000 at $z = 0$, 1 and 3 for $m = 10^{-6}$ eV, and by more than 10% below $m \approx 3 \times 10^{-3}$ eV: the range into which the mass-sum penalty pulls the fits. We expected little change for MeV mediators, where the s-channel resonance dominates and its shapes do not depend on s; the validation below finds it large there too. Below $\sim$100 keV with $m_\mathrm{lightest} \lesssim 10^{-4}$ eV, the relic at rest even closes the pair channel ($s < 4M^2$) that thermal relics open.

Implementation.—With `dark_portal.thermal_products` on (`src/dp_si.py`, `thermal="full"`; off by default, on in the campaign), every product is averaged. The at-rest products sit on one lattice in $\ln\mu$ with the energy bins' spacing; its spectra depend on neither the relic nor the mediator mass, so one table serves every run. Per relic state and redshift step, ρ becomes weights on every s-th node (cubic interpolation in $\ln\mu$, s from 1 to 8 nodes as ρ widens), and a bin's products are short sums over the table. The resonance factors are not smooth on the lattice: $\langle s/\mathrm{BW}\rangle$ and $\langle (s-M^2)/\mathrm{BW}\rangle$ are the semi-analytic kernel's exact thermal averages, the interference shapes enter as $G_q(1)\langle (s-M^2)/\mathrm{BW}\rangle$ plus the lattice average of the smooth $(s-M^2)/\mathrm{BW}\,[G_q(\mu)-G_q(1)]$, and the s-channel shapes do not depend on μ. There are no switches between treatments.

Validation.—A reference implementation averages every product as a mixture of relics at rest with masses σ and weights ρ by direct quadrature, each exact in time within a redshift step (`tools/thermal_reference.py`; checked against brute-force integrals over momentum and angle, converged to 2×10^{-5}). At the analysis grid, at 31.6 keV and 4.22 MeV and $m_{\mathrm{lightest}} = 10^{-6}$, 10^{-3} and 0.03 eV, the fluxes agree with it to 2×10^{-4} between 10 TeV and 10 PeV; the resonance-only averaging is off by 1–17%, and by a factor ~ 100 inside one absorption dip (4.22 MeV, 0.03 eV). Along the way, the semi-analytic kernel's momentum integral proved too coarse where the resonance sits at $\sigma\approx m$ for a light relic (1.3% off at $m/T\sim 0.03$, 0.2% at 10^{-6} eV); panels at $2m$, $20m$, $200m$ and $2000m$ bring it to 3×10^{-9} of direct quadrature (used by the new treatment only).

Cost.—A transport takes 1.1 ($m_{\mathrm{lightest}} = 10^{-6}$ eV) to 2 (0.03 eV) times as long. The campaign's MeV planes cost 1.5 to 2.5 times as much per point (pilot of 2026-10-18), and the campaign ~12 h on 176 cores instead of ~6.

13.6 Dirac relics in motion: their helicity (2026-10-21)

The secret-interaction planes use the same switch (`secret_interactions.thermal_products`); for Majorana relics nothing changes from Sec. 13.5. A Dirac relic is different: it keeps the helicity it had when it decoupled, $-1/2$ for ν and $+1/2$ for $\bar\nu$. At rest, averaged over directions, it is unpolarized, which is what the at-rest treatment assumes. In motion its spin is tied to its momentum, and since σ selects the direction, the beam sees a polarization that varies with σ. Mauricio asked for the exact treatment (Option 1).

The polarization the beam sees.—In the relic's rest frame (the pure boost along its momentum keeps the spin direction), the spin of a ν of helicity $-1/2$, or of a $\bar\nu$ of $+1/2$, along the incoming neutrino is

$$\pi = \frac{E_p\sigma - m^2}{p\,\sigma}, \tag{17}$$

from -1 at $\sigma = E_p - p$ (the relic moving with the beam) to $+1$ at $\sigma = E_p + p$ (against it); with the flux factor its average over directions is the speed v.

The amplitudes see two massless relics.—Write the relic's spinor as its left- and right-chiral parts. The left-chiral part enters every bilinear like a massless particle of momentum $(p_{\mathrm{r}} - mS)/2$, S the spin vector; the right-chiral part like one of momentum $(p_{\mathrm{r}} + mS)/2$. With the final neutrinos massless, each final helicity configuration picks one part, so the two never interfere. The transport needs only the final energies, so it averages over the final azimuth about the collision axis; then $mS \to -\pi\, p_{\mathrm{r}}$ up to terms of order m^2/s ($\sim 10^{-15}$), and

$$|\mathcal{M}|^2 = \frac{1+\pi}{2}\,|\mathcal{M}_L|^2 + \frac{1-\pi}{2}\,|\mathcal{M}_R|^2, \tag{18}$$

with $\mathcal{M}_{L,R}$ those of a massless relic of either chirality at the same s and angle (for the $\bar\nu$ the weights apply to its v spinor's chirality, opposite to its spin). The brute force (`tests/amplitudes_bruteforce.py`, massive spinors of any spin) confirms this for all six Dirac processes with complex couplings, the deviations scaling as m^2/s.

Which part takes part where.—The vector couples only to the left-chiral part: a Dirac relic offers it in the share $(1+\pi)/2$, which is 1/2 at rest (the factor κ of Sec. 7.2) and $(1+v)/2$ on average. Every scalar vertex flips chirality. The left-chiral part takes part in $\nu\nu\to\nu\nu$ through t and u together (both final neutrinos sterile), in $\nu\bar\nu\to\nu\bar\nu$ through t alone (both sterile), and in double-scalar production. The right-chiral part takes part in $\nu\nu\to\nu\nu$ through t or u apart (the relic's partner visible) and in $\nu\bar\nu\to\nu\bar\nu$ through s and t. The incident's own daughter is sterile on either. So the scalar's s-channel resonance needs the right-chiral part, whose share $(1-\pi)/2 \approx m^2/4p^2$ is tiny on a relativistic relic, except for relics moving almost along the beam. The split of the at-rest tensors by the relic's chirality was derived from this and agrees with the brute force to 10^{-15}, including parity: an incident sterile neutrino on one part gives the visible one's secondaries on the other, with visibility swapped.

The distributions.—The chiral shares replace ρ by

$$\rho_L(\sigma) = \rho\,\frac{1+\pi}{2} = \frac{1}{3\zeta(3)T_{\mathrm{r}}^3}\int_{p_X}^{\infty} f\,\frac{(E_p+p)(\sigma-E_p+p)}{2E_p}\,dp, \tag{19}$$

$$\rho_R(\sigma) = \rho\,\frac{1-\pi}{2} = \frac{1}{3\zeta(3)T_{\mathrm{r}}^3}\int_{p_X}^{\infty} f\,\frac{(E_p-p)(E_p+p-\sigma)}{2E_p}\,dp, \tag{20}$$

both with a kink at $\sigma = m$. With the flux factor, $(1-v\cos\theta)(1\pm\pi)/2 = (1\pm v)(1\mp\cos\theta)/2$, so the resonance kernels' exact thermal averages are the semi-analytic kernel at $v=\pm1$ times $(1\pm v)/2$.

Transport.—The products come in two sets of relic slots, weighted by ρ_L and ρ_R (`src/dp_si.py`, `model_spec(polarized=True)`). The vector runs Majorana's tensors on ρ_L alone. For the scalar, a polarized background breaks the parity that let visible and sterile neutrinos be run as sum and difference, so they are transported together, six states in one sweep. The option `"drop"` keeps the visible ones only.

Validation.—The reference (`tools/thermal_reference.py`) weights its mixture of relics at rest by $\rho(1\pm\pi)/2$ written out directly. At the analysis grid, at 31.6 keV and 4.22 MeV and $m_{\text{lightest}} = 10^{-6}$, 10^{-3} and 0.03 eV, the Dirac fluxes agree with it to 9×10^{-6} (31.6 keV) and to 3×10^{-4} of the free flux (4.22 MeV; 1.4×10^{-3} of the flux itself at one bin inside an absorption dip); relics at rest are off by factors up to 4×10^3. With the chiral shares forced to $\rho/2$ each, an unpolarized background, the polarized transport reproduces the parity runs to 10^{-11}: the chiral tensors, the six-state sweep and the slots' bookkeeping. The Majorana and dark-portal paths are unchanged (bit for bit against the campaign's code). A Dirac scalar transport costs 3.4–3.9 times one with the relics at rest, a vector 1.6–2 times.

Effect.—Large (Fig. 27). For the scalar the s-channel resonance on the lightest relic nearly disappears when it is relativistic: at 31.6 keV and $m_{\text{lightest}} = 10^{-6}$ eV the flux at 10 TeV keeps 86% where the unpolarized treatment absorbs it all, and at 4.22 MeV the dip near 10 PeV goes. For the vector the same relic absorbs more, its left-chiral share being near one rather than one half: at 4.22 MeV the dip near 10 PeV deepens from 0.30 to 0.19 of the free flux. With $m_{\text{lightest}} = 0.03$ eV the changes reach 0.14 (scalar) and 0.33 (vector) of the free flux at 4.22 MeV.

A numerical note.—Far below the resonance ($s\sim10^{-4}M^2$) the semi-analytic kernel's closed-form angular integral cancels to 10^{-3}–10^{-2} relative. For the chiral shares the angular integral is taken by 8-point quadrature there (exact to rounding). For ρ (Majorana, and the dark portal) the closed form stays until the running campaign ends, so that its planes stay reproducible; its effect on the fluxes is within the 2×10^{-4} agreement with the reference.

13.7 Soft scatterings on the relics (2026-10-22)

Below $m_{Z_D} = 1$ keV the cosmological phase used to carry the resonance only: the resonance energy $m_{Z_D}^2/2m_\nu$ falls far below the band there, and the t channel was left out on the grounds that it matters only for $g_D\sum|U|^2 \gtrsim 10^{-7}$ (so the split variants stopped at 10^{-8}). That reckoning left out the coupling the Z mixing gives every neutrino, $\kappa = \epsilon' g/c_W = 0.74\,\epsilon'$, which in the potential planes reaches 0.74. The vector's t channel is forward-peaked: on a relic of mass state r at rest, $\sigma \simeq \sum_c |G_{ca}|^2 (G'G'^\dagger)_{rr}/(4\pi M_t^2)$ for each of the relic's two roles (profiles 2 and 6 of dp_si), with $M_t^2 = m_{Z_D}^2 + m_D^2$, m_D the relics' Debye mass (`dp_absorption.debye_mass_sq`; Majorana, whose charge is v-suppressed). Nearly all of it moves the neutrino at its own energy;

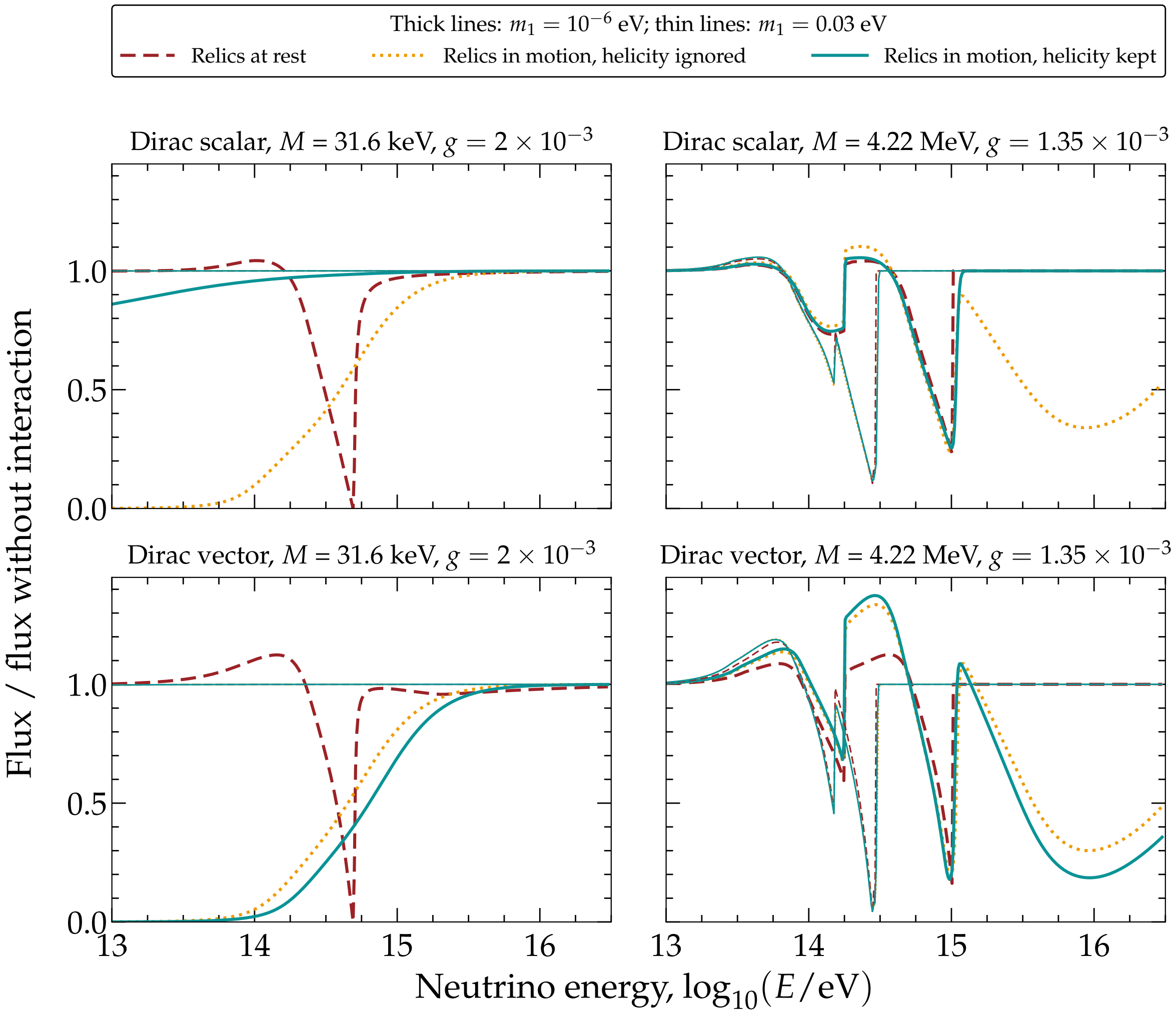


Figure 27: Dirac relics' helicity: the flux at Earth over the free-streaming one (texture e, pion sources) with the relics at rest, in motion ignoring their helicity (each chiral share $\rho/2$), and in motion with it, for $m_\mathrm{lightest} = 10^{-6}$ eV (thick) and 0.03 eV (thin) (`tools/dirac_relic_polarization.py`; drawn in `notebooks/paper_figures.ipynb`).

the rest (the logarithmic tail) costs it energy, at a rate that no longer depends on m_{Z_D} once $m_{Z_D}^2 \ll 2m_\nu E$.

The soft part: jumps between eigenstates.—Each scattering acts on the neutrino's flavor state as G (the scattered amplitude is $G|\psi\rangle$): the Lindblad equation $\dot{\rho} = -i[H,\rho] + \Gamma_\mathrm{u}(G\rho G - \{G^2,\rho\}/2)$, $\Gamma_\mathrm{u} = 2n_\mathrm{hel}(1+z)^3 \sum_r \eta_r (G' G'^\dagger)_{rr}/(4\pi M_t^2)$. Eliminating the coherences between the eigenstates a, b of H (Hubble-slow; they turn at ω_{ab} and are damped at $D_{ab} = \Gamma_\mathrm{u}[(G_{aa}^2 + G_{bb}^2)/2 - G_{aa}G_{bb}]$) leaves rates between populations $R_{ba} = \Gamma_\mathrm{u}|G_{ba}|^2\,\omega_{ab}^2/(\omega_{ab}^2 + D_{ab}^2)$: the secular rates where $D \ll \omega$, the Zeno-suppressed ω^2/D where not (Stodolsky's damping). They are applied per redshift step in the dark-portal phase (`dp_cosmo._soft_jumps`, exactly, in closed form on the complement of the uniform mode; Sec. A.17) wherever they add up to 10^{-4} over the path. Fast, they equalize the flavors (the states they connect end with equal shares): the flavor fractions go to 1/3 each, a shift of only ~ 0.01 from the standard ones for pion sources ($|\Delta\chi^2| < 1$ alone, fitted: ≤ 0.1 MESE, ≤ 0.75 HESE+DR2, -0.47 the projection, at 100 eV). But at $g_D \sum |U|^2 = 10^{-8}$, where the relics screen the local potentials and only the cosmological one is left, they erase its effect below $m_{Z_D} \sim 10^{-6}$ eV: a toy with both (one energy, adiabatic eigenstates, these rates) moves the rise of f_μ from $P/m_{Z_D}^2 \simeq 10^5$ to $10^{6.5}$ eV^{-2} at 10^{-8} eV and removes it at 10^{-20} eV; the solver, with the Z mixing's κ in the relics' factor, equalizes further still. The spin (magnetic) part of the exchange is not statically screened; its cutoff is dynamical, at a comparable or smaller scale, and where it matters the rates exceed the expansion rate by $\sim 10^{10}$, so nothing depends on it.

The hard part.—Below 1 keV, dp_si's transport (now with M_t) carries the cosmological phase wherever the couplings exceed 10^{-4} (the largest |eigenvalue| of G; the energy losses are $\sim 10^{-6}$ there, 1.5% of the flux at 10^{-3}, 40% at 3×10^{-3}) and the potential is negligible. Where it is not (`'combined'`), the dark-portal phase's eigenstate populations are rescaled per energy to dp_si's flux: exact when the hard scatterings do not depend on the flavor, as for κI, which dominates them in the potential planes. Checked against a coarse transport with all three (the potential, the jumps, the energy losses with every scattered neutrino projected on the eigenbasis at its new energy; textures e, μ, 10^{-10}–10^{-6} eV, $\epsilon' = 10^{-3}$–10^{-2}): the total to three digits, the flavor fractions to 10^{-3} near the edges (6×10^{-3} where 2% of the flux survives).

What it changes.—The top edge of every potential plane below 1 keV, from $\epsilon' \sim 5 \times 10^{-3}$ to $\sim 2 \times 10^{-3}$ (at $\epsilon' = 3 \times 10^{-3}$ the flux between 10 TeV and 3 PeV falls to 0.77); the 10^{-8} potential planes' diagonal edges below $\sim 10^{-6}$ eV; the dark portal's ultralight planes ($\epsilon' = 0$) below 1 keV; possibly the $L_e - L_\mu$, $L_e - L_\tau$ ultralight planes near 1 eV–1 keV. Left out: the scatterings while crossing the Galaxy (where they could act the cosmological phase has already equalized, or the potential suppresses them), the u-channel interference ($\sim \ln(1/\mu)$ against $1/\mu$), and the hard scatterings of a non-universal coupling where a potential matters too (only where the potential alone already excludes).

The projections.—Their Asimov data are the Standard Model's (Mauricio's choice): generated with the scatterings left out, since at a split such as 10^{-8} they act even at $\epsilon' \to 0$; each mass column is read against its own best fit, which absorbs that offset.

The two transports' normalizations.—With no scattering they differ by 0.7% at every energy: the dark-portal phase's 200 linear redshift steps overcount the sources by 0.5% (against 800), dp_si's steps, uniform in $\ln(1+z)$, undercount by 0.17%. The flux normalization floats in every fit, so no χ^2 moves, also where a plane switches transport.

13.8 Relic mass changes, neutrino decay, and where the models hold (2026-10-24)

Neither transport contains two processes that the couplings allow once the mediator is light. Both come from the off-diagonal entries of the couplings in the mass basis, $G' = U^\dagger G U$, which every model here has: the dark portal ($G' = g_D\, w w^\dagger + \kappa I$, $w = U^\dagger u$) and $L_e - L_\mu$, $L_e - L_\tau$ ($G' = g\, U^\dagger \mathrm{diag}(1, -1, 0)\, U$). Mauricio agreed to implement both (2026-10-24). The same couplings also set where the models can be trusted at all (item 3), and they rule out the potential planes (item 4). The MeV planes are untouched by all of this.

1. A soft scattering that changes the relic's mass state.—A scattering can leave the relic in another mass state, $r \to s$, with amplitude $\propto G'_{ca} G'_{sr}/(t - M_t^2)$. Its forward peak survives the mass change. For a beam along z, $t \to 0$ conserves the relic's light-cone momentum $p^- = E - p_z$, so the relic leaves with $p_s^- = p_r^-$ and the beam gives up

$$\omega_{rs} = \frac{m_s^2 - m_r^2}{2\, p_r^-} \qquad \Big(\text{relic at rest: } \frac{\Delta m_{sr}^2}{2 m_r}\Big). \tag{21}$$

The relic current's light-cone component is unchanged, $J^- = 2\sqrt{p_s^- p_r^-} = 2 p_r^-$, so the $1/M_t^2$ peak of the cross section is the elastic one. These events are already inside the jumps' Γ_u (the sum over s in $(G' G'^\dagger)_{rr}$) and inside dp_si's quasi-elastic conversions, but at no energy cost. Weighted with the flux factor $v_\mathrm{rel} = p_r^-/E_r$, the cost per unit time is exact over the relic's directions, $\langle v_\mathrm{rel}\, \omega_{rs} \rangle = \frac{1}{2}(m_s^2 - m_r^2)\langle 1/E_r \rangle$. So a neutrino in state a loses energy at a rate that does not depend on its energy:

$$\frac{dE}{dt} = -\sum_{r\neq s} \frac{2n_{\rm hel}(1+z)^3\,\eta_r\,(GG^\dagger)_{aa}\,|G'_{sr}|^2}{4\pi M_t^2}\;\frac{m_s^2-m_r^2}{2}\,\Big\langle \frac{1}{E_r}\Big\rangle_{T(z)}. \tag{22}$$

Up-conversions cost energy and down-conversions return it; the light relic, moving with $E_r \sim 3T$, costs the most. Relative to E the loss grows toward low energies ($\propto 1/E$). Estimate for the μ texture (G' and m_D from the solver at $g_D = 10^{-4}$, both scaled $\propto g_D$; lightest mass 10^{-6} eV; $z \leq 4$). The table gives the energy lost before arrival as a fraction of E at 100 TeV, to first order (1 or more: the beam is drained):

g_D	$m_{Z_D} = 10^{-20}$	10^{-12}	10^{-9}	10^{-6}	10^{-3} eV
10^{-4}	1.4×10^6	1.4×10^6	1.0×10^6	6	6×10^{-6}
10^{-5}	1.4×10^4	1.4×10^4	5.6×10^2	6×10^{-4}	6×10^{-10}
10^{-6}	1.4×10^2	1.4×10^2	0.06	6×10^{-8}	—
10^{-7}	1.4	1.0	6×10^{-6}	—	—

Below $m_{Z_D} \approx m_D \propto g_D$ the loss scales as g_D^2, and above it as $g_D^4/m_{Z_D}^2$. If this holds, the ultralight exclusion reaches $g_D \sim 10^{-7}$ at the smallest masses.

2. Neutrino decay into the mediator.—For $m_{Z_D} < m_i - m_j$ the heavier mass states decay, $\nu_i \to \nu_j + Z_D$, both in the beam and among the relics. For a vector coupled to a current that the neutrino masses do not conserve, the rest-frame rate of a Dirac neutrino with a left-handed coupling is the one of $t \to bW$, $|G'_{ji}|^2 m_i^3(1+2m_{Z_D}^2/m_i^2)(1-m_{Z_D}^2/m_i^2)^2/(32\pi m_{Z_D}^2)$: $|G'_{ji}|^2 m_i/16\pi$ from the transverse polarizations plus the longitudinal part $\propto m_i^3/m_{Z_D}^2$. Our neutrinos are Majorana: the vertex is $\gamma^\mu(G'_{ji}P_L - G'^*_{ji}P_R)$, the two chiralities do not interfere for $m_j \to 0$, and

$$\Gamma_{i\to j} \simeq \frac{|G'_{ji}|^2\,m_i^3}{16\pi\,m_{Z_D}^2}\Big(1+\frac{2m_{Z_D}^2}{m_i^2}\Big)\Big(1-\frac{m_{Z_D}^2}{m_i^2}\Big)^2, \tag{23}$$

twice the Dirac rate, as for heavy Majorana neutrinos. In flight it is slower by m_i/E. For $\nu_3 \to \nu_1$ at 100 TeV, with $|G'_{31}| \simeq 0.3\,g_D$, the rate over the Hubble rate is:

g_D	transverse	$m_{Z_D} = 10^{-12}$	10^{-8}	10^{-4}	10^{-2} eV (longitudinal)
10^{-8}	6×10^{-3}	8×10^{18}	8×10^{10}	8×10^2	0.08
10^{-6}	62	8×10^{22}	8×10^{14}	8×10^6	8×10^2
10^{-4}	6×10^5	8×10^{26}	8×10^{18}	8×10^{10}	8×10^6

Even the transverse part alone makes the beam arrive as ν_1, plus the daughters at lower energy, for $g_D \gtrsim 10^{-7}$: a large flavor signal, which IceCube already constrains. Half of the daughters keep the parent's helicity (the P_L piece); for those, the fraction z of the parent's energy follows $dN/dz = 2(1-z)$ from the transverse part and $2z$ from the longitudinal part (checked below). The other half have the opposite helicity (the P_R piece: a neutrino gives an antineutrino), with the mirrored spectra: $2z$ from the transverse part and $2(1-z)$ from the longitudinal part. Where $m_{Z_D} \ll m_i$ (the longitudinal part dominates), a neutrino in the beam thus gives a hard neutrino ($2z$) or a soft antineutrino ($2(1-z)$) with equal rates; an antineutrino, the CP mirror. The mediator is taken as invisible: it decays back to neutrinos only for $m_{Z_D} > 2m_1$, and then into the softest part of the spectrum. The relics' ν_2 and ν_3 decay at rest, with rates above H_0 already for $g_D \gtrsim 10^{-14}$. They are gone long before today, so every relic effect (the Debye screening, the absorption by the relics' flavor content, the soft scatterings, item 1) acts with ν_1 relics only.

3. Where the models hold.—A massive vector coupled to a current that the masses do not conserve has a longitudinal mode that, well above m_{Z_D}, acts as its Goldstone boson ϕ

(equivalence theorem), coupled through the current's divergence. For the neutrinos' current with Majorana masses, $\partial_\mu J^\mu = i\sum_{ij} G'_{ij}(m_i\chi_i\chi_j - m_j\chi_i^\dagger\chi_j^\dagger)$, so that

$$-\mathcal{L} = \tfrac{1}{2}\, Y_{ij}\,\phi\,\chi_i\chi_j + \text{h.c.}, \qquad Y_{ij} = i\,\frac{G'_{ij}m_i + G'_{ji}m_j}{m_{Z_D}}. \tag{24}$$

Diagonal entries count too: through the masses, the axial part of a purely left-handed current is not conserved even for a flavor-universal coupling. The tree-level calculations here hold where these couplings respect perturbative unitarity: the largest eigenvalue of the partial-wave matrix at most 1/2, built from the amplitudes of Allwicher, Arnan, Barducci, Nardecchia (JHEP 10 (2021) 129, Eqs. 2.5, 2.6, A.10). For $J = 0$, the two-fermion states $|ij\rangle$ with helicities $++$ and $--$: $T_{++\to++} = -Y^*_{ij}Y_{kl}$ (the s channel) and $T_{--\to++} = Y_{ik}Y_{jl} + Y_{il}Y_{jk} + Y_{ij}Y_{kl}$ (the s, t, u channels), with $a = T\,c_{ij}c_{kl}/16\pi$ ($c = 1/\sqrt{2}$ for identical fermions); for $J = 1/2$, $a = -3(YY^\dagger)/32\pi$ (`src/dp_unitarity.py`). It reproduces their bounds: $|Y|^2 < 4\pi$ for one Majorana flavor (their Eq. 6.2), $8\pi/3$ for a Dirac pair (4.7), $16\pi/3$ at $J = 1/2$ (6.4) (`tests/test_unitarity.py`). Decided (Mauricio, 2026-10-24): the planes keep their mass and coupling ranges, and the region beyond the bound is marked as not allowed in the figures, with the masses at the lightest mass 10^{-6} eV (the largest splittings, so the widest marked region). Where the bound holds, m_{Z_D} lies above the Debye mass ($m_D \approx 10^{-5}g_D$ eV), so the vacuum decay rates apply there.

4. The potential planes are dropped.—Decided (Mauricio, 2026-10-24). In the potential planes the mediator also mixes with the Z (ϵ', Table 1 of the theory notes, "free"). The theory notes define ϵ' without a mechanism; the model they follow (Jana and Swain, arXiv:2607.04441) produces it in two ways. (a) At tree level, from a Higgs doublet φ with dark charge (their Eq. 6): $\epsilon' \simeq 2g_D v_\varphi^2/[(g/c_W)v^2]$. Because φ carries dark charge, its vacuum value also gives the Z_D part of its mass, and diagonalizing the (Z, Z_D) mass matrix gives

$$\epsilon' \leq \frac{v_\varphi}{v_H}\,\frac{m_{Z_D}}{m_Z} \tag{25}$$

for a dark boson much lighter than the Z (v_H the vacuum value of the Higgs doublet without dark charge, which gives the quarks and charged leptons their masses, so that v_φ/v_H is at most a few). Checked in 50-digit arithmetic over 20 000 random versions of their model with $m_{Z_D} < m_Z/1000$, including a dark singlet: the ratio reaches 1.006 at most (`tools/check_zd_mixing_bound.py`). Their paper does not state this; it follows from their Eq. 6 and the gauge structure. (b) At one loop (their Eq. 20), $\epsilon_{\text{loop}} = gg_D\, m_4^2|U_{\gamma 4}|^2/(32\pi^2 c_W m_Z^2)$. In their model the only entry connecting active and dark neutrinos is $y_\nu v_\varphi$ (their Eq. 3), so $\sum_k |U_{\gamma k}|^2 m_k^2 = |y_\nu v_\varphi|^2$ exactly (the ν row of $MM^\dagger$), and with the doublet's share of the Z_D mass, $\epsilon_{\text{loop}} \leq (y_\nu^2/16\pi^2)(v_\varphi/v_H)\, m_{Z_D}/m_Z$: smaller still. Every ultralight exclusion that the 18 potential planes held needs ϵ' 8 to 30 orders of magnitude above this bound (the stored limits, per mass column; e.g. $\epsilon' \sim 5\times 10^{-3}$ at split 10^{-8}, against $\leq 1.1\times 10^{-10}\, v_\varphi/v_H$ at $m_{Z_D} = 10$ eV): in the model the analysis follows, those mixings cannot occur. The planes span 10^{-30}–10^{4} eV, so all 18 are dropped (per data set); the paper states why. Not re-derived here: their Eq. 20, and the bound on v_φ/v_H; neither can close a gap of eight orders of magnitude.

Where the effects fall (2026-10-24).—`tools/check_decay_cut.py` (the stored limits per mass column, pion and fitted, IC26) with the bound of item 3 and the decay rate of item 2, the identity part of G rotated exactly (otherwise $U^\dagger(\kappa I)U$ leaves $10^{-16}\kappa$ off the diagonal, which looks like decay; the implementation needs the same care). Fig. 28. $L_e - L_\mu$, $L_e - L_\tau$: the stored limits run along $g \propto m_{Z_D}$ below $\sim 10^{-13}$ eV, and the region beyond the bound starts 0.7 decade below them, so the whole stored exclusion there lies where the model does not hold. Below 0.03 eV the beam decays far below the stored limits (down to $g \sim 10^{-32}$ at 10^{-30} eV). Dark portal, ultralight: beyond the bound at every coupling of the plane below $\sim 10^{-9}$ eV,

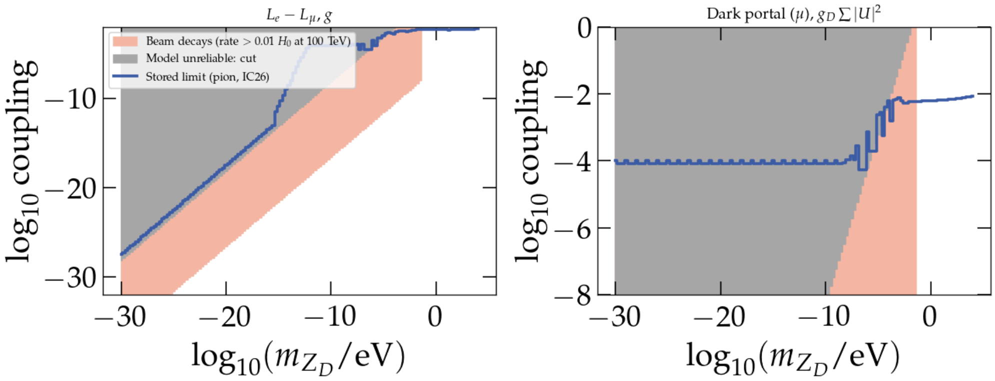


Figure 28: Where neutrino decay (rate above $0.01H_0$ at 100 TeV, red) acts and where the models do not hold (item 3, gray), against the stored limits (blue; pion, IC26; the dark portal's flat edge at 10^{-4} is the bug of Sec. A.17).

and up to $\sim 10^{-4}$ eV at the largest couplings; decay everywhere else below 0.03 eV. The MeV planes: neither.

Decay in the code (2026-10-24).—`src/dp_decay.py`, called once per redshift step after the collisions, in both transports: the dark-portal phase (eigenstate populations through their mass content, $|\langle i|a\rangle|^2$, tabulated on the σ nodes like the jumps' tables; each mass component decays on its own, and a daughter in mass state j is split over the eigenstates by the same content) and dp_si (mass states; one Majorana flux holding both kinds, so the two daughter halves add to a flat spectrum). The decay widths also damp the coherences in the jumps, $D_{ab} \to D_{ab} + (\Gamma_a + \Gamma_b)/2$ (lab frame): a state that decays fast is Zeno-protected from the conversions into it. Switch: `dark_portal.neutrino_decay` (default on); the projections' Asimov data have it off (the Standard Model). Where the potential dominates, the eigenstates are not mass states and the incoherent split is an approximation; those points lie beyond the models' range (item 3). A consequence: where the soft scatterings convert ν_1 back into ν_2, ν_3 faster than the expansion and those decay, every conversion–decay cycle costs the beam energy. At $m_{Z_D} = 10^{-6}$ eV, $g_D = 10^{-6}$ (the μ texture) the flux between 10 TeV and 3 PeV falls to 0.015 of the no-effect flux with both (both transports agree: 0.0145 and 0.0149), to 0.50 with the decay alone, all of it ν_1 at Earth.

Checked against exact results (2026-10-24).—With the relic scatterings off nothing decays into ν_3, so its share at Earth is its survival averaged over the sources (star-formation evolution), $\langle \exp[-\Gamma_3 m_3/E \int_0^z dz'/((1+z')^2 H)]\rangle$. At $m_{Z_D} = 10^{-2}$ eV, $g_D = 10^{-7.5}$ (μ texture), 100 TeV and 1 PeV: exact 0.5266 and 0.9337; dp_si 0.5266 and 0.9337; the dark-portal phase 0.5257 with its 200 redshift steps (0.5263 with 800). Two fixes came out of it: the decays are applied in two halves around each step's collisions and injection (Strang; what a step injects decays over half of it on average, not all of it), and the dark-portal phase takes each step's optical depth $\int dz/((1+z)^2 H)$ exactly (its end-point $dt/(1+z)$ was first order: 0.5148 with 200 steps). The energy loss of item 1, decay off: at $m_{Z_D} = 10^{-9}$ eV, $g_D = 10^{-6}$, the flux at 100 TeV falls to 0.9458 against 0.9455 for the same source average of $(1+\Delta E(z)/E)^{-p}$ with the states' mean $(GG^\dagger)_{aa}$ (the scatterings equalize them); the shifts the transport applies match $\int C(z)(GG^\dagger)_{aa}\, dz/((1+z)^2 H)$ to 0.2%. Tests (`tests/test_decay.py`): the jumps' protection by the decay widths ($\omega^2/(\omega^2 + D^2)$ with D grown by them), the Asimov data without decay, the solver's relics after their decays ($\eta' = (3, 0, 0)$ where they all decay).

How they enter the code.—Item 1 as a drift in $\ln E$, per relic pair with $\langle 1/E_r\rangle$: in dp_si's step, and through dp_si wherever the drift exceeds 10^{-4} of E over the path at the band's lowest energy (the routing extended below the present 10^{-4} coupling threshold). Item 2 as a loss of ν_i into ν_j at lower energies (both helicities, with their spectra) in the cosmological phase, plus ν_1-only relics wherever the relics decay. Item 3 as a marked region on the planes' figures, from `src/dp_unitarity.py`. Done (2026-10-24): item 2 (`src/dp_decay.py`, both transports, and the relics' composition after their decays by $z = 1$, `dp_decay.relic_composition`: time-dilated rest rates, the daughters by branching; the code ties a state's density to its temperature, so the grown ν_1 population is a hotter Fermi–Dirac gas, an approximation); item 1 (`src/dp_relic_loss.py`: $dE/dt = -C(z)(GG^\dagger)_{aa}$, a conservative shift of each bin's content per step, in both transports; at $m_{Z_D} = 10^{-9}$ eV, $g_D = 10^{-6}$, decay off, the flux at 100 TeV falls to 0.946 and 0.940 in the two transports, at 10 TeV to 0.624 and 0.619); item 3 (`analysis.exclusion.not_allowed_boundary`, `draw_not_allowed`: the coupling matrix is linear in the plane's coupling, so the boundary is the line $\log_{10} g = \log_{10} m_{Z_D} + c$, with $c = 2.46$, 1.76, 1.74 for the dark portal's e, μ, τ textures and 1.76, 1.75 for $L_e - L_\mu$, $L_e - L_\tau$).

Checks (2026-10-24).—Item 1, exact $2 \to 2$ kinematics in 60-digit arithmetic (beam at 100 TeV, relics in motion, up- and down-conversions, $|t| = 10^{-24}$ eV^2): ω_{rs} as above to 2×10^{-11}. The direction average $\langle v_{\rm rel}\, \omega_{rs} \rangle = \frac{1}{2}(m_s^2 - m_r^2)/E_r$: exact at $|p| = 10^{-5}$–10^{-2} eV. The matrix element from the Dirac traces with explicit γ matrices (left-handed currents at both vertices): $\sum |\mathcal{M}|^2 = 16\,(k \cdot p)(k' \cdot p')\, g^4/(t - M^2)^2$, with $k' \cdot p' = k \cdot p - \Delta m^2/2$, so $r \to s$ equals $r \to r$ at the same t up to $\Delta m^2/s$ (0.99986 and 1.000024 at $E = 1$ keV for up- and down-conversion). For Majorana relics the right-chiral piece of the current gives $16\,(k \cdot p')(k' \cdot p)$, the same in the forward limit; the mixed terms vanish there ($\propto m_r m_s\, t$). Item 2, the trace with the massive polarization sum, $-g^{\mu\nu} + q^\mu q^\nu/M^2$, for a Dirac neutrino with a left-handed coupling: the Dirac rate to six digits at $m_{Z_D} = 10^{-2}$–10^{-4} eV, a left-chiral daughter with angular distribution $1 + \cos\theta_s$ (transverse) and $1 - \cos\theta_s$ (longitudinal) about the parent's spin, which for a left-handed parent in flight gives the dN/dz above. The Majorana vertex's P_R piece: the same rate to six digits (so twice the Dirac rate in all), a right-chiral daughter, and the mirrored distributions, $1 - \cos\theta_s$ (transverse) and $1 + \cos\theta_s$ (longitudinal). (An earlier version of this subsection read that check as covering Majorana neutrinos, with no second channel; corrected above.) Item 3, the tests above; the Goldstone couplings match the decay rate ($|Y_{31}|^2 m_3/16\pi$ for $m_1 = 0$ is the Majorana rate's longitudinal part). Item 4, the mass-matrix check above (`tools/check_relic_mass_change.py`, `tools/check_decay_rate.py`, `tools/check_zd_mixing_bound.py`).

The secret interactions (2026-10-24).—Their planes span $m_\phi = 1$ keV–1 GeV and $g_\phi = 10^{-6}$–1 (`tools/check_si_new_physics.py`). The decay into the mediator needs $m_\phi < m_i - m_j \leq 0.05$ eV: closed everywhere. The models' range: the vector's longitudinal coupling is $g\,\Delta m/m_\phi \leq 5 \times 10^{-5}$ at 1 keV, the scalar's Yukawa at most 1, both far within the unitarity bounds. The relic mass-change energy loss: the scalar's chirality-flipping vertices give $|\mathcal{M}|^2 \propto t^2$, no forward peak; the vector's, as an upper bound (the whole path from $z = 4$, no screening), reaches 1% of the energy at 100 TeV only for $g \geq 0.45$–0.72 at 1 keV (e, μ, τ and the pairs; none for the universal texture) and only above $g = 1$ from 10 keV up – deep inside the region the quasi-elastic conversions already exclude (4.4×10^{-4} for ν_e at 1 keV, Sec. 7.6). So the secret-interaction transport needs neither; the redo's checks include them (their fluxes unchanged by the new code, and the pilot).

Exact decay kinematics, the steps, and two numerical guards (2026-10-24, evening; the transport reviews, docs/dev/TRANSPORT_REVIEW.md).*—(i) The daughters' spectra above took the daughter and the mediator as massless, so that the daughter's share z of the parent's energy ran over $[0, 1]$. With every mass kept it runs over $[z_-, z_+] = [(E^* - p^*)/m_i, (E^* + p^*)/m_i]$ (rest-frame E^*, p^*; the mediator takes the rest), which near the thresholds and for nearly degenerate masses (the lightest mass is profiled up to 0.1 eV) is narrow and close to 1: $[0.80, 1.00]$ at $m_i = 0.112$, $m_j = 0.1$, $m_{Z_D} = 10^{-3}$ eV. For a parent in flight helicity is light-front helicity, so each (parent, daughter) helicity rate is a covariant trace with the light-front spin vector $s = k/m - m\,n/(n \cdot k)$, $n = (1, 0, 0, -1)$, evaluated in the parent's rest frame with $z = k^+/p^+$ (`dp_decay.daughter_spectra`, `tools/derive_decay_kinematics.py`). On the window $d\Gamma/dz = a + bz + c/z$ exactly (fitted at nine angles, residual 10^{-15}; 10^{-7} for $m_j \sim 10^{-6}$ eV), the four helicity rates add up to Γ_{ij} of the formula above (to 10^{-6}), the massless limit is the earlier kernels, and averaged over the parent's helicity $\langle z \rangle = E^*/m_i$ (a test). With massive daughters the kept share is not 1/2: 1/3 at $m_i = 0.112$, $m_j = 0.1$ eV for real couplings, and it depends on the coupling's phase. Both transports deposit a parent's daughters over the bins its window covers: running sums of the spectrum's three moments for the bins wholly inside (entered at the window's top, removed below its bottom), the cut bins and the own bin by their exact integrals; against a brute-force sum over all bin pairs 10^{-14} (10^{-11} for degenerate masses), and dp_si's decay channel with every collision off moves the populations as `decay_step` to 10^{-6} (tests). (ii) The new processes' rates over a step were taken at its low-redshift end: first order, and where the soft scatterings and the decays compete the Zeno-suppressed conversions fall steeply with redshift; at 200 steps a drained flux came out up to 17% off (the redshift steps ×16 as reference; dp_si, converged, within 1% of the latter). Now every one is the step's trapezoid average with the step's exact time ($\int dz/((1+z)H)$ for the jumps, $\int dz/((1+z)^2 H)$ for the decays and the loss) and the widths' $(1+z)$ and the tables' σ at the midpoint. (iii) The loss's beam factor has the Z mixing's κ: $((G+\kappa)(G+\kappa)^\dagger)_{aa}$ (none below 1 keV in the planes). (iv) Deep in the not-allowed corner ($\gtrsim 8$ decades of coupling beyond the line) the decay reached 10^{65} per dp_si step and overflowed the bin's exponential: a decay beyond 10^9 per step and 10^7 times the state's other events is kept there (the state drains within the step, the decay takes its events); with it the results there agree for caps 10^7–10^{13} and move by 0.5% between ratios 10^7 and 10^9 (by 50% at 10^5), and at the allowed points tested it does not act (bit for bit). (v) Below the break, the upper piece of a broken power law now has no flux at energies its daughters cannot reach; the analysis interpolates the solver's runs in the spectral index with a spline in $\log F$, and a zero next to 10^{-39} made it peak at e^{-14}: elements whose $\log F$ bends by more than 5 across the index nodes are now interpolated linearly (`analysis.model.LOG_SPLINE_BEND_MAX`), and `decay_step` leaves no rounding residue (a state's survivors are put back as $N(1 - \sum_i w_i(1 - e^{-\lambda_i}))$, never below zero).

The Galactic component (2026-10-24).—The three processes act on the neutrinos emitted in the Galaxy too, over their 0.05–30 kpc to the Sun, at $z = 0$: `src/dp_galactic_path.py`, called by `_galactic_emission_populations` below 1 keV wherever one of them acts over 30 kpc at the grid's lowest energy (jumps above `SOFT_JUMP_MIN`, decays and losses above 10^{-6}). Each line of sight (the work map's 3072 pixels) is swept from its farthest emission bin to the Sun, over each of the 97 distance bins in turn: the bin's emission (the emission's populations, gas and sources weighted as before) is a constant source over the bin's length, and the populations evolve over it in the eigenstates of the local potential (the bin's density), each energy bin solved exactly over the bin with the jumps, the decays and the source together (`dp_sweep.step_full` at $z = 0$, as the cosmological phase where the neutrinos decay; 2026-10-25: until then the emission was added at the bin's centre and the jumps and the decays alternated, first order where they cycle) and the energy loss in two halves around it; then over the last 0.05 kpc to the Sun; both kinds together, since a flipped daughter changes kind. The relics' composition is the one after their decays by today. Where the potential rotates the eigenstates by less than 10^{-6} along every line of sight (the dark-portal planes, and the lepton-number planes at the smallest couplings), every line of sight evolves alike but for its length: the emission, scaled per energy by the sky-mean intensity and split into a few spectral shapes (an SVD over the pixels, kept to 10^{-10} of the largest singular value), is evolved once through the distance bins (a constant source from the Sun outward: the differences of the populations at consecutive edges are what each bin's emission sends to the Sun) and combined per pixel with the distance weights, at the output resolution (`vacuum_populations`; 1 s against 80–150 s on one core). Checks: the two paths agree to 8×10^{-6} (dark-portal points; an unscaled SVD lost 3×10^{-4} at the faint highest energies, and a step that skipped bins whose spectral shapes summed to zero had made them differ by a factor of two at a strongly processing point; both fixed); with the segments cut into 16 pieces as reference, the default is within 1.3×10^{-5} at moderate points and within 0.6–3% where the processes drain the Galactic flux (first order before the alternation was made symmetric: 4%); tests `tests/test_galactic_path.py` (the fast path equals the sweep to rounding with equal segments; the jumps keep the number).

Decay in a medium (2026-10-25; `src/dp_medium.py`, docs/dev/DECAY_IN_MEDIUM_2026-10-25.md).—In a potential the neutrinos travel as the eigenstates a of H, not as mass states (Sec. 3), and the decay acts between those eigenstates: its loss and gain terms in Eqs. (3), (8) and (9) are the rates below. Until 2026-10-25 each eigenstate decayed at the mean of its mass content's vacuum rates, $\sum_i T_{ia}\Gamma_i$, with vacuum kinematics, and each daughter mass state was split over the eigenstates by its content. That includes a decay that energy conservation forbids: the most stable eigenstate, mostly ν_1 with a small heavier admixture ϵ^2, decayed into itself at $\epsilon^2\Gamma_3$. Over a Hubble time $\Gamma_3 t_H$ reaches 10^{15}, so even $\epsilon^2\sim10^{-10}$ slid the flux down in energy. The decay is now done between the eigenstates, with their own kinematics.

States. With $D=\mathrm{diag}(m_i)$ and the texture on the mass basis, $S'=U^\dagger SU$, the effective masses squared $\mu_a(\sigma)$ are the eigenvalues of $D^2+\sigma S'$ (antineutrinos: $D^2-\sigma S'^*$), the energies $E_a=p+\mu_a/2p$. With $c_{ia}=\langle\nu_i|a\rangle$, $\mu_a=\langle a|D^2|a\rangle+\sigma\langle a|S'|a\rangle$. A state in flight is $\sum_i c_{ia}$ times the light-front spinor of mass state i at the same momentum: the potential changes the spinors' small components only by $V/E\lesssim10^{-20}$.

Amplitude. The vertex is the Lagrangian's, $\bar\nu\gamma^\mu(G'P_L-G'^*P_R)\nu\,Z_\mu$ (Majorana), between the parent's state at momentum p and the daughter's at zp, each with its own eigenvector (the daughter's at $\sigma'=z\sigma$):

$$\mathcal{A}_{a\to b}(z)=\sum_{ij}c'^*_{jb}(z\sigma)\,c_{ia}(\sigma)\,\mathcal{V}_{ji}(z,k_T),\tag{26}$$

$\mathcal{V}_{ji}$ the vacuum light-front helicity amplitude of $\nu_i\to\nu_jZ_D$, the masses m_i, m_j in its spinors. Energy conservation in light-front form fixes the daughter's transverse momentum,

$$k_T^2=(1-z)\big[z\mu_a(\sigma)-\mu'_b(z\sigma)\big]-zm_{Z_D}^2\ \geq\ 0.\tag{27}$$

The longitudinal mediator carries the Goldstone-like enhancement. Its polarization is $\epsilon_L=q/m_{Z_D}-(m_{Z_D}/q^+)\,n$, and between on-shell spinors the Dirac equation turns $q\cdot J$ into $m_i\bar u_j(G'P_R-G'^*P_L)u_i-m_j\bar u_j(G'P_L-G'^*P_R)u_i$: the Lagrangian masses. What is left of $q\cdot J$ is the pair's vacuum energy mismatch, $D_{ij}J^+/2p^+$ with $D_{ij}=(k_T^2+m_{Z_D}^2)/(1-z)+(k_T^2+m_j^2)/z-m_i^2$. Summed over the polarizations with the in-medium q and written in k_T (which carries the in-medium kinematics), these terms cancel against the mass terms identically, for any texture: the rates below equal the explicit light-front spinors' amplitude to 10^{-9} (both kinds, Majorana phases, a coupling that does not commute with the texture included; `tests/test_medium.py`). (Until 2026-10-25 these notes argued that the mismatch sums to $\sigma\langle b|[G',S']|a\rangle$ and needed $[G',S']=0$; the review's spinor computation showed that the cancellation needs no such condition.) So a potential is not a mass in the amplitude. It enters through the eigenvectors and the kinematics only.

Rates. Summed over the mediator's polarizations, with $G_s=G'$ for a neutrino parent and G'^* for an antineutrino (the light-front spinors and polarizations behind these, checked against `dp_decay._msq_lf`'s traces to 10^{-13}):

$$\begin{aligned}\text{kept:}\quad |\mathcal{A}|^2&=\frac{2k_T^2(z+1/z)}{(1-z)^2}|\alpha|^2+\frac{1}{m_{Z_D}^2z}\left|\frac{zm_{Z_D}^2-k_T^2}{1-z}\,\alpha+(1-z)\,\beta\right|^2, &&\alpha=c'^\dagger G_sc,\quad \beta=c'^\dagger DG_s^*D\,c,\\ \text{flipped:}\quad |\mathcal{A}|^2&=\frac{2}{z}|\alpha+z\beta|^2+\frac{k_T^2}{m_{Z_D}^2z}|\alpha+\beta|^2, &&\alpha=c'^\dagger DG_sc,\quad \beta=c'^\dagger G_s^*D\,c,\end{aligned}\tag{28}$$

the kept daughter of the parent's kind, the flipped one of the other, and $d\Gamma_{a\to b}/dz=|\mathcal{A}|^2/16\pi E$ on the window of Eq. (27), E the parent's physical energy; $d\Gamma/dE=(1/E')\,d\Gamma/dz$ in Eq. (8). Written in $Q=z\mu_a-\mu'_b$ every weight is $(Az+B+C/z)/m_{Z_D}^2$, the $1/(1-z)$ poles cancelled.

Properties. (i) With the mass states as eigenstates ($c = 1$) Eq. (28) is $\Gamma^{\rm rest}_{ij} m_i/E$ times `daughter_spectra`'s kept and flipped spectra: the totals to 5×10^{-8}, every bin's share to 3×10^{-6} (both the old fits' own accuracy). (ii) A kept decay $a \to a$ is closed: by Weyl's inequality $\mu'_b(z\sigma)/z \ge \mu'_b(\sigma)$ (since $D^2/z \ge D^2$), so Eq. (27) needs $\mu_a > \mu_b + m^2_{Z_D}$: a kept decay goes to a lower state only, and the lowest state of a kind has none. (iii) A potential opens flips $\nu_i \to \bar\nu_i$ of the same mass state, closed in vacuum, through the axial coupling $2m_i G'_{ii}/m_{Z_D}$, where it splits ν_i from $\bar\nu_i$ by more than $m^2_{Z_D}$. Where the vacuum term dominates, the window is $1 - z \lesssim 2\sigma|S'_{ii}|/m_i^2$ and $E\Gamma \simeq |G'_{ii}|^2\sigma^3|S'_{ii}|^3/(3\pi m^2_{Z_D} m_i^2)$ (checked against the exact integral); where the potential dominates, $k_T^2 \simeq 2z(1-z)\sigma|S'_{ii}|$ and the longitudinal term gives $E\Gamma \simeq |G'_{ii} m_i/m_{Z_D}|^2\sigma|S'_{ii}|/4\pi$ (until 2026-10-25 quoted a factor 4 too small). (iv) Where the potential dominates the eigenstates tend to those of G and $\langle b|G|a\rangle \to 0$: the decays left go through the masses (β, the flips). (v) Each decay keeps one neutrino.

Where this holds. Fermi's golden rule between eigenstates needs the width small against the splittings of the parent's kind, $\zeta_a = 2E\Gamma_a/\min_b|\mu_a - \mu_b| \ll 1$; otherwise $H - i\Gamma/2$ mixes the eigenstates. The ratio depends on σ alone; every run returns the largest one it meets (`meta['zeta']` of `propagate_sky(_basis)`). In vacuum $\zeta \simeq 2\Gamma^{\rm rest} m/\Delta m^2 \sim 10^{-4}(g/m_{Z_D})^2$ (eV units); where the potential dominates $\zeta \sim (G'm/m_{Z_D})^2/2\pi$, the Goldstone Yukawa squared, bounded by perturbative unitarity. At the validation points below $\zeta \le 1.5 \times 10^{-2}$, and at the lepton-number limits (the pilot of 2026-10-25, $g_D/m_{Z_D} \approx 10^{-3}$–$1\ \mathrm{eV}^{-1}$) $\zeta \sim 10^{-4}$: no map over the planes is needed.

Numerics. $E\Gamma$ depends on (E, z) through σ only. So every parent's plan is built once per run on a ladder of σ values with the energy grid's bin ratio: a parent at rung r has its daughters k bins below at rung $r - k$. A plan holds the total rate, the own bin's share, the cut bins (exact integrals over the window, the parent spread over its bin as `window_plan`) and the runs of bins wholly inside the window (moments, deposited by running sums). The ladder reaches down to where the eigenstates are the vacuum ones (10^{-6} of the smallest splitting); the daughters below the grid enter the total and leave. A step interpolates in $\log\sigma$ between the two rungs around it: a parent's decays go $(1 - w)$ by the lower rung's plan and w by the upper's, each keeping the number. A Galactic segment takes the nearest rung instead (σ within the grid's half-bin ratio, 1%). Within a daughter bin the daughter's eigenvector is held at the bin's centre and its μ' follows $\langle D^2\rangle + \sigma_D\langle S'\rangle$ across the bin (exact to first order in the bin width; holding μ' itself broke the cancellation between equal potential slopes, ν_μ-like $\to$ $\bar\nu_e$-like). Cut bins near $z = 1$ are integrated in $1 - z$ (the spectrum there is a small difference of large terms). Against the exact formula (eigenvectors at every $z\sigma$, fine grid): the widths to 2×10^{-4} at the validation points, every bin to 10^{-8}. Code: `dp_sweep.step_full` (cosmology; the bin solved exactly with the jumps, the absorption and the injection), `dp_medium.decay_sweep` (the Galactic segments), tests `tests/test_medium.py`.

The other processes in a potential. The soft scatterings' jumps use $\langle b|G|a\rangle$ at the beam's own σ, right: the momentum transfer is soft, and the energy the eigenstate change needs, $(\mu_a - \mu_b)/2E$, is absorbed by the relic's recoil ($s \sim 2Em_r \gg \Delta\mu$), so nothing restricts it; only their Zeno damping took the old widths, now the eigenstates' in-medium ones. The relic mass-change loss costs the beam the relic's $\Delta m^2/2p_r^-$, its own state entering only through $(GG^\dagger)_{aa}$: right as it is. The relics' composition is unaffected: the relics are non-relativistic, their splitting $\Delta m \gg V$, and a vector potential on a Majorana neutrino at rest is suppressed by its velocity; wherever the potential could matter ($g/m_{Z_D} \gtrsim 10^{-3}\ \mathrm{eV}^{-1}$) their own vacuum decays are complete by $z = 1$ (exponent $\gtrsim 10^{20}$), and the flips do not change the occupation per mass state.

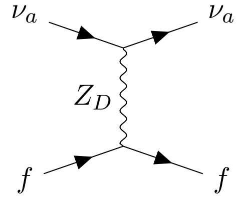
The drift pieces are gone. The dark-portal phase used to cut a step into pieces where an eigenstate's energy slid; that slide was the old model's self-decay, and nothing slides now. The slides left are physical: the jumps and the decays cycling (a stable state converted into a decaying one, again and again). Each bin's step solves them together exactly, and what is left of the step dependence is first order in the redshift step (the sources from the bins above held over a step), so more pieces per step do not help and more steps do.

Validation. On Tycho, the cosmological phase alone, against 3184 redshift steps; the shape is the ratio with its median divided out, the flavor the largest change of a flavor fraction, both over 10^{13}–10^{16} eV where the flux is at least 10^{-3} of the standard one. Where the potential sets the eigenstates (le_lmu_ul at $(m_{Z_D}, g) = (10^{-20}, 10^{-20})$, $(10^{-16}, 3 \times 10^{-17})$, $(10^{-12}, 10^{-11})$ eV, and five more), 200 steps are within 1.3×10^{-4} in shape and 3×10^{-3} in flavor (mostly $\leq 10^{-7}$), and twice the energy bins within 5×10^{-4}, the grid's own. Where the decays and the jumps cycle among nearly vacuum eigenstates and drain the flux to 10^{-3}–10^{-2} of the standard one (dp_e_ul and le_lmu_ul at $(10^{-8}, 10^{-8})$ eV, $m_l = 0.1$ eV), 200 steps leave 3–6×10^{-3} in shape and 400 steps $\leq 1.7 \times 10^{-3}$ (between the two regimes, $(10^{-9}, 10^{-9})$ eV at a tenth of the standard flux, 1.3×10^{-3} at 200 steps); where the potential is negligible the phase agrees with dp_si to 1.7×10^{-3}. A cosmological phase takes 4–8 s at 200 steps on one thread (600–750 s with the drift pieces). On the Galactic lines of sight, halving the segments changes the Galactic flux by at most 1.7×10^{-8} and the nearest rung differs from the interpolation by at most 4×10^{-5} (le_lmu_ul, three points); there the old model had drained the Galactic flux to 10^{-4}–3×10^{-2} of the standard one, the new one leaves 0.56–0.58 of it.

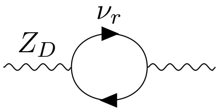
Decays the transport leaves out. With the exact widths, on the le_lmu_ul planes at 3×10^{16} eV ($g/m_{Z_D} \leq 60$ eV^{-1}): above the vacuum band ($m_{Z_D} > m_3 - m_1$, no decays in the code) a decay depth above 10^{-3} appears only at $m_{Z_D} = g = 0.1$ eV. From the Sun on, the Sun's potential (1 AU) needs $\theta = \sigma/\Delta m^2_{21} \geq 840$ of the same potential and the Earth's $\theta \geq 3.4 \times 10^5$: there the potential's coherent effects already change the flux completely, and the lowest states do not decay at all. The extragalactic beam crossing the Galaxy (up to 30 kpc, taken as coherent and decay-free in the transit, App. A.12) is the one place left, and there the decays matter. Measured (2026-10-25, `tools/galaxy_crossing.py`): the beam's populations at $z = 0$ (all spectral pieces) carried along four lines of sight (toward the Galactic centre and anticentre, a pole, $l = 90°$, $b = 30°$) through the local eigenstates with the in-medium decays, then through the Sun and the Earth, against the decay-free crossing, at $m_l = 0.01$ eV and $m_{Z_D} = 10^{-24}$–10^{-8} eV (up to 10^{-12} eV the same at fixed g/m_{Z_D}; at 10^{-8} eV and $g/m_{Z_D} \geq 1$ the cosmology drains the flux below 10^{-3} of the standard one, and the crossing still takes 27–46% of the beam's populations): at $g/m_{Z_D} = 0.1$ eV^{-1} the flux at 10^{13}–10^{16} eV drops by 13–41% at the top, its shape changes by 11–34% and a flavor fraction by up to 0.5; at $g/m_{Z_D} = 1$ by 1–17%, 1–16% and 0.15–0.19 (most toward the Galactic centre); at $g/m_{Z_D} = 10$ by 10^{-8}. So the beam does not arrive in the lowest states at $g/m_{Z_D} \lesssim 1$. Since 2026-10-25 the transport follows the crossing (`dark_portal_solver._flux_runs`): the populations at $z = 0$ of all the runs of a point, as the columns of one sweep, are carried along every line of sight of the Galaxy (the Galactic component's lines and processes, each energy bin exact over each distance bin), averaged per declination band, and handed to the Sun and the Earth band by band through the handover's responses to the three unit inputs (the handover is linear in its input); the whole sky is the bands weighted by their widths in $\cos\theta_z$, as the handover's own. Without processes this is the transit as before to rounding (`tests/test_galactic_path.py`); the decays alone keep the number.

13.9 Processes below 1 keV: the catalogue (2026-10-24)

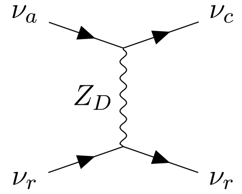
Every process the ultralight planes involve, with its diagram, what it does to the beam, where it lives in the code, and its status. Neutrino lines are mass states unless a flavor is written; G' are the couplings in the mass basis; the relics are Majorana. For the potential planes (dropped, Sec. 13.8) the matter lines would also couple through the Z mixing.



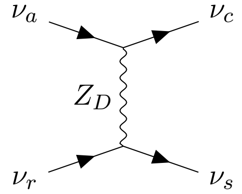
(a) Coherent forward scattering on matter ($t \to 0$). The potential from neutrons (dark portal through ϵ'; dropped) or electrons ($L_e - L_\mu$, $L_e - L_\tau$) in the Earth, Sun, Galaxy and the cosmos; flavor oscillations. *In the code*: dp_spatial, dp_cosmo (the cosmological potential), dp_transit.



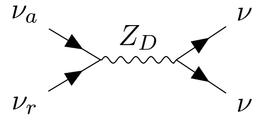
(b) Debye screening by the relics. The relic loop gives the mediator the mass $m_D \propto g_D T$ (Majorana: v-suppressed for slow relics); it cuts the range of (a) and the forward peak of (c). *In the code*: dp_absorption.debye_mass_sq.



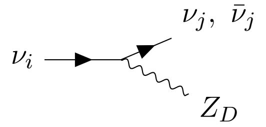
(c) t-channel scattering on a relic, $r \to r$. Soft ($|t| \lesssim M_t^2$, $\sim 10^{19}$ events at 10^{-9} eV, $g_D = 10^{-4}$): jumps between eigenstates, flavors equalized, no energy cost (Sec. 13.7). Hard: energy loss, mass-independent. Majorana relics add the crossed diagram, not forward-peaked. *In the code*: dp_cosmo._soft_jumps; dp_si.



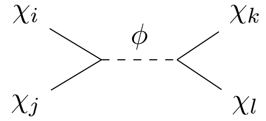
(c′) The same with a mass change, $r \to s$. Same forward peak, but the beam pays $\omega_{rs} = (m_s^2 - m_r^2)/2p_r^-$ per event: energy loss $\propto 1/E$ (Sec. 13.8, item 1). *Status*: **new**, to implement (dp_si drift; routing).



(d) s-channel absorption on a relic. The resonance at $s = 2Em_r = m_{Z_D}^2$: in the band only for $m_{Z_D} \gtrsim$ MeV; below 1 keV its energy is far below the band. *In the code*: dp_cosmo (resonant absorption), dp_si; the MeV planes.



(e) Decay, $\nu_i \to \nu_j + Z_D$ ($m_{Z_D} < m_i - m_j$). Majorana vertex $\gamma^\mu(G'_{ji}P_L - G'^*_{ji}P_R)$: half the daughters keep the helicity, half flip it. In the beam: the flux arrives as ν_1 for $g_D \gtrsim 10^{-7}$; among the relics: ν_1 only. The longitudinal mode dominates for $m_{Z_D} \ll m_i$ (Sec. 13.8, item 2). *Status*: **new**, to implement.



(f) The longitudinal mode as a Goldstone boson. Couplings $Y_{ij} = i(G'_{ij}m_i + G'_{ji}m_j)/m_{Z_D}$; $\chi\chi \to \chi\chi$ in the s, t, u channels sets where the models hold (perturbative unitarity, Sec. 13.8, item 3). Not a propagation process: it marks the not-allowed region. *Status*: `src/dp_unitarity.py`; marking on the plots to implement.

Left out, with reasons: the mediator's own decay back to neutrinos (only for $m_{Z_D} > 2m_1$, into the softest part of the spectrum; taken as invisible), relic–relic scattering (no effect on the beam beyond the relics' composition, which (e) sets), and processes through the Z mixing (the potential planes are dropped).

14 Open items

Residual in the phase-noise regime (resolved).—At "cosmoX2" the chord count moved the averaged fractions by $\sim 6 \times 10^{-4}$, and the Earth step count by 2.9×10^{-3}. We attributed this to phases that double precision cannot follow; it was the propagator's non-unitarity. Since 2026-10-12 the results with 12, 24 and 50 chords, and with 8, 25 and 100 Earth steps, agree to 10^{-14}

(Appendix A.10).

Analyses to rerun.—Profiles and limits computed before 2026-10-12 used a solver whose cascade ran away and whose absorption did not depend on the couplings (Appendices A.8 and A.9); they are void, not merely shifted. The new analysis package replaces the old pipeline, and its first production run is planned for Tycho.

Effect sizes to re-measure.—The effects quoted in Appendices A.1 to A.7 were measured with the solver of the time, before the corrections of Appendices A.8–A.10. They describe the change each correction made then; their size on the corrected solver is not known. [**Re-measure**]

Run time.—The production defaults ran in 0.32 s on 8 cores on 2026-07-29; on 2026-10-12 the same machine took 0.80 s before the solver corrections (commit `135ed02`) and 0.71 s after them (1.04 s on one core). [**VERIFY: find what changed between July and** `135ed02`]

Thermal-rate tails.—Far from resonance, where the rates are 10^{-6} to 10^{-10} of their peak, the semi-analytic kernel's results are dominated by rounding: a change in how numba compiles it (2026-10-13, numpy's error model for the fastmath kernels) moved them by up to 70%, while within 10^{-3} of the peak they agree to 3×10^{-12}. At "strong" this moves the flux in its absorption dip by 7×10^{-6} and the flavor fractions by 5×10^{-10}; the same happens between machines. [**Check whether those cells' optical depths matter anywhere in the planes**]

Numba on Tycho's Zen 4 nodes.—Compiled for the AMD EPYC 9554 with every fastmath flag, numba's division-by-zero check in the Earth's Yukawa evaluation fired spuriously (the division itself was correct), failing the first potential-plane jobs. The fastmath kernels now use numpy's error model (`src/dp_core.py`); the solver's own finiteness checks remain.

Up-going only (resolved 2026-10-13).—The average now covers the whole sky with equal weights (Sec. 4.5); the per-direction output of Sec. 13 will add the detector's acceptance.

Implementation queue.—The core-count fix is done (2026-10-12): pools are sized from the process's CPU affinity capped by the SLURM allocation, not from the machine's core count. Tycho is set up (2026-10-13, `slurm/README.md`): an environment pinned to the laptop's versions, job scripts for the tests, the four-point validation (`tools/validate_record.py`) and the analysis; the test suite passes there. Next, in order: the per-direction solver output (the whole-sky average is done), the muon channel, the starting-event channel with the systematic uncertainties, then the statistical tests of Sec. 13.3. The details are in `docs/dev/WORK_PLAN.md`.

A Corrections

A.1 The relic-neutrino temperature (2026-10-12)

The default $T_{\nu,0}$ was 1.95×10^{-4} eV, which is 2.26 K: the value "1.95 K" had been entered as if it were in eV. The correct value is 1.676389×10^{-4} eV, so the relic neutrinos were 16% too hot (commit `5ecf10e`). We measured the effect on the absorption rates directly. Above every resonance, where the relics act as if at rest, the old temperature raised the rates by a factor of 1.553–1.570 across the energy range at the "below" point, close to $(1.95/1.676389)^3 = 1.574$. Across a resonant dip it raised the energy-integrated absorption by 1.573, the depth of the dip by 1.35–1.38 (against $(1.95/1.676389)^2 = 1.353$) and its width by 5–16%. Over the energies and redshifts of the "inband" grid, where the shapes of the dips also depend on the temperature, the ratio ranged from 1.33 to 5.4.

Effect on the fluxes.—At the benchmark points the fluxes changed by up to a factor of ~ 130 at individual energies. At "below" and "cosmoX2", where the flux is strongly absorbed across the energy range, the median change was a factor of ~ 30, because a 57% change of the optical depth τ changes $e^{-\tau}$ by large factors once τ exceeds a few. Absorption features above the resonances moved down in energy, by 0.195 decades at "below" and "cosmoX2" (the relic-density factor, because the optical depth there scales as density over energy) and by ~ 0.15 decades at "inband". The resonant dips stayed at their energies (E_{res} does not depend on the temperature) but became shallower: the flux at the deepest "inband" dip rose by a factor of 3.6, at the deepest "strong" dip by 36%. The energy-averaged flavor fractions moved by at most 3.8×10^{-4} (below), 1.8×10^{-4} (inband), 7×10^{-9} (strong) and 6.7×10^{-5} (cosmoX2). Therefore, results of the flavor-ratio analysis obtained before the correction are almost unchanged, while spectra and event rates change substantially.

A.2 The cosmological and Galactic neutron densities (2026-10-12)

Cosmological.—The default $n_{n,0}$ was 2.5×10^{-7} cm^{-3}, documented as the neutron density but equal to the total baryon density (2.515×10^{-7} cm^{-3}, PDG 2024). The potential (T9) is sourced by neutrons alone: the vector couplings of the Z to protons ($\frac{1}{2} - 2\sin^2\theta_W$) and to electrons ($-\frac{1}{2} + 2\sin^2\theta_W$) cancel in neutral matter. The mediator Z_D inherits them through the mass mixing. Neutrons sit almost entirely in helium-4, which holds a fraction $Y_p = 0.2458 \pm 0.0013$ of the baryons (LBT Y_p project, 2026), half of them neutrons; deuterium adds $\sim 2 \times 10^{-5}$ per baryon. Therefore, $n_{n,0} = n_b Y_p/2 = 3.091 \times 10^{-8}$ cm^{-3}, and the cosmological potential was 8.1 times too strong.

Galactic.—The Galactic model converted the stellar mass density at half a neutron per nucleon (the factor $20.235 = 0.5 \times 40.464$ nucleons cm^{-3} per $M_\odot$ pc^{-3}) and the hot-halo density of Miller and Bregman (2013), which is an electron density, at 0.5 neutrons per electron. We now use 0.20 neutrons per nucleon for stars. Locally, stars and remnants amount to $33.4\,M_\odot$ pc^{-2}, of which 81% are visible stars, 15% white dwarfs and 4% brown dwarfs (McKee, Parravano and Hollenbach 2015). With $(Y+Z)/2 = 0.146$ for disc stars ($Y \approx 0.275$, $Z \approx 0.017$), 0.5 for carbon–oxygen white dwarfs and 0.142 for brown dwarfs, the weighted fraction is 0.20. For the hot gas ($Y \approx 0.26$, $Z \approx 0.3\,Z_\odot$, fully ionized) we use $(Y+Z)/2$ neutrons per $1-(Y+Z)/2$ electrons, i.e., 0.15. The Galactic potential was thus 2.5 times (stars) and 3.3 times (gas) too strong.

Effect.—At the four benchmark points neither change alters the results beyond their floors: "below", "inband" and "strong" agree to $\leq 4 \times 10^{-11}$ in flux and $\leq 10^{-13}$ in flavor fractions; "cosmoX2" moves by 1.4×10^{-3} pointwise and 1.0×10^{-4} in the energy-averaged fractions, its known sensitivity to the last digit. At mediator masses of order MeV the coherent potentials are $\sim 10^{-22}$ of the vacuum term, so only the resonant absorption acts. (Correction, 2026-10-13: this holds for the cosmological and Galactic potentials changed here. The Earth's own potential, sourced by densities 24 orders of magnitude higher, is $\sim 10^{-8}$ eV at the benchmark couplings, far above the vacuum term; see Sec. 6.) The corrections matter for light mediators, where the potential grows as $1/m_{Z_D}^2$; we compared the old and new code at three such masses, with the default couplings. At $m_{Z_D} = 10^{-2}$ eV the total flux changes by up to 32%, the flavor fractions at single energies by up to 0.074 and the energy-averaged fractions by up to 0.013 (f_τ: $0.315 \rightarrow 0.328$). At 10^{-5} eV the changes are 69%, 0.144 and 0.004. At 10^{-8} eV, where the potential is so large that the flavor evolution is fully averaged, they are 3%, 0.008 and 10^{-4}.

A.3 Neutron fractions along Earth chords (2026-10-12)

Along Earth chords the code counted half of the nucleons as neutrons in every PREM layer (3.011×10^{23} g^{-1}). For neutral matter the neutrons per nucleon are $1 - Y_e$, where Y_e is the electron (equal to proton) fraction. We now use $Y_e = 0.4656$ in the iron–nickel core and 0.4957 in the silicate mantle and crust, the values used with PREM in neutrino-oscillation tomography, and the composition of water (8/18.015 neutrons per molecular mass) for PREM's top layer. The core was thus 6.9% low, the mantle 0.9% low and the water layer 12.6% high. The regional boundaries coincide with PREM layer boundaries, so the phase-exact Earth steps stay exact (test `test_earth_neutrons_per_gram_by_layer`).

Effect.—The four benchmark points move: the energy-averaged ν_e and ν_μ fractions change by 0.025 (below), 0.026 (inband), 0.009 (strong) and 0.032 (cosmoX2), with the ν_τ fraction unchanged, and the flux at single energies by up to 21%. A change of a few per cent in the potential shifts the chord phases by $\sim 10^6$ rad, which redraws the residual conversion left by the 12-chord average. Adding one chord to the old code moves the same fractions by comparable amounts (Sec. 14). At light mediators the changes are smaller: up to 3.8×10^{-3} in the energy-averaged fractions at $m_{Z_D} = 10^{-2}$ eV and below 10^{-5} at 10^{-5} and 10^{-8} eV.

A.4 Point sources and cold disc gas (2026-10-12)

The Sun, the Moon and the Earth enter the Milky Way and Earth phases as point sources of a Yukawa potential, which reaches the detector only when the mediator's range, $1/\sqrt{m_{Z_D}^2 + m_D^2}$, spans the distance to the body. Their neutron numbers were inconsistent: 0.248 neutrons per nucleon for both the Sun and the Moon, 0.496 for the Earth. The Sun's present bulk composition is the protosolar one (X, Y, $Z = 0.7154$, 0.2703, 0.0142; Asplund et al. 2009) plus the $\sim 0.038\, M_\odot$ of hydrogen it has burned into helium, which we estimate from its luminosity integrated over 4.57 Gyr; that gives $Y = 0.308$ and $(Y + Z)/2 = 0.161$ neutrons per nucleon, 1.929×10^{56} in all. The Moon is silicate rock with a small iron core, 0.505 per nucleon, 2.232×10^{49}. For the Earth we integrate PREM with the per-layer fractions of the chords (Appendix A.3); PREM's mass comes out at 5.973×10^{27} g, within 0.02% of the Earth's, with 1.849×10^{51} neutrons (0.514 per nucleon).

The Galactic model had no cold gas. We add the HI and H_2 discs of McMillan (2017, arXiv:1608.00971), $\rho = \Sigma_0/(4z_d)\, e^{-R_m/R - R/R_d}\, \mathrm{sech}^2(z/2z_d)$ with $(\Sigma_0, R_d, R_m, z_d) = (53.1\, M_\odot\, \mathrm{pc}^{-2}, 7, 4, 0.085 \text{ kpc})$ and $(2180\, M_\odot\, \mathrm{pc}^{-2}, 1.5, 12, 0.045 \text{ kpc})$. Their surface densities already include helium and metals; at the Sun they give 10 and $2\, M_\odot$ pc^{-2}, against the $13.7\, M_\odot$ pc^{-2} that McKee et al. (2015) measure in all gas. We convert them at $(Y+Z)/2 = 0.142$ neutrons per nucleon (protosolar composition).

Effect.—At the four benchmark points nothing changes beyond the floors ($\leq 4 \times 10^{-11}$ in flux; "cosmoX2" 9×10^{-4} pointwise), because at MeV masses the point sources are out of range and the Galactic potential is negligible. At light mediators the changes appear at individual energies: at $m_{Z_D} = 10^{-2}$ eV up to 18% in flux and 0.064 in the flavor fractions, at 10^{-8} eV up to 3% and 0.008. We also ran $m_{Z_D} = 10^{-20}$ eV, where the bodies are within range: with Debye screening off the flux changes by up to 27% and the flavor fractions by up to 0.13; with $g_D = 10^{-12}$, by 1.6% and 0.004. The energy-averaged fractions move by at most 1.2×10^{-3} in all these runs.

A.5 The length of the Earth chords (2026-10-12)

A chord from the detector, at radius $R - d$, meets the surface after a length u that solves $u^2 + 2(R-d)\cos\theta_z\, u - d(2R-d) = 0$. The function `get_baseline_earth` dropped the terms in d^2, so for $d = 1.5$ km every chord but the vertical one started above the ground: by 8.1 m for the horizontal chord, 1.9 m at $\cos\theta_z = -0.09$ and less than a metre for steeper ones. For 10 of the 12 default chords that vacuum stretch was shorter than the margin with which the layer-crossing search discards the entry point (10^{-6} of the chord), so the step integration extended the top layer to the start of the chord. The phase-exact scheme remained step-count independent, since the same stretch was added for every step count, but it integrated up to 0.93 m too much water, a phase of up to 0.4 rad at "below". We now solve the quadratic exactly, in a form without cancellation for either sign of $\cos\theta_z$; every chord starts on the surface to 3×10^{-12} km (test `test_exact_chord_baseline_starts_on_the_surface`).

Effect.—The energy-averaged flavor fractions change by 9.1×10^{-4} (below), 4.5×10^{-5} (inband) and 1.5×10^{-5} (strong), the flux at single energies by at most 8.7×10^{-3}, 4.1×10^{-4} and 4.5×10^{-5}, with the dips unchanged. "cosmoX2" moves by 4.1×10^{-3} in the averaged fractions, a redraw of its phase noise. Among the light mediators only $m_{Z_D} = 10^{-2}$ eV, also phase-noise dominated, moves visibly (8.5×10^{-3}); the ultralight points do not change.

A.6 The Earth's own potential (2026-10-12)

Until 2026-10-12 the Earth chords used the local density, the limit of Eq. (1) when $1/m$ is far below every length over which PREM varies. A mediator with a range of 200 m ($m \approx 10^{-9}$ eV) already sees the layer boundaries smoothed; once the range rivals the Earth, the potential is set by the enclosed neutrons and the local form overstates it by $\sim 1/(mR)^2$, 10^{13} at $m = 10^{-20}$ eV.

Evaluation.—Writing $f = rn$, Eq. (1) becomes $n_{\text{eff}} = (m/r)\,[P + (1 - e^{-2mr})Q/2]$, with $P(r) = \int_0^r f(r')\,\frac{1}{2}(1 - e^{-2mr'})\,e^{-m(r-r')}dr'$ and $Q(r) = \int_r^R f(r')\,e^{-m(r'-r)}dr'$, where every exponential decays; e^{mR} itself would overflow at any MeV mass. For $mR < 50$ we accumulate P outward and Q inward on subintervals no longer than $1/m$, with 10-point Gauss–Legendre quadrature, which is exact for PREM's polynomials to rounding. For $mR \geq 50$ we integrate by parts, again exactly for polynomials: inside a shell $(a, b]$, $n_{\text{eff}} = n + (f''/m^2 + f''''/m^4)/r + (m/r)[\delta\,e^{-m(r-a)} + \gamma\,(1 - e^{-2mr})\,e^{-m(b-r)}/2]$, with one pair of constants δ, γ per shell. The last term carries the boundary layers; away from them every exponential underflows and n_{eff} equals n to $(mR)^{-2}$. At the benchmark masses the result is the local density bit for bit.

Tests.—The suite checks the two evaluations against the closed form for a uniform ball from $mR = 10^{-7}$ to 10^8 (relative 10^{-12} for the tabulated one, 10^{-13} of the density for the other), against direct quadrature of the shell kernel for PREM (ranges from $R_\oplus$ to 200 m, 10^{-9}), against each other across the switch, and against the enclosed-charge limit for $mR = 10^{-12}$. Outside, the integral is a point source of $4\pi \int r^2 n\, \sinh(mr)/(mr)\, dr$ neutrons, which equals the Earth's 1.849×10^{51} to 10^{-3} where the Milky Way phase can reach it. Along the chords, the step-summed integral is the same for 8, 25 and 1000 steps to 10^{-12}, with ranges from 200 m to 2×10^6 km. The boundary layers carry a floor of their own. Near a layer the radius is known only to $\sim 10^{-12}$ km, so a layer of thickness $1/m$ is integrated to a relative accuracy of $\sim m \times 10^{-12}$ km (5×10^{-6} at 10^{-3} eV), an error below 10^{-12} of the step it lies in. Layers thinner than 10^{-10} of a step are left unresolved, an error below 10^{-10} of that step's phase.

Effect.—At the four benchmark points the fluxes are identical, bit for bit, to those of the chord-length fix alone, as are the step densities of all 12 chords. At light mediators, $m_{Z_D} = 10^{-2}$ eV moves by 2.7×10^{-3} in the averaged fractions (phase noise), 10^{-5} eV by 1.6×10^{-6} and 10^{-8} eV by 5×10^{-6} (flux at single energies up to 2%). The ultralight points do not change: there the Galactic potential is already so large that the state reaching the Earth is fully averaged in the basis of the potential, whose texture the Earth's potential shares. To see the effect where the Earth matters, we ran three points with ranges of $R_\oplus$, 100 km and 1 km at fixed $g_D\epsilon'/m^2$ ($m_{Z_D} = 3.1 \times 10^{-14}$, 1.97×10^{-12} and 1.97×10^{-10} eV; $g_D\epsilon' = 10^{-49}$, 4×10^{-46}, 4×10^{-42}). In the local form all three accumulate 52 rad along the vertical chord and 1.6 rad at $\cos\theta_z = -0.09$; the Yukawa integral reduces these to 8.7 and 0.4 rad at the range $R_\oplus$, to 52.0 and 1.0 rad at 100 km (the grazing chords run through the thin top layers), and leaves them at 1 km. With no potential the averaged fractions are $(0.3037, 0.3543, 0.3420)$; the local form shifts f_e by $+0.0224$ at all three, because it depends only on $g_D\epsilon'/m^2$. The Yukawa integral gives $+0.0240$, $+0.0207$ and $+0.0224$: the Earth's effect changes by 7–8% once the range reaches 100 km. At these points 25 and 200 steps agree to 1.8×10^{-10} in flux and 2.6×10^{-11} in the fractions.

A.7 The zenith average of the Earth chords (2026-10-12)

The flux at the detector is the average over the up-going hemisphere, uniform in $\cos\theta_z$, which we take from a finite set of chords. Wherever the dark-portal phase across the Earth matters, the potential dominates the vacuum term: a phase of 1 rad over the Earth needs $V \sim 10^{-14}$ eV, against $\Delta m^2/2E \lesssim 10^{-16}$ eV above 10 TeV. Therefore a chord evolves as $U(c) = \sum_g e^{-i\lambda_g \Psi(c)} P_g$, with P_g the eigenprojectors of the texture (eigenvalues λ_g) and $\Psi(c) = \text{Coef} \int n\, dl$ the chord's column of potential, and each block of the evolved density matrix is $P_g \rho P_h = A_{gh}(c)\, e^{-i\omega_{gh}\Psi(c)}$, $\omega_{gh} = \lambda_g - \lambda_h$, with an amplitude A_{gh} that varies slowly with c (the vacuum term's own phases across the Earth are below 10^{-2} rad). At MeV masses Ψ reaches 10^5–10^8 rad, so the cross blocks have effectively random phases from chord to chord, and the equal-weight mean we used kept them as sampling noise. Adding a thirteenth chord to twelve moved the averaged fractions by up to 0.048 (commit `0db8111`), $\sim 10^{-2}$ at the analysis's 50 chords, which exceeds the projected 2050 width of f_μ (≈ 0.005). The phase does not depend on energy, so the noise did not average out over the spectrum.

Method.—We now integrate each block against its known phase. A_{gh} is interpolated linearly between the chords, and the oscillatory factor is integrated exactly on a fine grid of $\Psi(c)$ with the phase linear on each subinterval (a Filon-type rule). The result is a complex weight per chord and block, $w_k^{gh} = \int \mathrm{hat}_k(c)\, e^{-i\omega_{gh}[\Psi(c)-\Psi(c_k)]}\, dc$, the same for every energy. The fine grid has 4096 chords, with points added toward the cosines at which a chord starts to cross a PREM boundary, where Ψ' diverges; their columns take ~ 10 ms once per process. For $\omega = 0$ the weights are the trapezoid rule. Where the phase turns many times between neighbouring chords they fall to $\sim 1/(\omega\Psi')$, and the average becomes the decoherent $\sum_g P_g \rho P_g$ that the continuum of zeniths produces. In between they resolve the oscillation that a chord sum aliases. The rule needs the phase-exact steps: under the historical left-endpoint scheme, which keeps the equal-weight mean, each chord's phase carries a Riemann error of thousands of radians.

Tests.—For chords evolved exactly as $e^{-i\Psi S}$, the average of 12 or 13 chords matches the integral over $\cos\theta_z$ to $\lesssim 10^{-5}$ for Earth phases from 0.1 to 10^8 rad (a dense trapezoid over 200,001 chords up to 10^3 rad, the block-diagonal limit at 10^8 rad; at 10^5 rad the continuum keeps a cross term of 3×10^{-4}, which the weights reproduce to 10^{-5}). The equal-weight mean misses by 10^{-3}–10^{-1}. The suite also checks the trapezoid limit, Hermiticity for a full-rank complex texture, and the options.

Effect.—At production settings, 12, 13 and 25 chords agree with 50 to 6×10^{-9} (below), 1.4×10^{-7} (inband) and 5×10^{-8} (strong) in the energy-averaged fractions; with the equal-weight mean, 12 and 13 chords differed by 4.1×10^{-3}, 1.2×10^{-3} and 5.6×10^{-4}. The step-count certificate tightens to $\leq 1.2 \times 10^{-10}$ in flux. Against the equal-weight mean of 12 chords, the fractions move by 0.029 (below), 0.018 (inband) and 0.0057 (strong). At "below" they become $(0.3192, 0.3192, 0.3617)$, against $(0.2904, 0.3480, 0.3617)$: with $u_e = u_\mu$ the random Earth phases mix ν_e and ν_μ equally, and the asymmetry was noise. At $m_{Z_D} = 10^{-2}$ eV the chord count no longer matters at all (10^{-16}), and in the coherent window of Appendix A.6 the fractions move by 10^{-3}–4.5×10^{-3} and agree across chord counts to 5×10^{-9}. "cosmoX2" keeps a residual of $\sim 6 \times 10^{-4}$, down from 5×10^{-3}, for the reason given in Sec. 14. A solver run costs the same as before. (The "cosmoX2" residual was the propagator's non-unitarity and is gone since commit `2066f3b`, Appendix A.10.)

Per declination band (2026-10-13).—The per-direction output (`docs/dev/DIRECTIONS_PLAN.md`) needs the flux per band of declination, which at the South Pole is a band of $\cos\theta_z = -\sin\delta$. A single chord is not a per-direction answer, since it carries its own fast Earth phase; the band's average is. The Filon weights above are built interval by interval between chord nodes, so the weights of a band are those of its intervals, divided by its width (`average_earth_chords_bands`); with the band edges on chord nodes (`earth_chord_nodes`: 20 bands of 0.1 in $\cos\theta_z$, two intervals each, 41 chords against 23) the bands weighted by their widths return the whole-sky average to rounding (10^{-14}, at Earth phases from 0.1 to 10^8 rad). Each band equals the exact integral over its own range of the potential-dominated evolution to 10^{-4} (dense trapezoid, 20,000 directions per band, phases up to 10^3 rad); at 10^8 rad the up-going bands reach the decoherent limit $\sum_g P_g \rho P_g$, while the bands near straight down, whose chords cross only 1.5–15 km of ice, keep a residual coherence of order the inverse of their phase spread (131–979 rad there), which the rule reproduces against the dense integral. Without the dark portal every band equals the whole sky (isotropy). `tests/test_directions.py`.

A.8 The cosmological cascade (2026-10-12)

Each absorption at an energy E_j re-emits a neutrino at a lower energy E with the spectrum $3E^2/E_j^3$. Within a redshift step the code held this regenerated source at its value at the start of the step, ΓF_n. Near a resonance the optical depth of one step, $\Gamma\Delta t$, reaches 10^{10}, and it exceeds 1 in 70% of the (z, E) cells at "inband". A bin then re-emitted up to $\Gamma\Delta t$ times more neutrinos than it absorbed. The excess compounded from step to step: at "inband" the energy reaching the Earth was 5×10^{36}, 8×10^{60} and 6×10^{66} times the no-interaction value with 100, 1000 and 10,000 redshift steps (at "strong", 6×10^{9} to 10^{15}). The midpoint weights of the kernel also re-emitted $1+1.5\,\Delta\ln E$ neutrinos per absorption, 1.2% too many per generation at 1000 bins. The flavor fractions, dominated by the runaway secondaries, stayed near 1/3 and looked plausible. Every earlier validation compared the code with a previous version of itself, which shared the error; the first check against conservation found it.

Fix.—We now sweep the energy bins of each step from high to low. A bin solves $dF/dt = -\Gamma F + S$ exactly over the step, with S the injection plus the regeneration from higher bins, computed from the number those bins actually absorbed during the step, $F_n(1-e^{-\Gamma\Delta t})+S\,[\Delta t-(1-e^{-\Gamma\Delta t})/\Gamma]$. A bin's regeneration into itself is solved implicitly. The kernel is integrated exactly over the bins (commit `7dc95e9`). A step conserves the number of neutrinos to rounding for $\Gamma\Delta t$ from 10^{-8} to 10^{10}, and the solver never returns more energy than it injects; tests check both.

Effect.—With the absorption of the time, the energy reaching the Earth fell to 3.0×10^{-3} (below), 3.3×10^{-3} (inband), 0.11 (strong) and 3.0×10^{-3} (cosmoX2) of the no-interaction value. Absorption that strong at couplings of 10^{-4} is implausible; we traced it to the normalization of Appendix A.9.

A.9 The normalization of the resonant absorption (2026-10-12)

The rates integrate a Breit–Wigner cross section over the relic distribution. The theory notes give its width, $\Gamma = (\Sigma G)M/(12\pi)$ (T11), and the couplings ΣG (T12), but not its numerator. The code used $\sigma_\alpha = 2\pi(\Sigma G)_\alpha s/[(s-M^2)^2+M^2\Gamma^2]$, with a single coupling factor. Its integral over s, $12\pi^2(\Sigma G)_\alpha/\Sigma G$, does not depend on the size of the couplings: over the benchmark grid the integrated rate was 1.870×10^{-17} eV whether ΣG was 7.6×10^{-4} or 3×10^{-19}. Any nonzero mixing, however small, absorbed as strongly as $u = 0.5$; only $u = 0$ exactly recovered standard propagation.

Fix.—We decided (2026-10-12) on the Breit–Wigner cross section of a vector resonance formed by $\nu_L + \bar\nu_R$, summed over final states,

$$\sigma_\alpha = \frac{48\pi\,\Gamma_{\mathrm{in},\alpha}\,\Gamma\,s}{M^2[(s-M^2)^2+M^2\Gamma^2]} = \frac{(\Sigma G)_\alpha(\Sigma G)\,s}{3\pi[(s-M^2)^2+M^2\Gamma^2]}, \qquad \Gamma_{\mathrm{in},\alpha} = \frac{(\Sigma G)_\alpha M}{12\pi}, \tag{29}$$

whose integral over s is $4\pi(\Sigma G)_\alpha$; for the Z it gives the classic $\langle\sigma\rangle = 2\sqrt{2}\pi G_F$. The rates are the old ones times $\Sigma G/(6\pi^2)$, applied where the grids leave `src/dp_thermal.py`, so the coupling cache keeps working on the unnormalized grids (commit `944360d`). We test that the integrated rate is proportional to ΣG at fixed flavor ratios and vanishes at $u = 0$, the narrow-width limit against an independent quadrature (to 2×10^{-3}), and that $u = 10^{-10}$ reproduces the $u = 0$ flux to 10^{-8}.

Effect.—With the cascade fixed, the energy reaching the Earth is 0.999 (below), 0.841 (inband) and 0.982 (strong) of the no-interaction value (Sec. 5). In July we had rejected rescaling the rate grid across coupling scans because the rates at the resonance did not scale with the couplings; that independence was this error.

A.10 The unitarity of the spatial propagator (2026-10-12)

The propagator evaluated $U = e^{-iH\Delta x}$ as a polynomial in the traceless part of H, through Newton divided differences of $e^{-i\lambda\Delta x}$ at the computed eigenvalues. A polynomial evaluated at rounded eigenvalues is unitary only to $\sim \varepsilon\, \Delta\lambda\, \min(\Delta x, 2/\delta\lambda)$, with $\Delta\lambda$ the spread and $\delta\lambda$ the smallest gap of the eigenvalues. Under the potential of "cosmoX2", $\sim$ 70 eV in the Earth's core, the error reached 5×10^{-3} per Earth step: the Milky Way and Earth transits changed the total flux of an energy bin by -0.5% to $+0.7\%$. The energy reaching the Earth exceeded the no-interaction value (1.0003), which absorption alone cannot do. A guard added in July for $m_{Z_D} \sim 10^{-6}$ eV, where the error grew to $\sim 10^{11}$ and overflowed to NaN, switched to an exactly unitary form only once the error exceeded 0.1. The polynomial also read both triangles of an H that is Hermitian only to rounding. For single-flavor mixing ($u_\mu = 0$ or $u_e = 0$) above $\sim 10^7$ eV, its basis for nearly degenerate pairs divided by zero.

Fix.—We now evaluate U as the spectral sum over an eigenbasis that stays orthonormal to rounding at any gap (Sec. 4; commit `2066f3b`). It is unitary to 2×10^{-15} over 12,880 production textures, with potentials from 10^{-24} to 10^{22} eV and steps up to 6×10^{32} eV^{-1}, and over 20,000 random spectra; the polynomial reached errors of order one in the same sweep. Against a 50-digit matrix exponential we find it as accurate as the polynomial where the latter was sound. A pair whose splitting is below $16\varepsilon(|q|+p)$, with q and p the mean and the scale of the eigenvalues, evolves as degenerate. Rounding when H is formed already splits an exact degeneracy by up to $3.1\varepsilon(|q| + p)$; evolving that splitting would rotate the pair by tens of radians per Earth step. The new form costs 0.6–0.7 times the old one per Earth step and 1.25 times per Milky Way step.

A stale compilation.—Our first check of the fix returned the old totals and the old division by zero. numba validates a cached function against its own source file only, so the spatial kernels, which call the propagator from another file, had kept its old machine code. We now key the cache by a digest of every source file (`src/dp_core.py`), so any edit recompiles everything.

Effect.—The spatial phase conserves every total to 10^{-14}. At "below", "inband" and "strong" the results are unchanged to five digits. At "cosmoX2" the energy reaching the Earth falls from 1.0003 to 0.99934 of the no-interaction value. The fractions move from $(0.32917, 0.32917, 0.34166)$ to $(0.32899, 0.32899, 0.34201)$. The point no longer depends on the last digit, on the number of chords or on the number of Earth steps (Sec. 5 and 14).

A.11 Both secondaries of every absorption (2026-10-13)

An absorption $\nu(E_j) + \bar\nu_{\text{relic}} \to Z_D \to \nu + \bar\nu$ emits two particles. The Z_D is polarized along the incoming neutrino, so the outgoing neutrino carries the spectrum $3E^2/E_j^3$ (on average $\frac{3}{4}$ of E_j) and the antineutrino $3(E_j - E)^2/E_j^3$ (the remaining $\frac{1}{4}$); an absorbed antineutrino gives the mirror image. The code kept only the secondary of the absorbed particle's own kind and propagated neutrinos and antineutrinos independently, as Eq. (T6) of the theory notes does, which regenerates ρ only from ρ. Half of the regenerated particles and a quarter of the absorbed energy were dropped.

Fix.—We now evolve neutrinos and antineutrinos together through the cosmological phase. Within a redshift step, each energy bin receives both kinds of secondaries from the absorptions above it; the second kernel, integrated exactly over the bins, separates into three running sums over the parents, and a bin's own secondaries of both kinds make each bin a 2×2 linear system, which we solve directly. The Milky Way and Earth transits stay separate for the two kinds. A step balances the number of particles to rounding for $\Gamma\Delta t$ from 10^{-8} to 10^{10}: every absorption adds one particle, less the secondaries that fall below the grid (a test). The analysis evolves the configured sources in one coupled run, and the responses to neutrino and antineutrino sources separately only when their ratio floats.

Effect.—At production settings the energy reaching the Earth rises from 0.841 to 0.941 of the no-interaction value at “inband” and from 0.982 to 0.99995 at “strong”, and the number from 0.955 to 1.092 and from 0.99993 to 1.00074: what is left is the energy of secondaries below 10 TeV. The energy-averaged fractions move by 1.8×10^{-3} at “inband” (f_τ: $0.33801 \to 0.33621$) and by $\leq 2\times10^{-5}$ at the other points.

On the theory notes.—Section 4 of the theory notes asks for a factor $\frac{1}{2}$ in the absorption rate for identical initial-state background particles. We do not apply it, and we think the sentence should be reworded: the rate of one high-energy neutrino counts each relic it can annihilate with once (one helicity state, 56 cm^{-3} per mass state), which is Eq. (T7) as written; a factor $\frac{1}{2}$ belongs to final states integrated over all angles, where it is part of the width (T11), and to collision rates within one gas of identical particles. Equation (T6) should include the regeneration of ρ from $\bar\rho$ described above.

A.12 The transit to the Earth (2026-10-13)

The potential planes of Sec. 6 scan the coupling product $P = g_D\epsilon'|U|^2$ down to ultralight mediators. Their lower-left corner scattered: at $(m_{Z_D}, P) = (10^{-27}\ \mathrm{eV}, 10^{-28})$ the χ^2 was 1.83 with the default six Milky Way directions, 14.49 with 50 and 15.07 with 200. We traced the scatter, and the physics around it, to four errors in the transit from the cosmological arrival to the Earth, and to a fifth in the cosmological potential.

Precision.—At large couplings the potential exceeds the vacuum term by up to 10^{27}. The rank-one texture $uu^\dagger$ has two zero eigenvalues, split only by the vacuum term, and any method that forms the Hamiltonian first (the closed-form eigenvectors, LAPACK) computes its eigenvalues with an absolute error $\varepsilon\sigma$: above $\sigma/M \sim 10^{13}$ the two states orthogonal to u were rounding noise. The noise entered the cosmological mixing table and the start of every Milky Way path, and the average over directions converged on the average of the noise. We now diagonalize by Jacobi rotations in the texture’s eigenbasis, where the large entries sit on the diagonal and the degenerate eigenvalues are exactly zero (`eigh_graded`): eigenvalues and eigenvectors agree with 60-digit arithmetic to 3×10^{-15} up to $\sigma/M = 10^{40}$, where LAPACK is wrong already at 10^{16} (a test, for rank-one, full-rank and degenerate textures).

The Galactic potential.—The Milky Way phase used the local density at every mediator mass. That is the Yukawa integral only for ranges short against every length over which the density varies; for longer ranges the potential at the Sun is set by the neutrons within the range, and the local form overstated it by 5×10^{6} at 10^{-30} eV, 750 at 10^{-28} eV, 19 at 10^{-27} eV and 3 at 10^{-26} eV. We now evaluate $n_{\mathrm{eff}}(m) = \int_0^\infty m^2 r\, e^{-mr}\, \bar n(r)\, dr$, with $\bar n(r)$ the model’s density averaged over the sky at distance r from the Sun, tabulated once by composite Gauss–Legendre quadrature graded toward the plane and the Galactic Centre (accurate to 4×10^{-9} against a finer rule; `tools/mw_sky_density.py`), plus the cosmic mean beyond the Galaxy, where the halo falls below it (336 kpc). The model’s exponential discs have a cusp at $z = 0$, where the Sun sits, so n_{eff} approaches the local density only as $1/m$; above 10^{-12} eV it is the local density itself, continuous to 1.5×10^{-14}.

Adiabaticity.—The Galactic potential changes over ≥ 10 pc, while the oscillation length is ≤ 0.005 pc even at 30 PeV, so the transit is adiabatic: each eigenstate keeps its population. The 25 left-endpoint steps of 4 kpc turned every step boundary into a sudden jump of the eigenbasis, an error that fell only as the inverse step count: at $(10^{-12}\ \mathrm{eV}, 10^{-24})$ the χ^2 was 6.6, 10.6 and 13.4 with 25, 200 and 1000 steps. We now map the arriving populations onto the local eigenstates directly, which is exact in the adiabatic limit and independent of the direction. At “cosmoX2” ($m_{Z_D} = 40$ eV, where the Galactic potential is up to 0.7% of the solar vacuum term) the old method needed 50 directions to converge onto the new result (4×10^{-6}); at its defaults it was 1.4×10^{-3} off pointwise.

The Sun and the Earth’s exterior.—Per unit P at the Earth, the Sun’s potential exceeds the Galaxy’s above $\sim 10^{-26}$ eV, as long as its range spans an AU ($\lesssim 10^{-18}$ eV), and the Earth’s own potential dominates above that. The Milky Way phase sampled the Sun only at its 4-kpc steps;

the Earth phase then switched it on as a constant along the chords, and the chords started at the Earth's surface, without its exterior. Both potentials turn on over distances comparable to the oscillation lengths (0.07–6600 AU between 10 TeV and 30 PeV; the Earth's $1/r$ tail reaches hundreds of Earth radii before it falls below the vacuum term near the planes' boundary), so their transit can be adiabatic, sudden or in between. We integrate it in the instantaneous eigenbasis. The eigenvectors of $(M+\sigma S)/2E$ depend on σ only, so one table serves every energy and path, with nodes no more than $\Delta\theta = 0.002$ of eigenbasis rotation apart; across a step the dynamic phases are integrated exactly and the non-adiabatic coupling K/h, $K = \log(W_n^\dagger W_{n+1})$, against the phase difference (first-order Magnus with a Filon rule, exact in the adiabatic and sudden limits). Against a fourth-order Magnus integration that resolves every oscillation (up to 7×10^6 steps) through a $1/r$ potential that is neither adiabatic nor sudden, the results agree to $\sim 10^{-4}$ at $\Delta\theta = 0.01$ and $\sim 10^{-5}$ at 0.002, falling linearly with $\Delta\theta$, in 0.2–1 ms per path instead of seconds (tests). The Sun's transit is averaged over eight Gauss–Legendre cosines of the angle to it; the Earth's exterior beyond $2R_\oplus$ is one radial path for all chords, inside it each chord's own line in eight steps, and each chord adds its line's extra column beyond $2R_\oplus$ as a phase. The Moon, at most 2×10^{-4} of the Earth's surface potential, stays a constant background.

The cosmological potential.—`calc_V_LRI_cosmo` and the mixing table used $1/\max(m_{Z_D}^2 + m_D^2, 10^{-25}\ \text{eV}^2)$, a guard against division by zero added on 2026-07-20 with the Frequentist-Analysis adaptation, which capped the cosmological potential for every mediator lighter than 3×10^{-13} eV (10^{35} times too weak at 10^{-30} eV). The decided replacement is the uniform medium's Yukawa integral cut at the Hubble radius, $(1-(1+x)e^{-x})/m^2$ with $x = mR_H(z)$: $1/m^2$ above $\sim 10^{-32}$ eV, $R_H^2/2$ below, with no division by zero.

Convergence.—On the $e\mu$ potential plane, at points chosen where each source dominates near the boundary of the excluded region and at the old scattered points, χ^2 moves by at most 0.03 from 8 to 32 directions to the Sun and from $\Delta\theta = 0.002$ to 0.001, by at most 0.05 from 400×200 to 800×400 energy and redshift bins, and by at most 0.002 when the split radius doubles and the exterior steps quadruple. The old scattered points are now plateaus: $\chi^2 = 57.19$ at both $(10^{-27}\ \text{eV}, 10^{-28})$ and $(10^{-25}\ \text{eV}, 10^{-28})$, and 14.36 at $(10^{-12}\ \text{eV}, 10^{-24})$, under every setting.

Effect.—At the benchmark points below, inband and strong the fluxes agree with the previous solver to 3×10^{-12}; at cosmoX2 they move by 1.4×10^{-3} pointwise and the fractions by 1.8×10^{-8}, the old method's own error (above). The potential planes are recomputed in Sec. 6.

What this does not model.—The cosmological phase uses the cosmic mean density everywhere, including at the sources, whose own environment would set the projection of the injected flavors for short ranges. The Local Group beyond the Galaxy (M31 adds $\sim 1\%$ for ranges beyond 1 Mpc) is not included. The Sun's and the Earth's averages are taken independently, as if the Sun's direction and the zenith angle at the South Pole were uncorrelated. Inside the Earth, under a potential 10^{14} times the vacuum term, the propagator still evolves the two states orthogonal to the potential as degenerate (Sec. 4), a phase error of at most 8×10^{-3} rad across the Earth at 10 TeV.

A.13 The sources' redshift weighting (2026-10-13)

Sources of comoving density $R(z)$ (the star-formation rate) that emit $\Phi_0 E_s^{-p}$ neutrinos per unit source energy and time give today

$$n(E) = \int_0^{z_{\max}} dz\, \frac{R(z)\, \Phi_0\, [E(1+z)]^{-p}}{H(z)}\,, \tag{30}$$

because the time element $dt = dz/[(1+z)H]$ and the energy interval, narrower by $1+z$ at arrival, cancel. Both transports work on a grid of observed energy, where the injection per unit observed energy must carry $dE_s/dE = 1 + z$. The main solver (`dp_cosmo.step_cosmological_pair`)

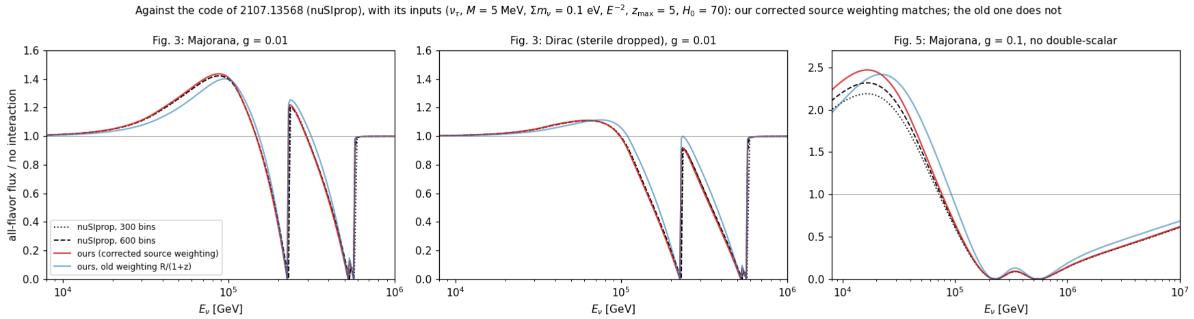


Figure 29: **Against nuSIprop**, the code of 2107.13568, with its inputs: its Fig. 3 (Majorana and Dirac, $g = 0.01$) and Fig. 5 ($g = 0.1$, without double-scalar production), nuSIprop at 300 and 600 bins against our transport with the corrected source weighting and with the old one.

had omitted it since its first version, and `src/dp_si.py` copied it, so a source at z was weighted by $R(z)/(1+z)$. Fixed in commit `124fd03` (the $1+z$ in both injections, their references and tests, and two new tests: the free-streaming flux against the closed form above, and the main solver's ratio of $z_{\max} = 4$ to $z_{\max} = 1$, 16% off before).

How it was found.—The 9% difference with Fig. 3 of 2107.13568 (Sec. 7.4) survived every check of the Dirac physics. Their code, nuSIprop, run on Tycho, reproduces their figure, and with the standard weighting (equivalent to our `rho_evol = 1` before the fix) our transport matches it, Majorana and Dirac alike (Table 9, Fig. 29).

Independent checks.—(i) A Monte Carlo of 2×10^7 neutrinos emitted in cosmic time at the rate $R(z)$ and redshifted one by one: the emission redshifts of those arriving in one energy bin follow the standard weighting ($\chi^2/\mathrm{dof} = 1.05$, against 1681 for the old one; mean emission redshift 1.066, against 1.065 and 0.823, for E^{-2}). (ii) Free streaming destroys nothing: the standard form returns every emitted neutrino (ratio 1 to 10^{-12}), the old one 55% of them. (iii) The transport equation of arXiv:2001.04994 (its Eq. 2, $\partial_t \tilde{n} = \partial_E(HE\tilde{n}) + \mathcal{L}$, $\mathcal{L}$ per unit source energy) integrates to the standard form. (iv) Waxman and Bahcall's ξ_z for E^{-2} sources, $\simeq 3$ for star-formation-like evolution: 2.33 with the standard weighting (our $R(z)$ to $z = 4$), 1.28 with the old.

Table 9: **Fig. 3 of 2107.13568** (ν_τ couplings, $M = 5$ MeV, $g = 0.01$, $\Sigma m_\nu = 0.1$ eV, E^{-2}, their inputs: $z_{\max} = 5$, $H_0 = 70$, NuFIT 5.0): all-flavor flux over the free one at its peak below the dips and just above the first dip.

	Majorana peak	Majorana, above dip	Dirac peak	Dirac, above dip
Paper (read off)	1.418	1.205	1.110	0.914
nuSIprop (their code, 300 and 600 bins)	1.423	1.207	1.109	0.917
Ours, weighting $R(z)/(1+z)$ (before)	1.408	1.241	1.114	0.993
Ours, standard weighting	1.442	1.204	1.111	0.911

At $g = 0.1$ (their Fig. 5; without double-scalar production) our corrected transport agrees with nuSIprop extrapolated in the bin width to 1% from 10 TeV to 10 PeV; the old weighting was off by up to 35%.

Effect.—Without interactions the flux is a pure power law either way, and the flavor composition does not depend on the weighting; with them, weight moves to higher redshift, where the relic density, the cosmological potential and the resonance crossings differ. Every plane was recomputed (Sec. 7.6 and Sec. 6); the secret-interaction limits became stronger by $\sim 10\%$ (the median ratio of new to old per mass column is 0.81–0.93 across the 19 planes; best limits, e.g., ν_e $1.7 \to 1.5 \times 10^{-5}$, universal $8.8 \to 8.0 \times 10^{-6}$), with single columns near the window's allowed channels moving by large factors either way; the dark portal's potential planes and its exclusion plane are unchanged in the median (single columns up to a factor 2), its MeV planes 3–23% stronger. `output/analysis/` (Tycho jobs 66307194–99, commit `124fd03`).

A.14 The Galactic gas at $N_{\rm side} = 64$ (2026-10-14)

The first track rates (Section 9) showed no Galactic flux for the dark portal and standard physics, while the secret-interaction case, whose Galactic flux needs no emission points, had it. The gas columns of the emission model (Söding et al.) come natively at $N_{\rm side} = 64$ in Galactic coordinates; the reduction averaged the native pixels whose centres fall in each equatorial pixel, which at the same resolution, rotated, leaves 12% of the pixels with none: 0/0. The check that every line of sight carries emission tested total ≤ 0, which NaN passes, and the emission populations then dropped every density node such a pixel touched, leaving 17 of 49 152 pixels with Galactic flux. Fixed: as fine as the native maps, the reduction interpolates the native columns at each pixel's centre (bilinear on the sphere); coarser (the committed $N_{\rm side} = 16$, unchanged bit for bit), it averages as before. Both checks now reject non-finite values (`tests/test_galactic.py`). The flavor fits, at $N_{\rm side} = 8$ from the $N_{\rm side} = 16$ data, were never affected.

A.15 The mixing-parameter priors (found 2026-10-12)

The Gaussian priors in the `fallbacks` block of the analysis configurations have never been applied, for three reasons. First, the configurations name the shape `"gaussian"`, while the code accepts only `"gauss"`. Second, the entries `dcp` and `dm31_sq` do not match the parameter names `d_cp_deg` and `dm31_sq`. Third, the prior centres disagree with the fixed values ($\delta_{\rm CP}$ 212° against 132°, Δm^2_{21} 7.537×10^{-5} against 7.41×10^{-5} eV^2, which is 3.4σ of the prior). In addition, the branch for angles in degrees converts a value that is already in degrees. The NuFIT 6.1 constraints of Sec. 13.2 will replace these priors. FrequentistAnalysis was retired on 2026-10-12; the analysis package that replaced it applies its priors and checks their names against the model's parameters (Sec. 6).

A.16 The thermal kernels' grid (2026-10-21)

What went wrong.—With the relics thermal in every collision (Sec. 13.5), the resonance factors' exact thermal averages are computed on a grid in $\ln(2E(1+z))$ and interpolated in their logarithm with four points. The grid merged a regular one with finer ones around slow relics' resonances, so two nodes could lie arbitrarily close together. Deep in the thermal tail the averages' quadrature is good to about 1%, and an interpolation through two nodes ϵ apart whose values differ by that noise overshoots by roughly 1% × spacing/ϵ: by e^{66} at $\epsilon \sim 10^{-6}$, to infinity below $\sim 10^{-7}$. An infinite kernel stopped a fit (a grid-extension task at 453 keV, and the potential planes' points at 1–10 keV, whose fits had failed); a large finite one absorbed isolated energies outright. In 48 random MeV points, two were off by 0.66 and 0.76 of the free flux, once inside the data range (2–18 TeV).

The fix.—Neighbouring nodes at least half the finest spacing apart. Against the reference, which uses no such grid, the worst points then agree to 3.4×10^{-5} of the free flux.

The campaign.—Every stored solution that went through this code was re-evaluated with the fix (125 458 evaluations; with the old grid, at a tenth of them, as a control: it gives back the stored χ^2 exactly for MESE and the projection), and refitted where χ^2 moved by more than 0.05 or the fit had failed. The fix moved χ^2 by more than 0.05 at 0.5–0.8% of points, by at most about 1.5. No robust 95% edge moved. One artifact went: in the MESE $L_e - L_\tau$ planes at 1 MeV, the bug had pulled one point below 3.84 among neighbours at 4.37, which made a thin excluded sliver from $10^{-4.96}$ to $10^{-4.6}$. A few columns where $\Delta\chi^2$ hovers within about 0.07 of 3.84 over a wide range of couplings changed their reading; and the potential planes gained their 1–10 keV columns. Along the way: the HESE+DR2 χ^2 is not exactly reproducible on re-evaluation at 0.6% of points (by up to about 1.5, its inner fit of the detector nuisances). **[Look at the HESE+DR2 reproducibility]**

A.17 Two rounding failures below 1 keV (2026-10-24)

Both came from the work of 2026-10-22, and both removed flux that the physics keeps. In both, the rates far exceed the losses, and a quantity that should vanish exactly (a rate matrix's zero mode) came out as 10^{-16} of the largest rate.

dp_si's bins.—Within a bin, the quasi-elastic conversions between mass states reach 10^{16}–10^{30} per step below $m_{Z_D} \sim 10^{-5}$ eV, against losses of order 1. In the diagonal, $-\mathrm{col} - \mathrm{loss}$ rounded to $-\mathrm{col}$, and the exponential kept only the rounding. The October 22 fix had removed the NaN and the overflow but left this loss. It reached 99.9% at $g_D \geq 10^{-4}$ (band-mean flux 0.001 at 10^{-30}–10^{-6} eV). Fix (`dp_si._bin_matrix`): beyond 10^8 events per step the conversions are scaled down to 10^6 per step, at least 10^7 times the bin's largest loss and 10^3 per state. That keeps their equilibrium within the bin, and the bin's loss rate on it, to about 10^{-7}. Below 10^8 per step, which covers every MeV plane, nothing changed.

The jumps' step.—Through `eigh`, the zero eigenvalue of the uniform mode (the total) came out as about -10^{-16} of the largest rate. Beyond rates $\times\, dt \sim 10^{16}$ it removed up to half of the flux (0.53 at 12 TeV for the μ texture at 10^{-8} eV, $g_D = 2 \times 10^{-5}$). That happens along a diagonal band, $g_D \propto m_{Z_D}^{1/2}$, where the Zeno suppression sets in and the effective rates peak. Fix (`dp_cosmo._soft_jumps`): the step is taken on the complement of the uniform mode, with the 2×2 eigensystem in closed form and its determinant, $3(r_{01}r_{02} + r_{01}r_{12} + r_{02}r_{12})$, taken from the rates without cancellation. CI had flagged the step on 2026-10-24: a conservation test lost 9×10^{-8} under CI's NumPy.

How they were found and checked.—The preview plots showed the dark portal's ultralight exclusion flat at exactly $g_D = 10^{-4}$, where the routing switches to dp_si. A transport audit followed (`tools/audit_transport.py`: 19 466 fluxes over every family, at 22 masses, about 15 couplings and each mode, plus the rerun's code at the 7 648 routed points, 2 000–3 200 cores for about 30 min). Checks: a synthetic bin against the fast-mixing limit $\pi e^{-\pi \cdot l} 1^T$ to 10^{-7} at 10^8–10^{30} per step; the jump step against `expm` to 10^{-10}, with the total kept to 10^{-14} at rates 10^{-3}–10^{30} (both now in `tests/test_relic_scattering.py`); at the worst points, flux 1.0000 where it had been 0.001 and 0.53.

What they touched.—The dark-portal ultralight planes for $g_D \gtrsim 10^{-7.3}$, at masses up to 0.1 eV, and the ultralight L rows with $g \geq 10^{-4.5}$, which are deep inside the excluded region. The potential planes and the MeV planes are unchanged, to 10^{-14}. The rerun and refinement there were stopped on 2026-10-24 and will be redone once Sec. 13.8 is settled.

A.18 Earlier corrections

Thermal kernel (July 2026).—The midpoint-rule momentum integral erred by 6–9% at on-resonance cells; validation against the previous code could not reveal this, since it measures only differences. A dense reference integration did. The semi-analytic kernel replaced it, which changed the fluxes at the dips by up to 15% with flavor ratios within 10^{-6} and dip energies unchanged (round 7).

Earth steps (July 2026).—The left-endpoint stepping scattered the results at the 10^{-2} level under any change of the step count (Sec. 4.5); in-band energy-averaged flavor ratios carried a 1.7×10^{-2} ambiguity at the analysis settings. The phase-exact scheme removed it.

Coupling cache (July 2026).—The cache never activated in real scans (Sec. 4.2); its narrower brackets fixed that without loss of accuracy.

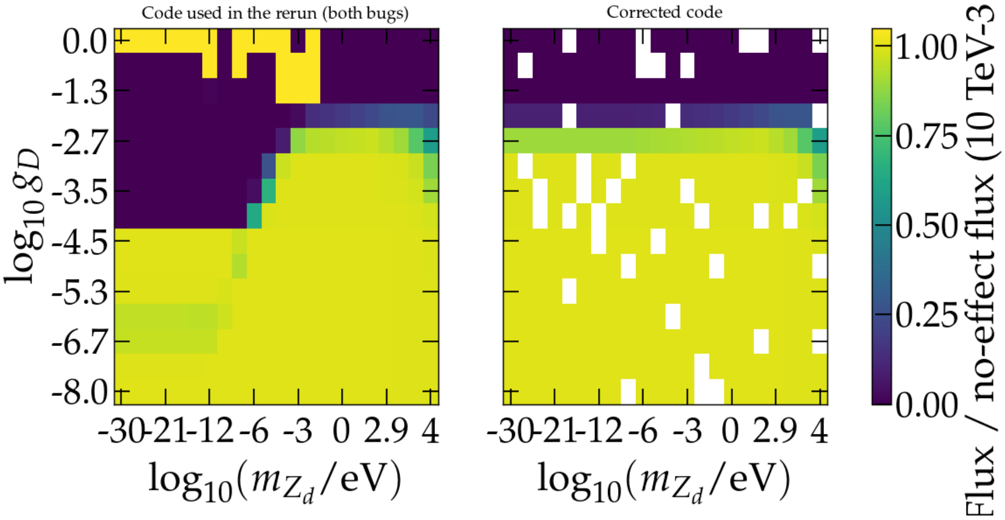


Figure 30: The dark portal's μ-texture ultralight plane: band-mean flux (10 TeV–3 PeV) over the no-effect flux. Left, the code the rerun used: the block at $g_D \geq 10^{-4}$ is dp_si's bins; the faint bands and the diagonal edge are the jumps' step. Right, both corrected. Only the true energy losses at $g_D \geq 10^{-2.7}$ remain. White: points not yet evaluated when the figure was made.